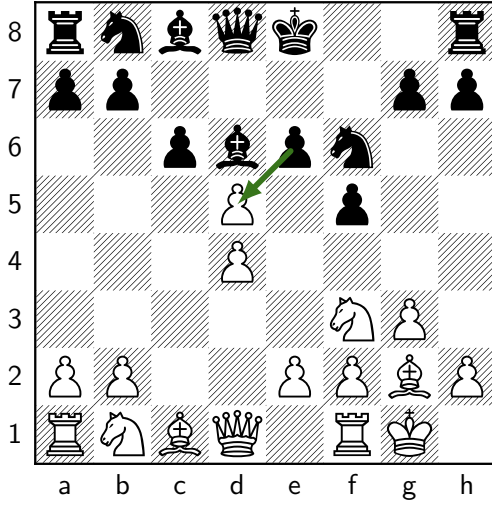


Melanie-BASE — Dutch, Stonewall

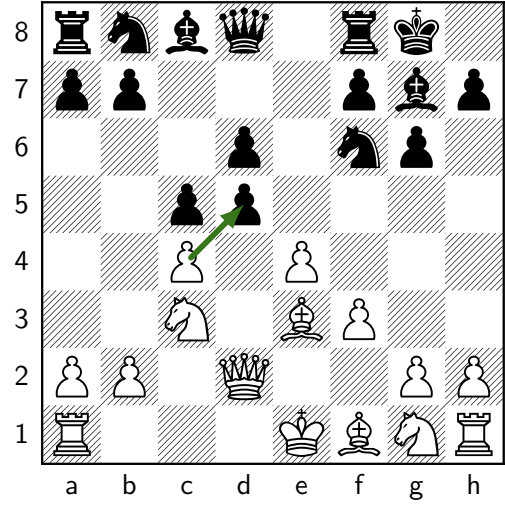
Melanie (Deep_Vision)

Abstract

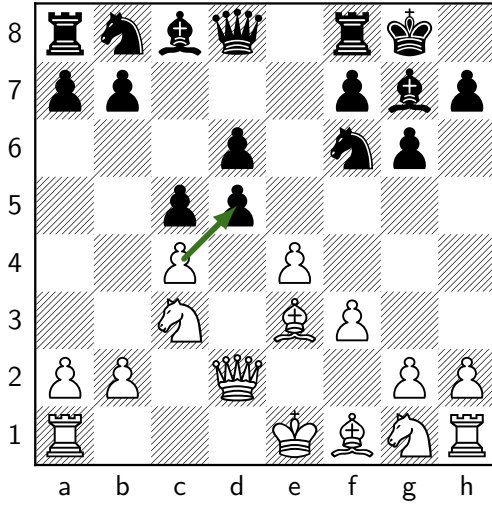
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie's move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen's role) are called out, as no material/eval engine explains a move that way.



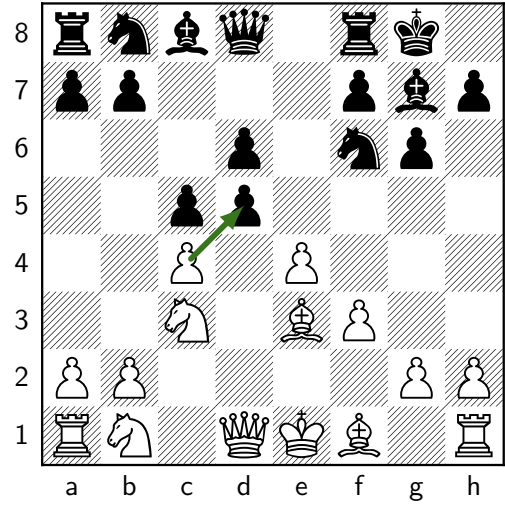
(a)



(b)

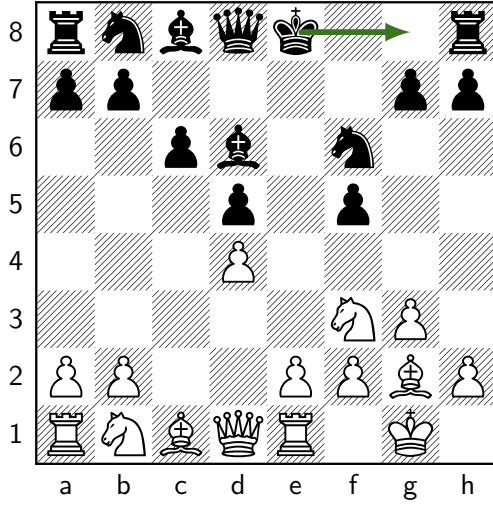


(c)

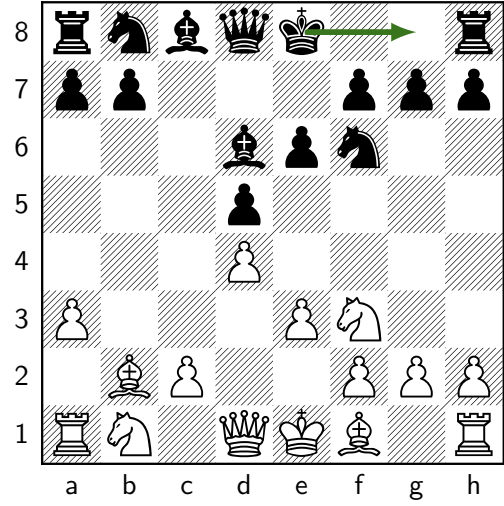


(d)

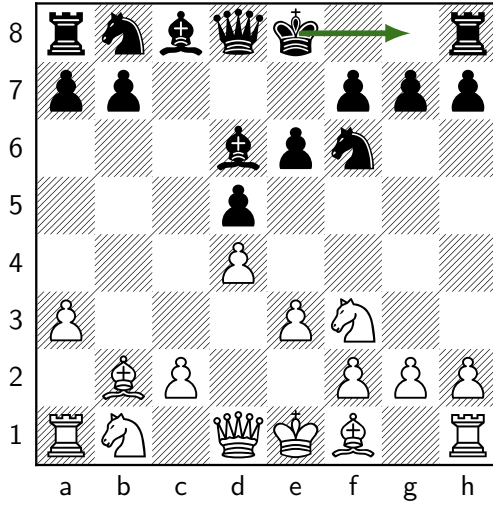
Figure 1: **Move 7, Black: exd5.** (a) Our position; Melanie plays exd5 (e6d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#1703 (ply 16) [structural distance $d=46$, QAP obj=175]: the master played cxd5; under μ our $Pe3$ plays the role of that game’s $Pc4$; cross-type roles: our $nf3$ as their $Bf1$, our $pb2$ as their $Ng1$; transferred move e6d5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#1704 (ply 16) [structural distance $d=46$, QAP obj=175]: the master played cxd5; under μ our $Pe3$ plays the role of that game’s $Pc4$; cross-type roles: our $nf3$ as their $Bf1$, our $pb2$ as their $Ng1$; transferred move e6d5 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#2850 (ply 16) [structural distance $d=45$, QAP obj=172]: the master played cxd5; under μ our $Pe3$ plays the role of that game’s $Pc4$; cross-type roles: our $Qd8$ as their $Ra1$, our $Bc8$ as their $Nb1$; transferred move e6d5 (**endorses this move**). Synthesis: 30 retrieved analogs transferred to exd5 (summed QAP objective 2947, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



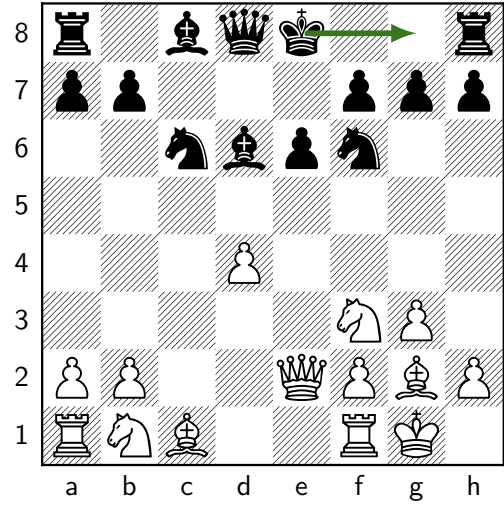
(a)



(b)

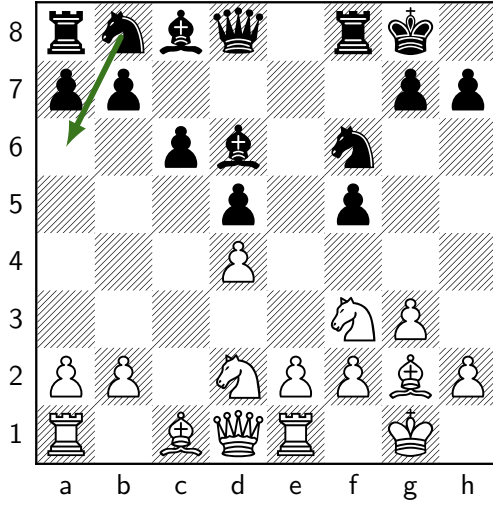


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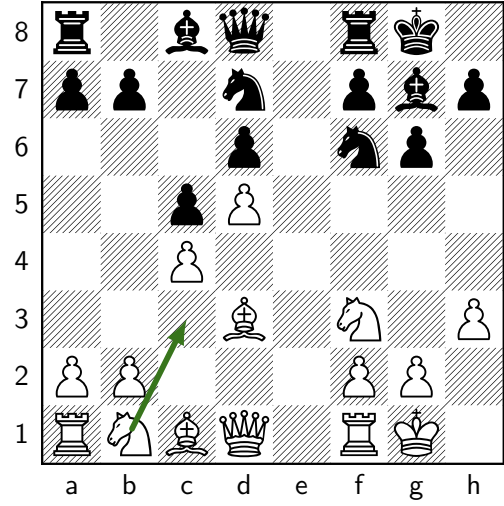


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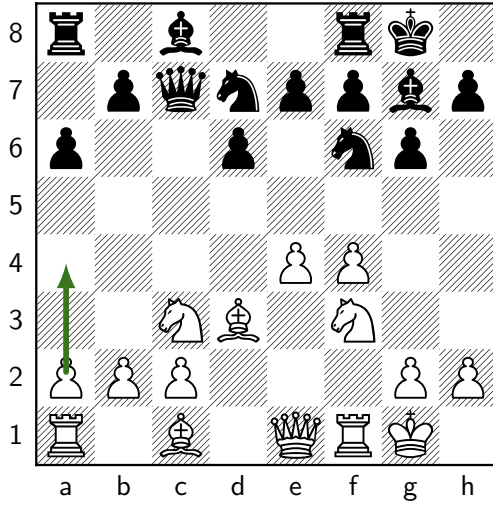
Figure 2: **Move 8, Black: 0-0.** (a) Our position; Melanie plays 0-0 (e8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2763` (ply 13) [structural distance $d=36$, QAP obj=42]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pd5* as their *Bc8*; transferred move e8g8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2764` (ply 13) [structural distance $d=36$, QAP obj=42]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Bc8* as their *Ra8*, our *Pd5* as their *Bc8*; transferred move e8g8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#163` (ply 19) [structural distance $d=48$, QAP obj=29]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Nb8* as their *Pa7*, our *nb1* as their *Pb7*; transferred move e8g8 (**endorses this move**). Synthesis: 21 retrieved analogs transferred to 0-0 (summed QAP objective 461, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



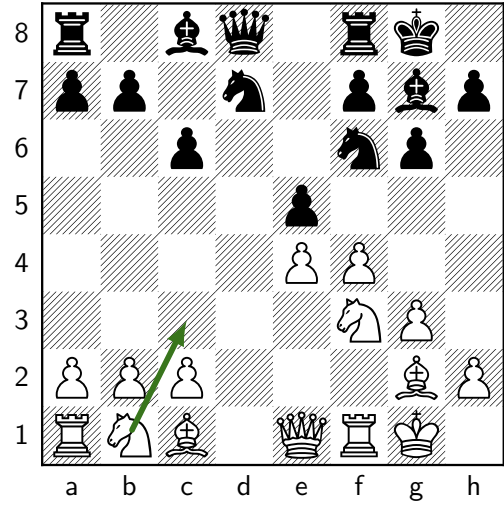
(a)



(b)

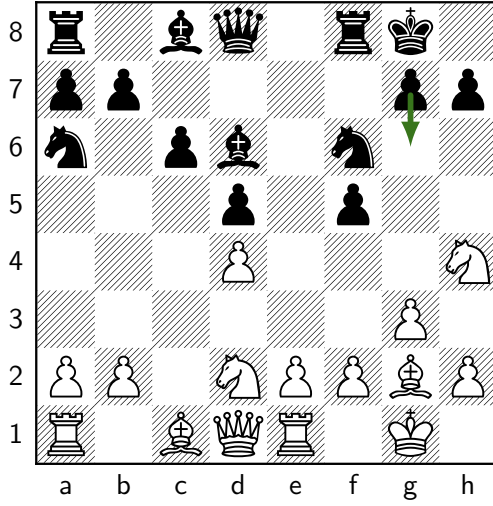


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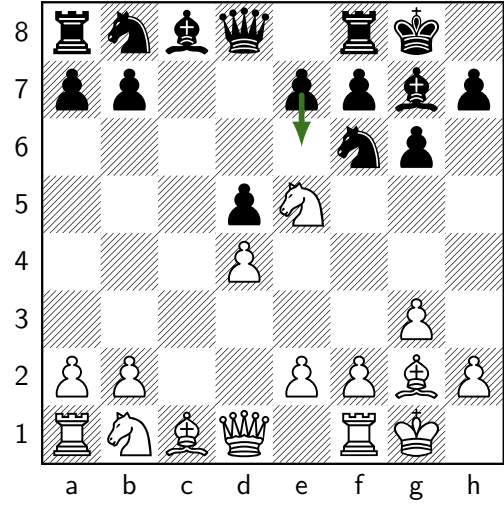


(d)

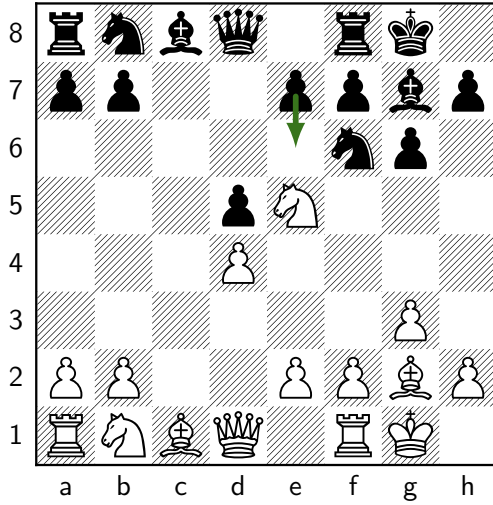
Figure 3: **Move 9, Black: Na6**. (a) Our position; Melanie plays Na6 (b8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#792` (ply 18) [structural distance $d=48$, QAP obj=26]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Pf5 as their Bc1, our pf2 as their Qd1; transferred move b8a6 (**endorses this move**). (c) Retrieved from `cctl_top_p0003#510` (ply 20) [structural distance $d=57$, QAP obj=17]: the master played a4; under μ our Nb1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our pf2 as their Qe1, our pg3 as their Nc3; transferred move b8a6 (**endorses this move**). (d) Retrieved from `cctl_top_p0002#2301` (ply 18) [structural distance $d=57$, QAP obj=16]: the master played Nc3; under μ our Nb1 plays the role of that game’s Nb1; cross-type roles: our Qd8 as their Bc1, our Nf6 as their Qe1; transferred move b8a6 (**endorses this move**). Synthesis: 8 retrieved analogs transferred to Na6 (summed QAP objective 120, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



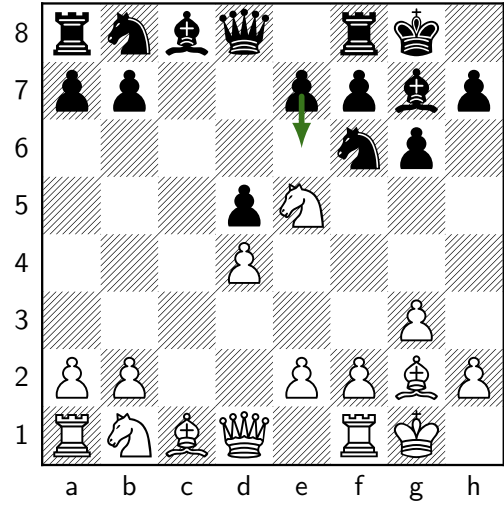
(a)



(b)

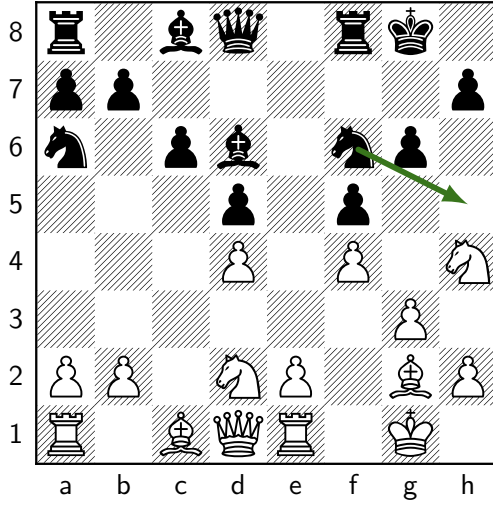


(c)

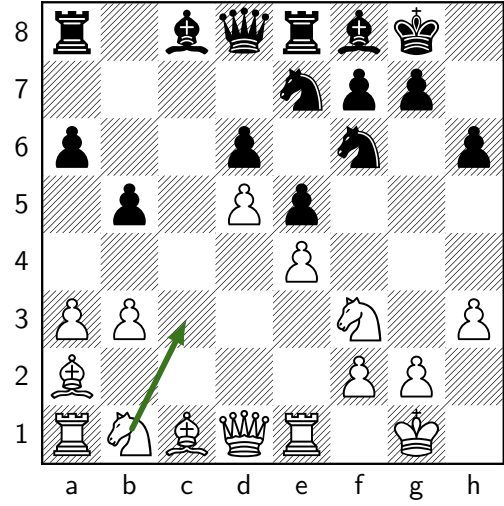


(d)

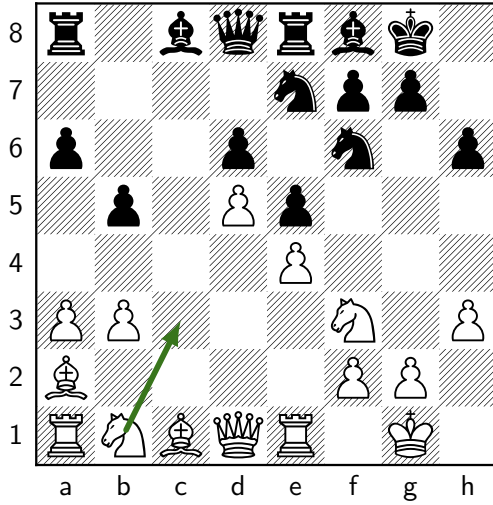
Figure 4: **Move 10, Black: g6.** (a) Our position; Melanie plays g6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0001#88` (ply 15) [structural distance $d=58$, QAP obj=53]: the master played e6; under μ our $Pg2$ plays the role of that game’s $Pe2$; cross-type roles: our $Bc8$ as their $Nb8$, our $Nf6$ as their $Bc8$; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top.p0001#89` (ply 15) [structural distance $d=58$, QAP obj=53]: the master played e6; under μ our $Pg2$ plays the role of that game’s $Pe2$; cross-type roles: our $Bc8$ as their $Nb8$, our $Nf6$ as their $Bc8$; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top.p0003#2420` (ply 15) [structural distance $d=58$, QAP obj=53]: the master played e6; under μ our $Pg2$ plays the role of that game’s $Pe2$; cross-type roles: our $Bc8$ as their $Nb8$, our $Nf6$ as their $Bc8$; transferred move g7g6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to g6 (summed QAP objective 212, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



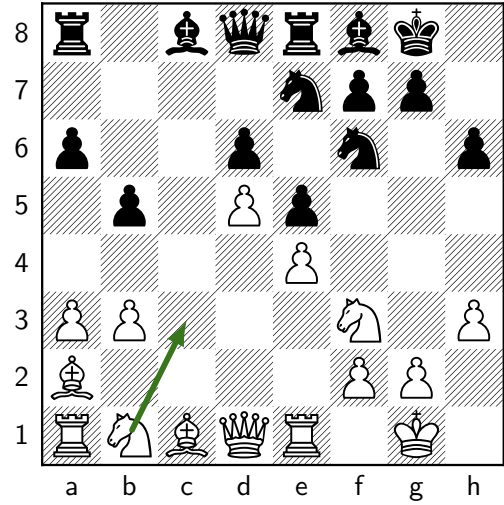
(a)



(b)

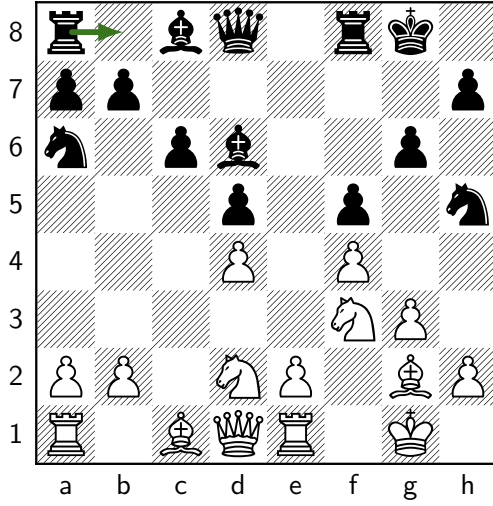


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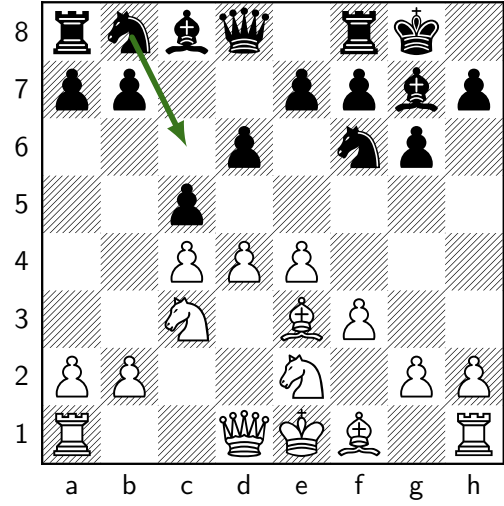


(d)

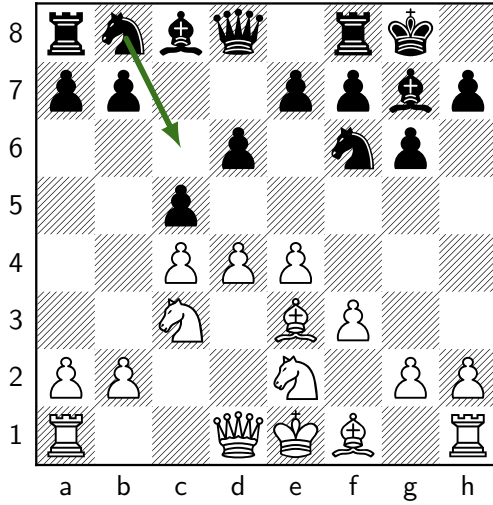
Figure 5: **Move 11, Black: Nh5**. (a) Our position; Melanie plays Nh5 (f6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#684 (ply 32) [structural distance $d=72$, QAP obj=100]: the master played Nc3; under μ our Nf3 plays the role of that game’s Nb1; cross-type roles: our Qd8 as their Ra1, our Ph7 as their Qd1; transferred move f6h5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2980 (ply 32) [structural distance $d=72$, QAP obj=100]: the master played Nc3; under μ our Nf3 plays the role of that game’s Nb1; cross-type roles: our Qd8 as their Ra1, our Ph7 as their Qd1; transferred move f6h5 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#3590 (ply 32) [structural distance $d=72$, QAP obj=100]: the master played Nc3; under μ our Nf3 plays the role of that game’s Nb1; cross-type roles: our Qd8 as their Ra1, our Ph7 as their Qd1; transferred move f6h5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nh5 (summed QAP objective 400, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



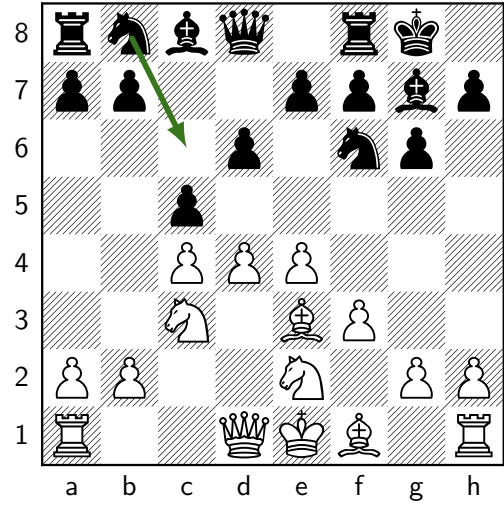
(a)



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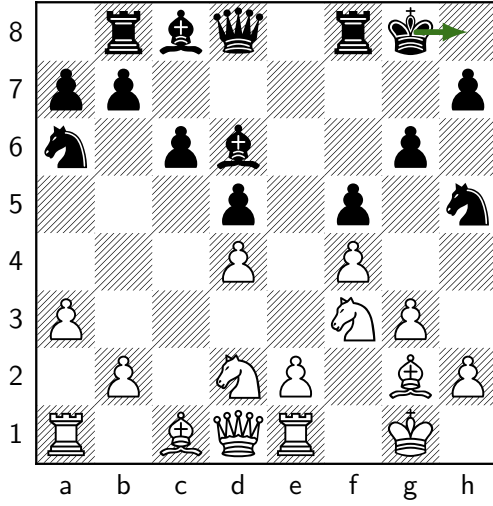


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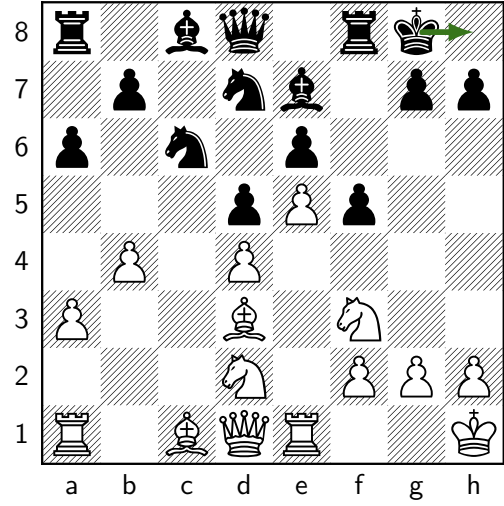


(d)

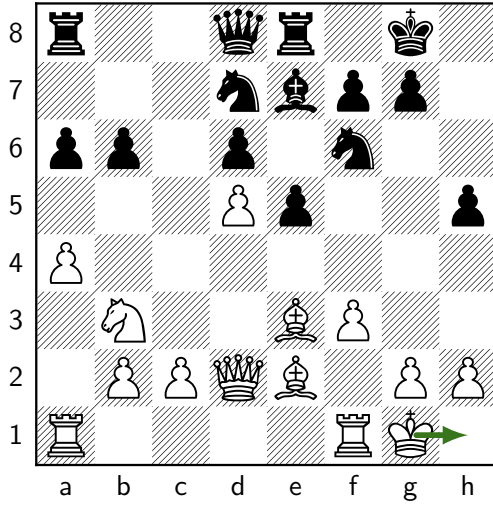
Figure 6: **Move 12, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1813` (ply 13) [structural distance $d=86$, QAP obj=23]: the master played Nc6; under μ our *Ra1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pc6* as their *Bc8*; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1814` (ply 13) [structural distance $d=86$, QAP obj=23]: the master played Nc6; under μ our *Ra1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pc6* as their *Bc8*; transferred move a8b8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3964` (ply 13) [structural distance $d=86$, QAP obj=23]: the master played Nc6; under μ our *Ra1* plays the role of that game’s *Nb1* (a **cross-type** correspondence); cross-type roles: our *Bc8* as their *Ra8*, our *Pc6* as their *Bc8*; transferred move a8b8 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Rb8 (summed QAP objective 224, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



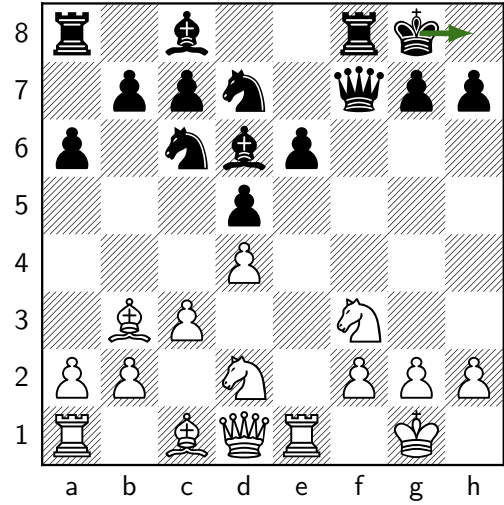
(a)



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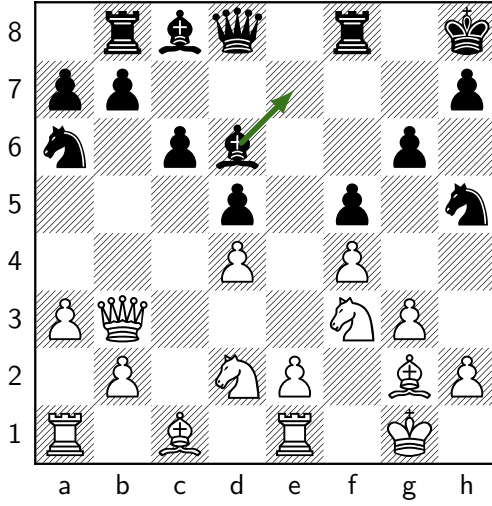


(c)

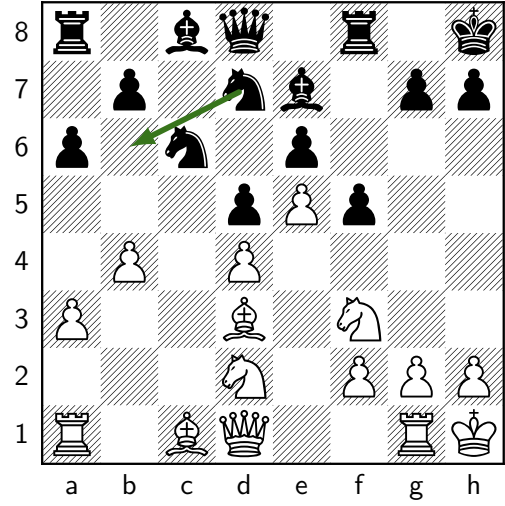


(d)

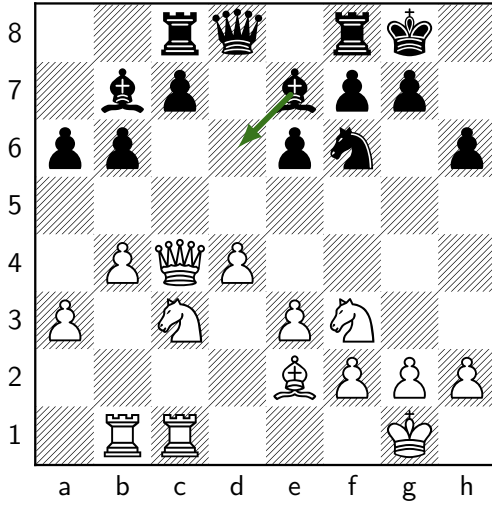
Figure 7: **Move 13, Black: Kh8.** (a) Our position; Melanie plays Kh8 (g8h8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1992` (ply 25) [structural distance $d=83$, QAP obj=148]: the master played Kh8; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Na6* as their *Bc8*, our *Bd6* as their *Nd7*; transferred move g8h8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2884` (ply 28) [structural distance $d=87$, QAP obj=48]: the master played Kh1; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Pd5* as their *Ra1*, our *Rb8* as their *Pg2*; transferred move g8h8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2006` (ply 23) [structural distance $d=85$, QAP obj=37]: the master played Kh8; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *Pb7* as their *Bc8*, our *Qd8* as their *Pc7*; transferred move g8h8 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kh8 (summed QAP objective 233, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



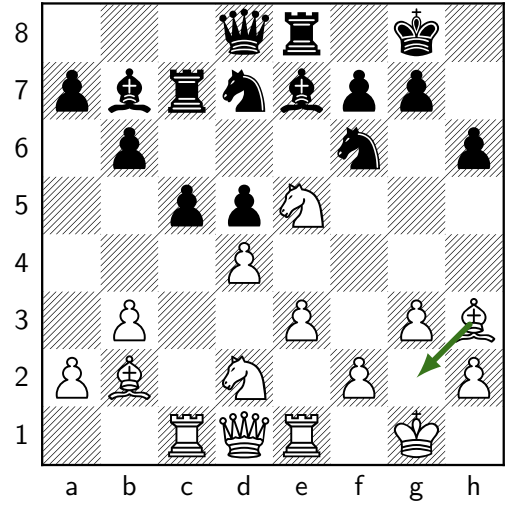
(a)



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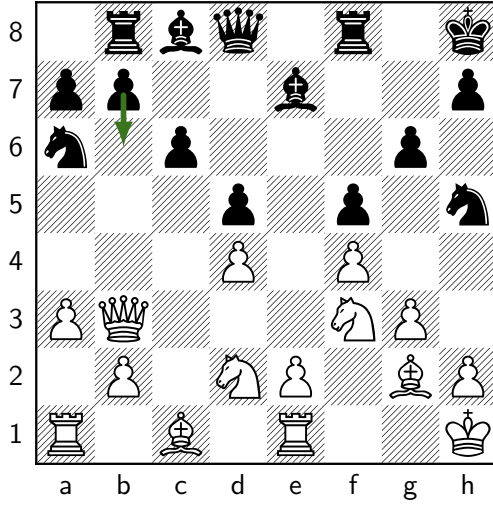


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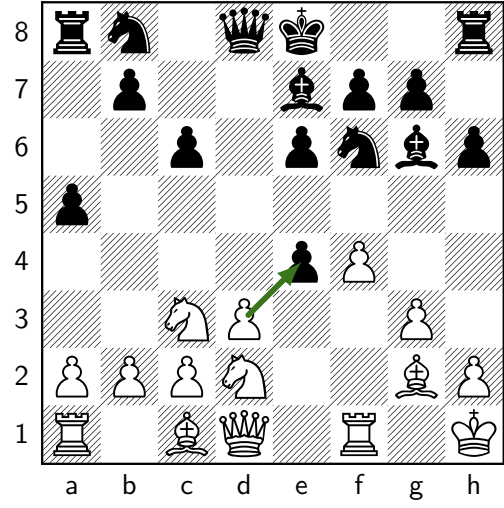


(d)

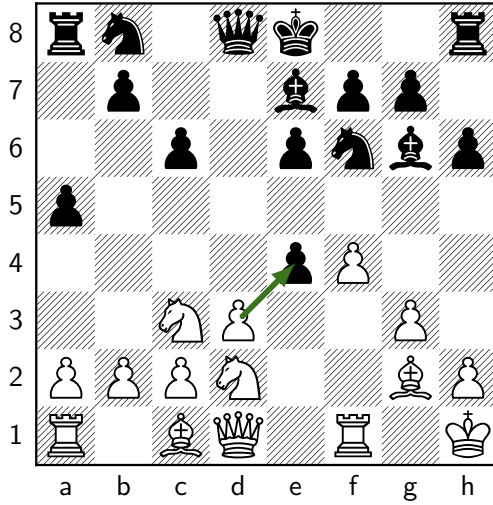
Figure 8: **Move 14, Black: Be7**. (a) Our position; Melanie plays Be7 (d6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1992` (ply 27) [structural distance $d=95$, QAP obj=143]: the master played Nb6; under μ our Bd3 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Ra8, our Pc6 as their Bc8; transferred move d6e7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1984` (ply 31) [structural distance $d=99$, QAP obj=108]: the master played Bd6; under μ our Bd3 plays the role of that game’s Be2; cross-type roles: our Bd6 as their Qd8, our Qd8 as their Rf8; transferred move d6e7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3001` (ply 30) [structural distance $d=95$, QAP obj=54]: the master played Bg2; under μ our Bd3 plays the role of that game’s Bh3; cross-type roles: our qb3 as their Rc1, our pb2 as their Qd1; transferred move d6e7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Be7 (summed QAP objective 402, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



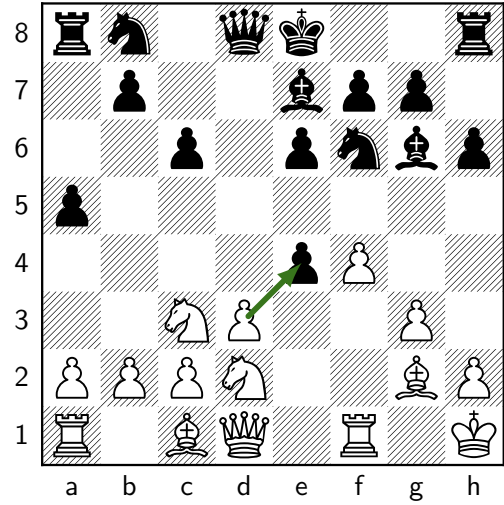
(a)



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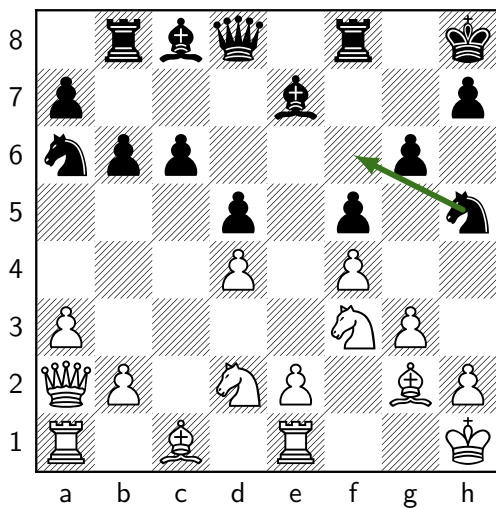


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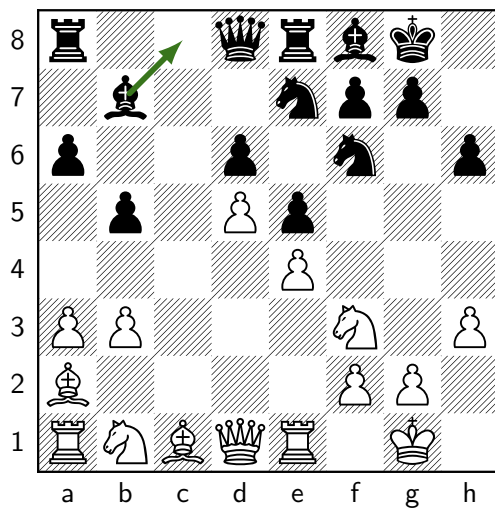


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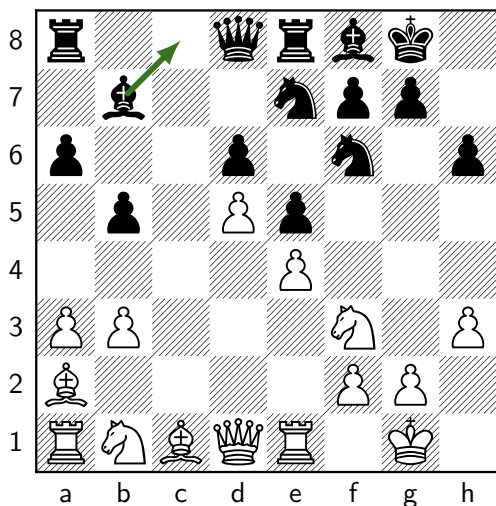
Figure 9: **Move 15, Black: b6.** (a) Our position; Melanie plays b6 (b7b6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#226` (ply 20) [structural distance $d=86$, QAP obj=45]: the master played `dxe4`; under μ our *Pb2* plays the role of that game’s *Pd3*; cross-type roles: our *Na6* as their *Ra1*, our *Pc6* as their *Bc1*; transferred move `b7b6` (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#227` (ply 20) [structural distance $d=86$, QAP obj=45]: the master played `dxe4`; under μ our *Pb2* plays the role of that game’s *Pd3*; cross-type roles: our *Na6* as their *Ra1*, our *Pc6* as their *Bc1*; transferred move `b7b6` (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3520` (ply 20) [structural distance $d=86$, QAP obj=45]: the master played `dxe4`; under μ our *Pb2* plays the role of that game’s *Pd3*; cross-type roles: our *Na6* as their *Ra1*, our *Pc6* as their *Bc1*; transferred move `b7b6` (**endorses this move**). Synthesis: 8 retrieved analogs transferred to b6 (summed QAP objective 206, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



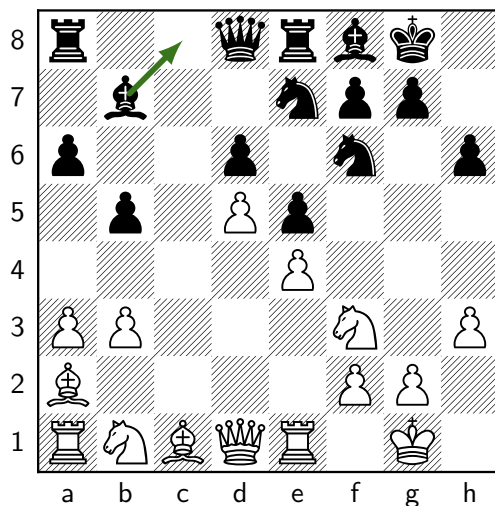
(a)



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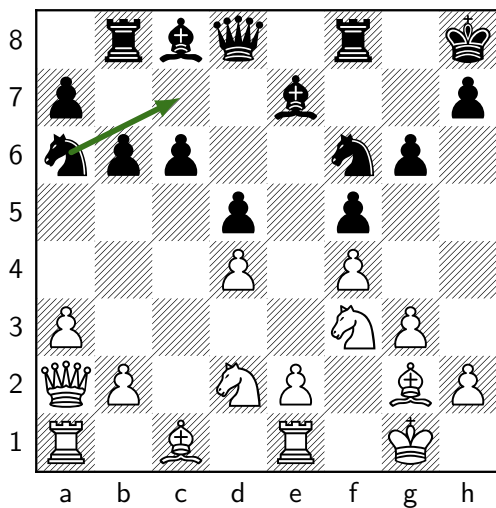


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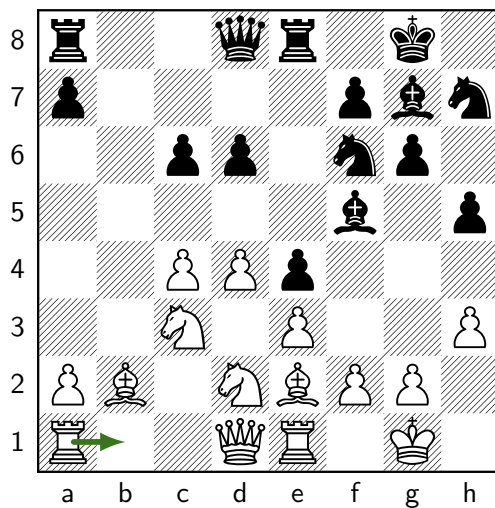


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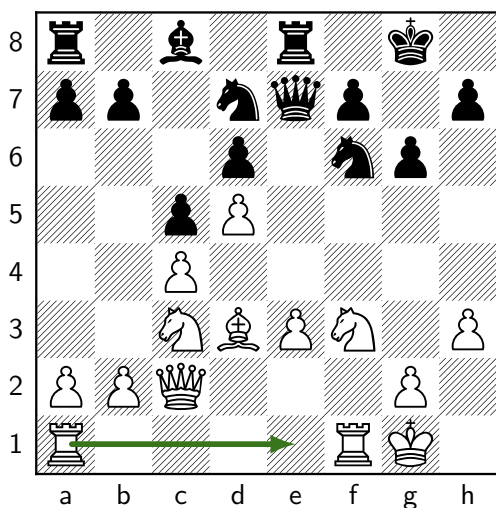
Figure 10: **Move 16, Black: Nf6**. (a) Our position; Melanie plays Nf6 (h5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#684` (ply 31) [structural distance $d=80$, QAP obj=110]: the master played Bc8; under μ our Nh4 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our nd2 as their Ra8, our pd4 as their Qd8; transferred move h5f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2980` (ply 31) [structural distance $d=80$, QAP obj=110]: the master played Bc8; under μ our Nh4 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our nd2 as their Ra8, our pd4 as their Qd8; transferred move h5f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3590` (ply 31) [structural distance $d=80$, QAP obj=110]: the master played Bc8; under μ our Nh4 plays the role of that game’s Bb2 (a **cross-type** correspondence); cross-type roles: our nd2 as their Ra8, our pd4 as their Qd8; transferred move h5f6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Nf6 (summed QAP objective 676, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



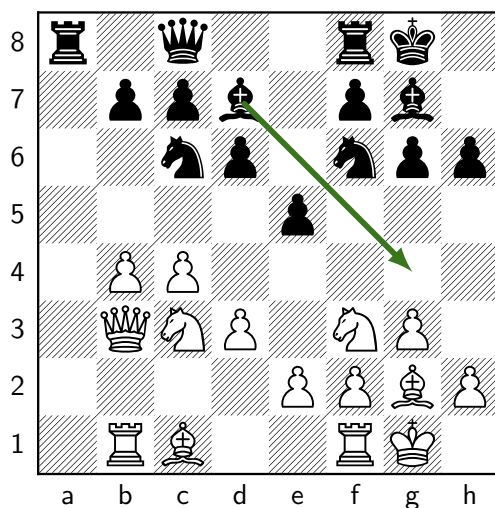
(a)



(b)

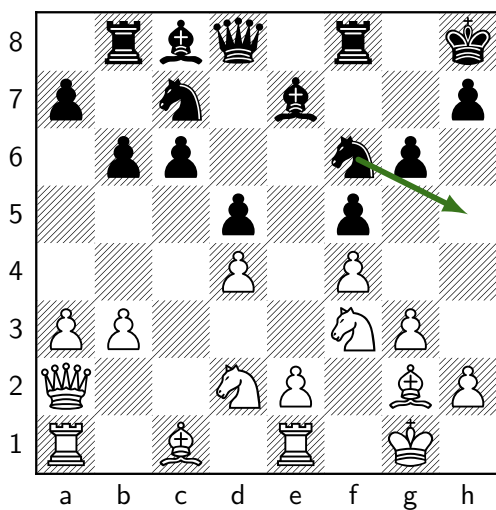


(c)

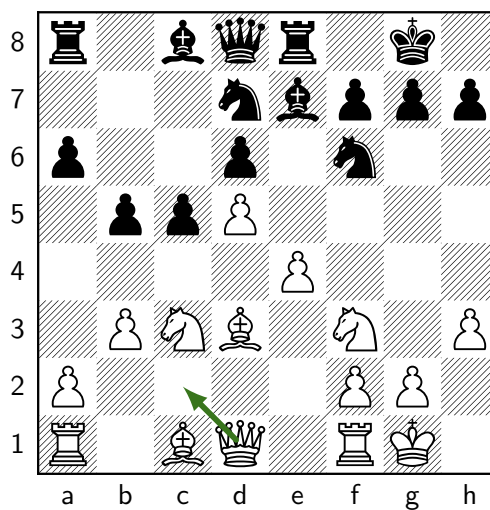


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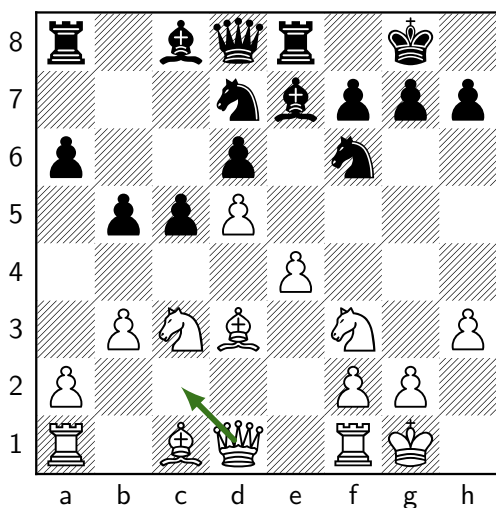
Figure 11: **Move 17, Black: Nc7.** (a) Our position; Melanie plays Nc7 (a6c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0001#3797` (ply 30) [structural distance $d=75$, QAP obj=45]: the master played Rb1; under μ our Na3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Pa2, our Pb6 as their Bb2; transferred move a6c7 (**endorses this move**). (c) Retrieved from `ccr1.top_p0000#1105` (ply 26) [structural distance $d=71$, QAP obj=32]: the master played Rae1; under μ our Na3 plays the role of that game’s Ra1 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Pa2, our Pg6 as their Nf3; transferred move a6c7 (**endorses this move**). (d) Retrieved from `ccr1.top_p0002#1700` (ply 25) [structural distance $d=76$, QAP obj=31]: the master played Bg4; under μ our Na3 plays the role of that game’s Bd2 (a **cross-type** correspondence); cross-type roles: our nf3 as their Ra8, our pf4 as their Qc8; transferred move a6c7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Nc7 (summed QAP objective 158, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



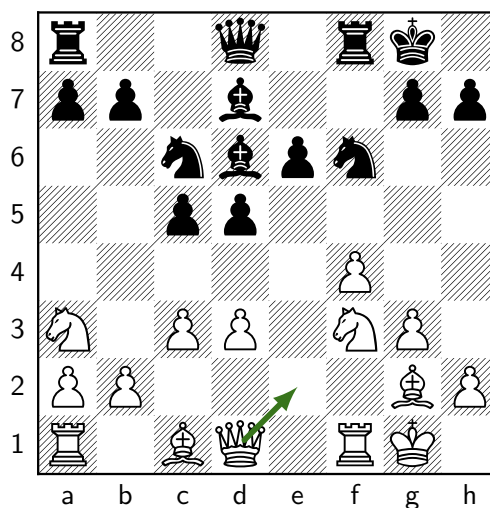
(a)



(b)

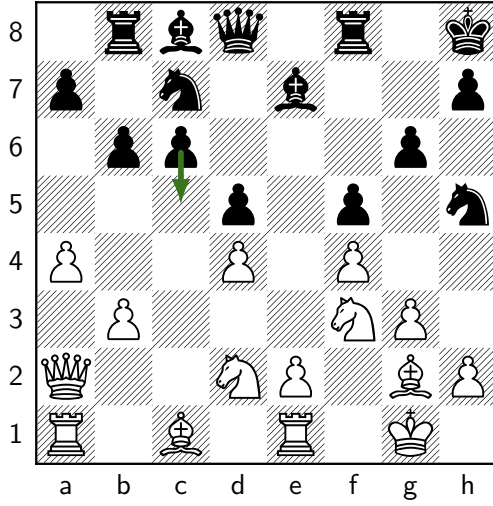


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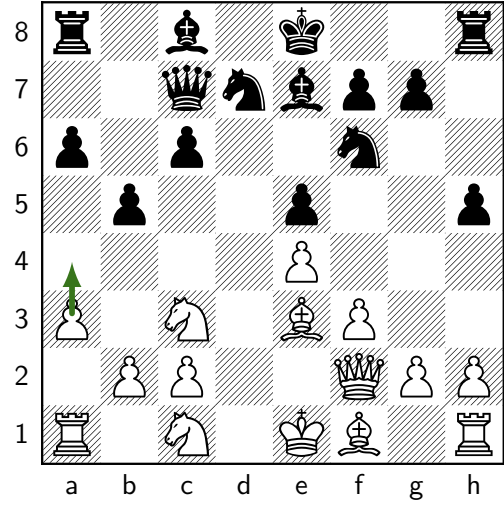


(d)

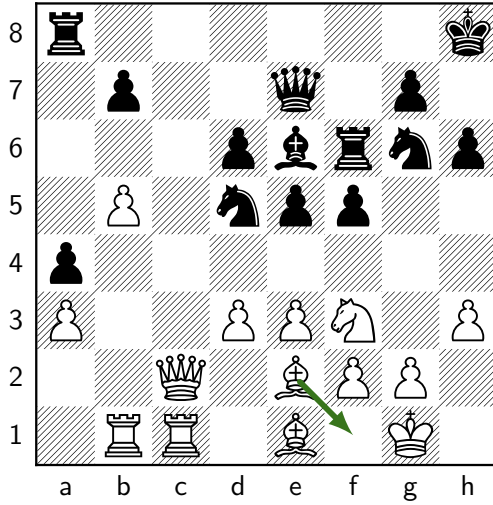
Figure 12: **Move 18, Black: Nh5**. (a) Our position; Melanie plays Nh5 (f6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#854 (ply 24) [structural distance $d=79$, QAP obj=28]: the master played Qc2; under μ our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pf4 as their Ra1, our pg3 as their Bc1; transferred move f6h5 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#855 (ply 24) [structural distance $d=79$, QAP obj=28]: the master played Qc2; under μ our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pf4 as their Ra1, our pg3 as their Bc1; transferred move f6h5 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#2251 (ply 22) [structural distance $d=70$, QAP obj=15]: the master played Qe2; under μ our Nf3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Nc7 as their Ra1, our qa2 as their Bc1; transferred move f6h5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Nh5 (summed QAP objective 70, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



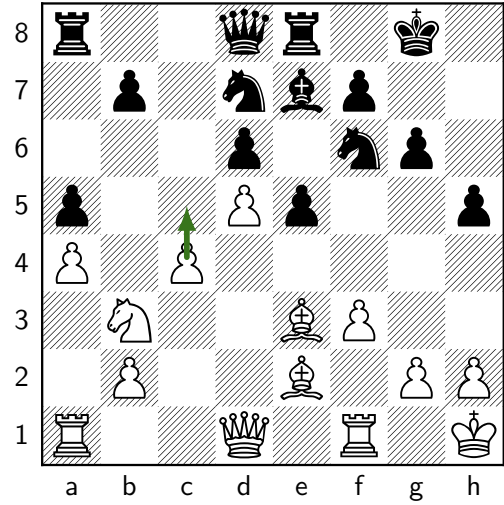
(a)



(b)

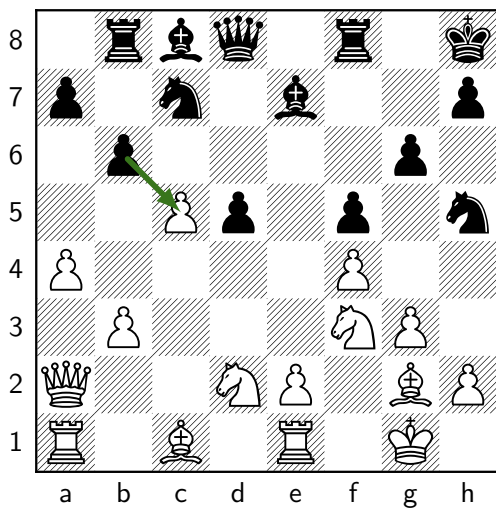


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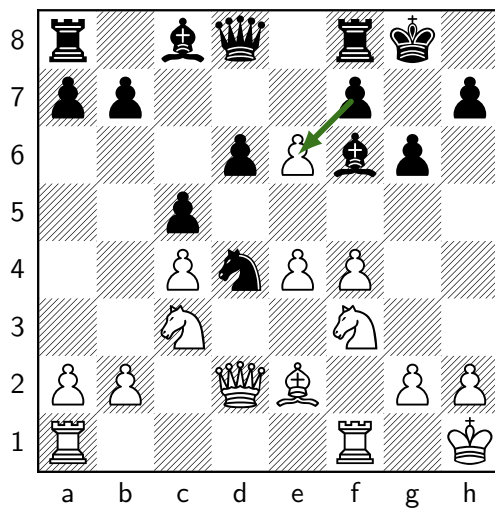


(d)

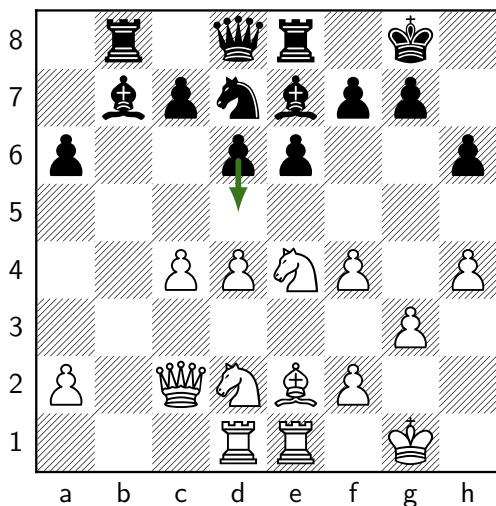
Figure 13: **Move 19, Black: c5**. (a) Our position; Melanie plays c5 (c6c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3868` (ply 22) [structural distance $d=77$, QAP obj=44]: the master played a4; under μ our $Pc3$ plays the role of that game’s $Pa3$; cross-type roles: our $pb3$ as their $Ra1$, our $Rf8$ as their $Nc1$; transferred move c6c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1917` (ply 46) [structural distance $d=91$, QAP obj=40]: the master played Bf1; under μ our $Pc3$ plays the role of that game’s $Be2$ (a **cross-type** correspondence); cross-type roles: our $bc1$ as their $Rb1$, our $Qd8$ as their $Rc1$; transferred move c6c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2958` (ply 30) [structural distance $d=86$, QAP obj=40]: the master played c5; under μ our $Pc3$ plays the role of that game’s $Pc4$; cross-type roles: our $Pf5$ as their $Ra1$, our $Nc7$ as their $Qd1$; transferred move c6c5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to c5 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



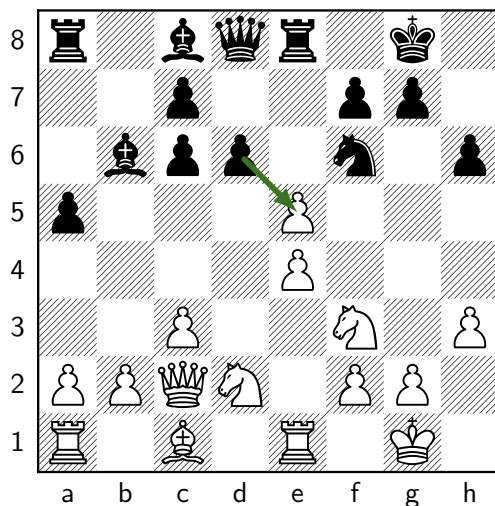
(a)



(b)

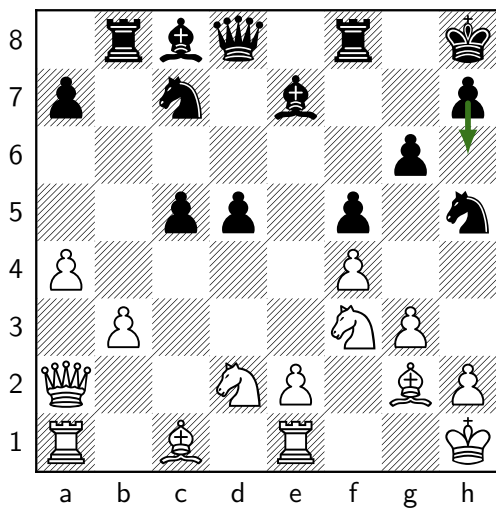


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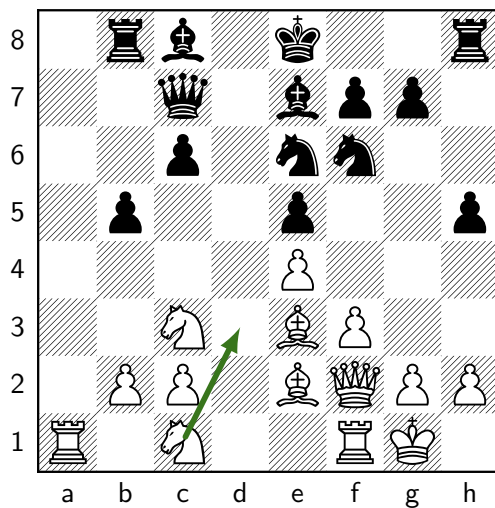


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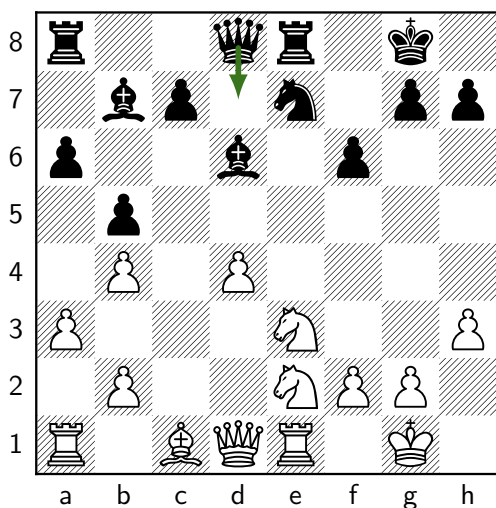
Figure 14: **Move 20, Black: bxc5.** (a) Our position; Melanie plays **bxc5** (**b6c5**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2220` (ply 27) [structural distance $d=70$, QAP obj=74]: the master played **fxe6**; under μ our *Pb3* plays the role of that game’s *Pf2*; cross-type roles: our *Pd5* as their *Ra8*, our *nf3* as their *Pb7*; transferred move **b6c5** (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#3474` (ply 37) [structural distance $d=80$, QAP obj=39]: the master played **d5**; under μ our *Pb3* plays the role of that game’s *Pd3*; cross-type roles: our *Ph7* as their *Qd8*, our *Pb6* as their *Bb7*; transferred move **b6c5** (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#3942` (ply 25) [structural distance $d=79$, QAP obj=31]: the master played **dxe5**; under μ our *Pb3* plays the role of that game’s *Pd3*; cross-type roles: our *pg3* as their *Ra8*, our *Pg6* as their *Bc8*; transferred move **b6c5** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **bxc5** (summed QAP objective 144, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



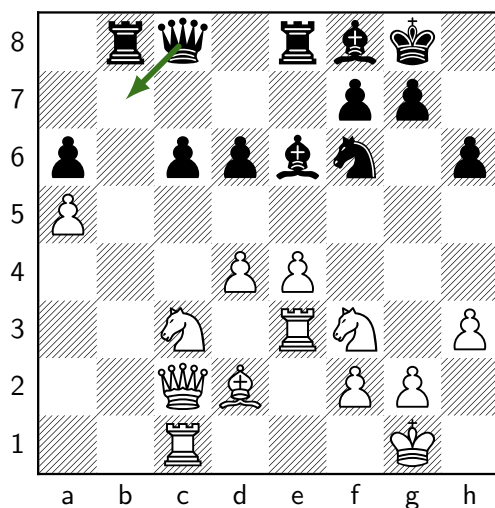
(a)



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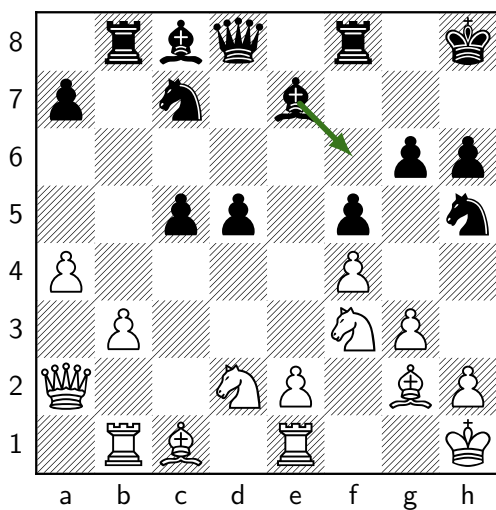


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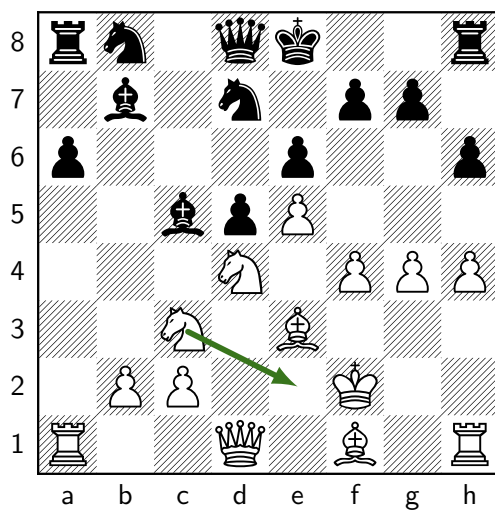


(d)

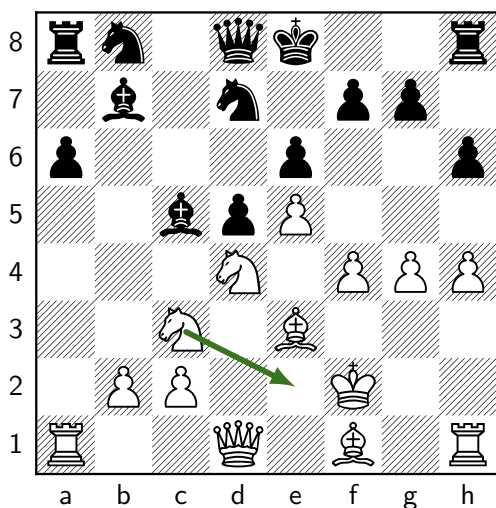
Figure 15: **Move 21, Black: h6**. (a) Our position; Melanie plays h6 (h7h6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#3868` (ply 30) [structural distance $d=84$, QAP obj=46]: the master played Nd3; under μ our *Ph2* plays the role of that game’s *Nc1* (a **cross-type** correspondence); cross-type roles: our *Be7* as their *Ra1*, our *Qd8* as their *Pc2*; transferred move h7h6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2641` (ply 37) [structural distance $d=73$, QAP obj=33]: the master played Qd7; under μ our *Ph2* plays the role of that game’s *Qd1* (a **cross-type** correspondence); cross-type roles: our *Nc7* as their *Ra8*, our *Rb8* as their *Bb7*; transferred move h7h6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1825` (ply 43) [structural distance $d=57$, QAP obj=32]: the master played Qb7; under μ our *Ph2* plays the role of that game’s *Qc1* (a **cross-type** correspondence); cross-type roles: our *nf3* as their *Rb8*, our *pb3* as their *Re8*; transferred move h7h6 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to h6 (summed QAP objective 156, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



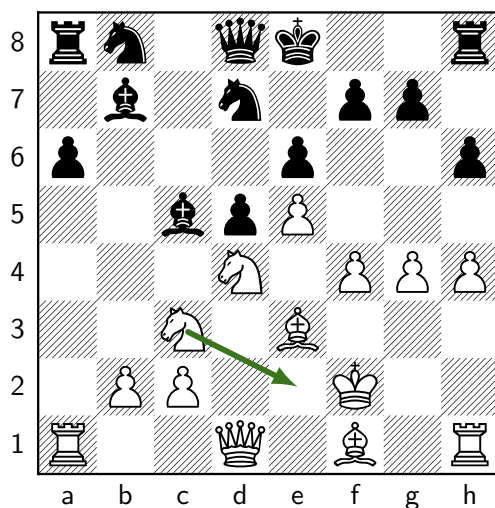
(a)



(b)

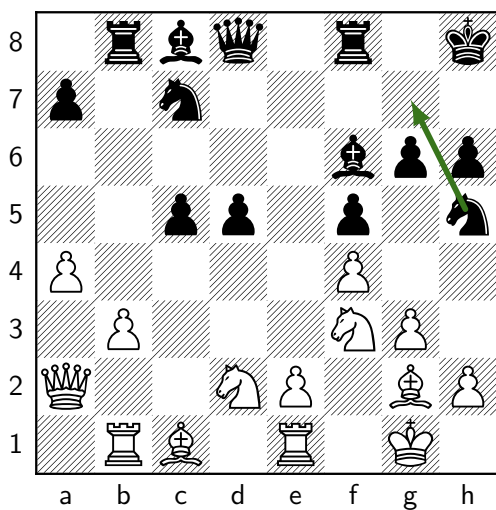


(c)

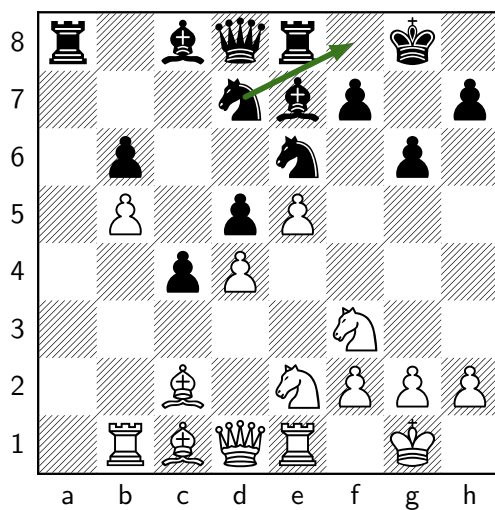


(d)

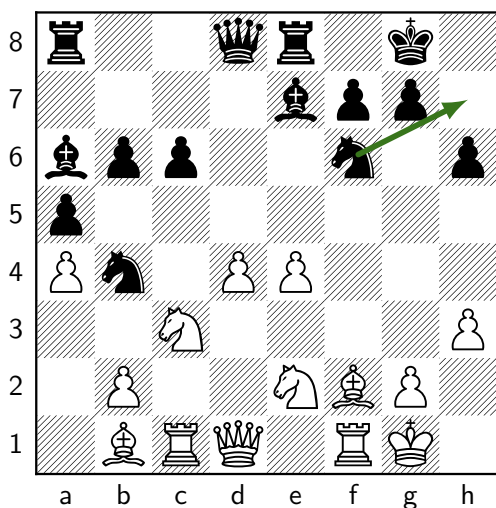
Figure 16: **Move 22, Black: Bf6**. (a) Our position; Melanie plays Bf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1439` (ply 28) [structural distance $d=102$, QAP obj=31]: the master played Nce2; under μ our Be2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Qd1, our Nc7 as their Bf1; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#2842` (ply 28) [structural distance $d=102$, QAP obj=31]: the master played Nce2; under μ our Be2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Qd1, our Nc7 as their Bf1; transferred move e7f6 (**endorses this move**). (d) Retrieved from `ccrl_top_p0000#2866` (ply 28) [structural distance $d=102$, QAP obj=31]: the master played Nce2; under μ our Be2 plays the role of that game’s Nc3 (a **cross-type** correspondence); cross-type roles: our Bc8 as their Qd1, our Nc7 as their Bf1; transferred move e7f6 (**endorses this move**). Synthesis: 9 retrieved analogs transferred to Bf6 (summed QAP objective 282, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



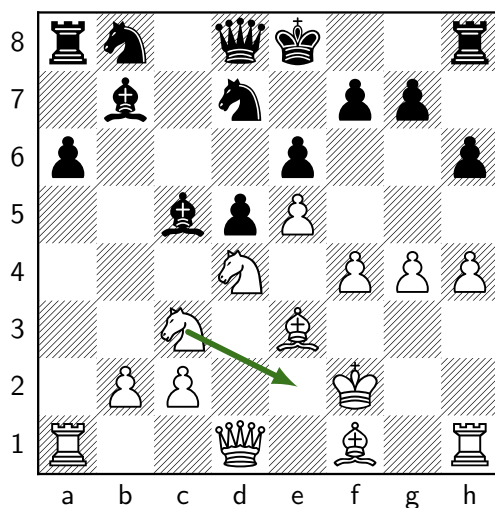
(a)



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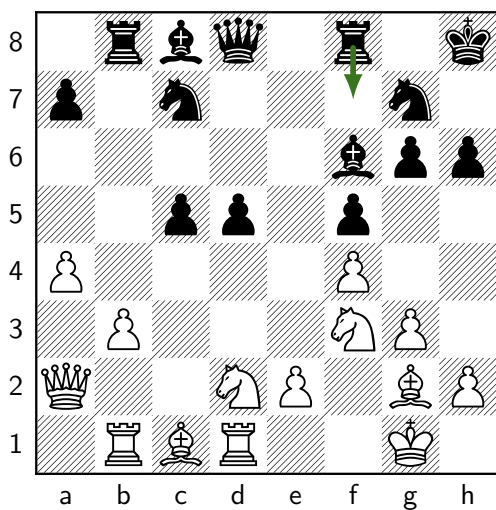


(c)

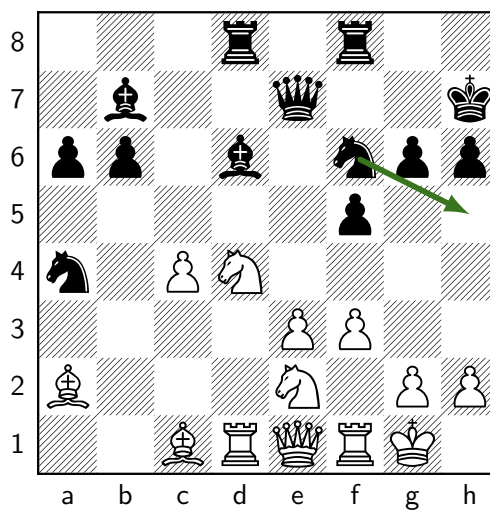


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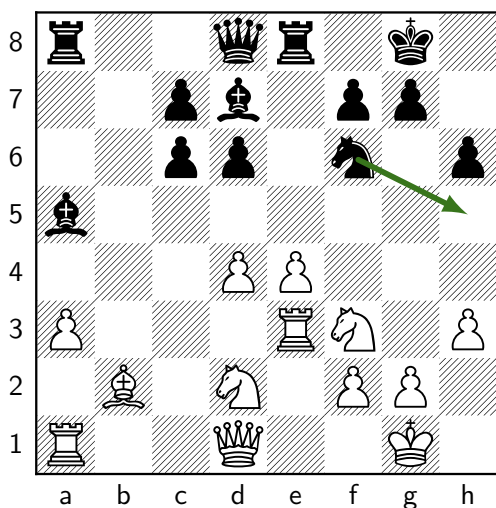
Figure 17: **Move 23, Black: Ng7**. (a) Our position; Melanie plays Ng7 (h5g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#81 (ply 37) [structural distance $d=105$, QAP obj=51]: the master played Ndf8; under μ our Nh4 plays the role of that game’s Nd2; cross-type roles: our Qd8 as their Ra8, our Pg6 as their Bc8; transferred move h5g7 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#937 (ply 35) [structural distance $d=123$, QAP obj=38]: the master played Nh7; under μ our Nh4 plays the role of that game’s Nf3; cross-type roles: our Ph6 as their Ra8, our Rb8 as their Qd8; transferred move h5g7 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#1439 (ply 28) [structural distance $d=118$, QAP obj=32]: the master played Nce2; under μ our Nh4 plays the role of that game’s Nc3; cross-type roles: our ph2 as their Qd1, our Pd5 as their Bf1; transferred move h5g7 (**endorses this move**). Synthesis: 11 retrieved analogs transferred to Ng7 (summed QAP objective 375, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



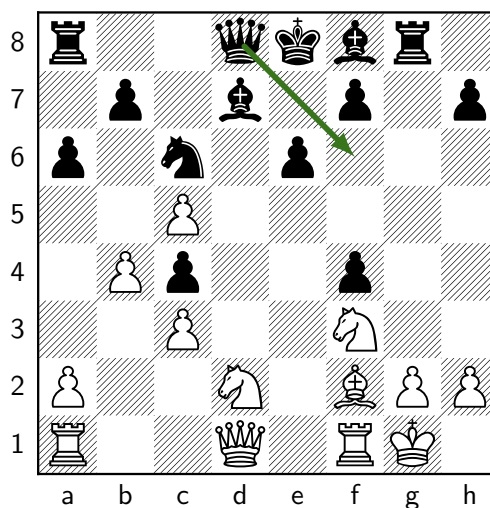
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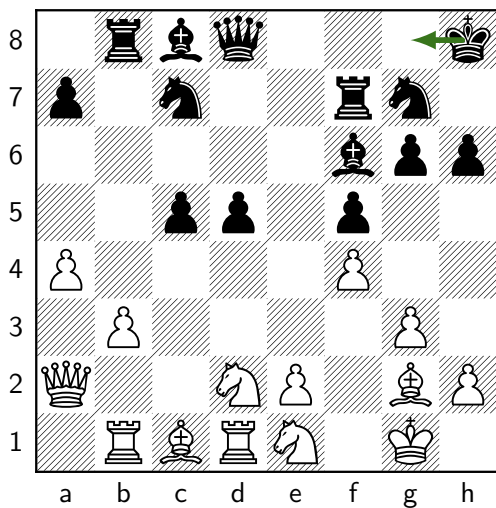


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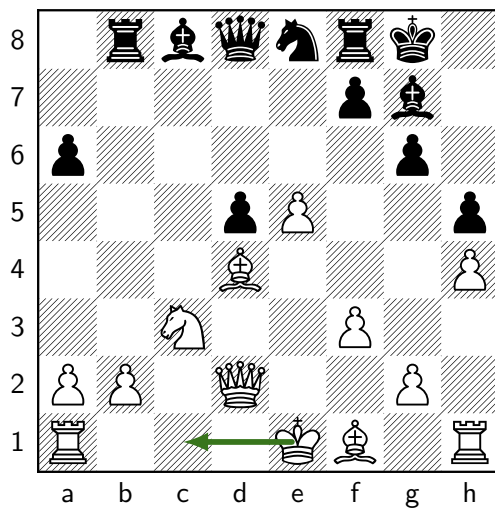


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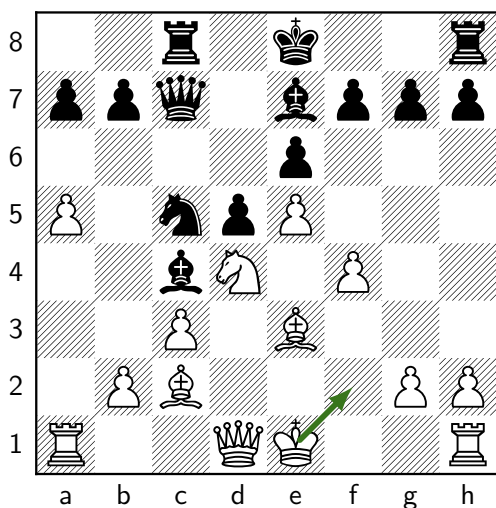
Figure 18: **Move 24, Black: Rf7**. (a) Our position; Melanie plays Rf7 (f8f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3193` (ply 51) [structural distance $d=113$, QAP obj=33]: the master played Nh5; under μ our Rf1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Rd8, our Bf6 as their Rf8; transferred move f8f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3943` (ply 33) [structural distance $d=116$, QAP obj=32]: the master played Nh5; under μ our Rf1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Ph6 as their Ra8, our Bc8 as their Qd8; transferred move f8f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3374` (ply 31) [structural distance $d=121$, QAP obj=31]: the master played Qf6; under μ our Rf1 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our qa2 as their Ra8, our Ng7 as their Pb7; transferred move f8f7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf7 (summed QAP objective 126, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



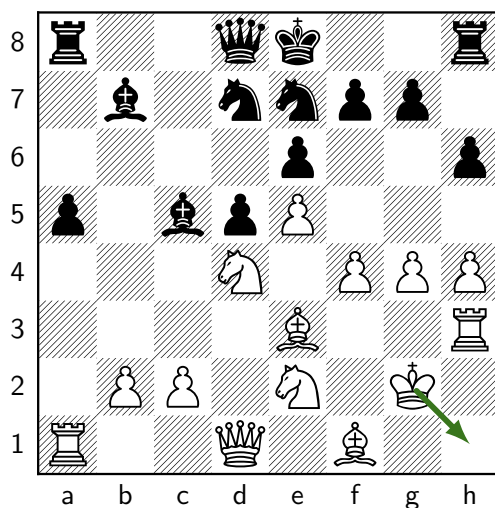
(a)



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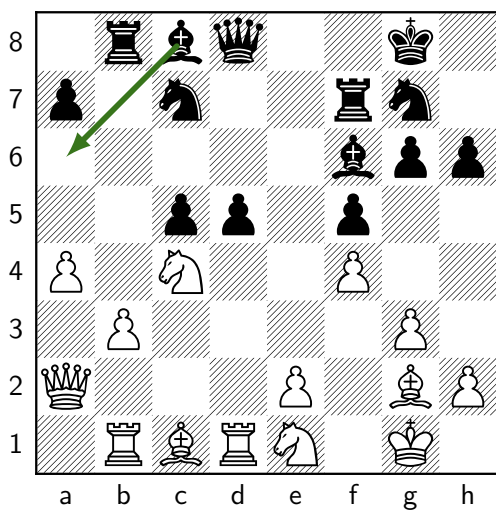


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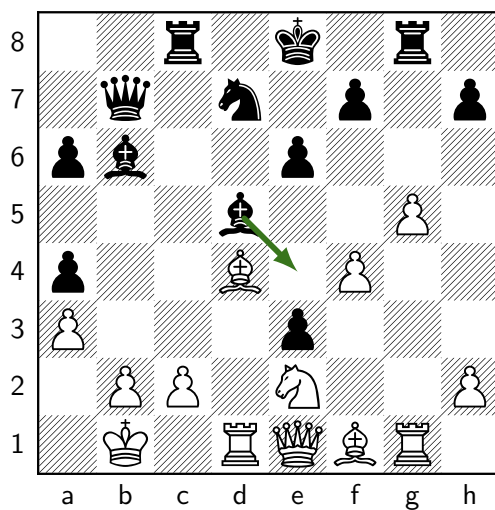


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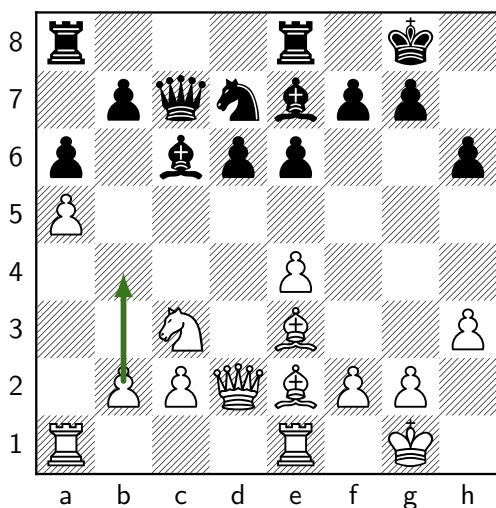
Figure 19: **Move 25, Black: Kg8.** (a) Our position; Melanie plays Kg8 (h8g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#233` (ply 34) [structural distance $d=122$, QAP obj=36]: the master played 0-0-0; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd8* as their *Ra1*, our *nd2* as their *Bf1*; transferred move h8g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#1225` (ply 32) [structural distance $d=124$, QAP obj=33]: the master played Kf2; under μ our *Kh1* plays the role of that game’s *Ke1*; cross-type roles: our *Qd8* as their *Ra1*, our *Ng7* as their *Qd1*; transferred move h8g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2842` (ply 34) [structural distance $d=104$, QAP obj=23]: the master played Kh1; under μ our *Kh1* plays the role of that game’s *Kg2*; cross-type roles: our *Nc7* as their *Qd1*, our *Qd8* as their *Bf1*; transferred move h8g8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg8 (summed QAP objective 115, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



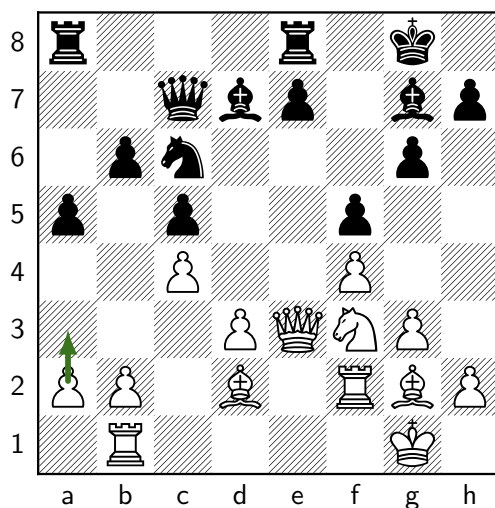
(a)



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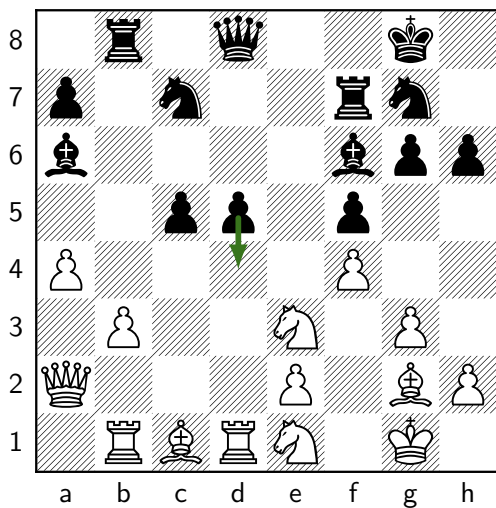


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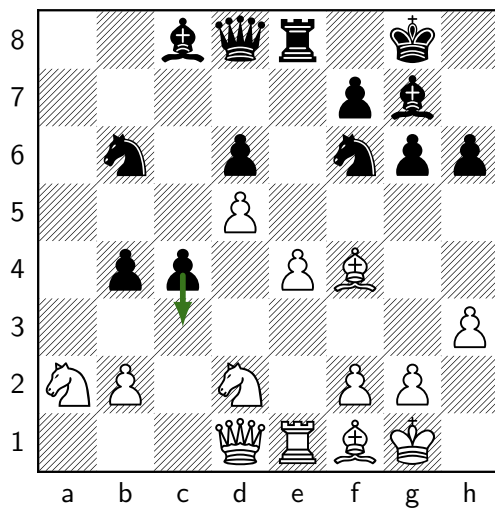


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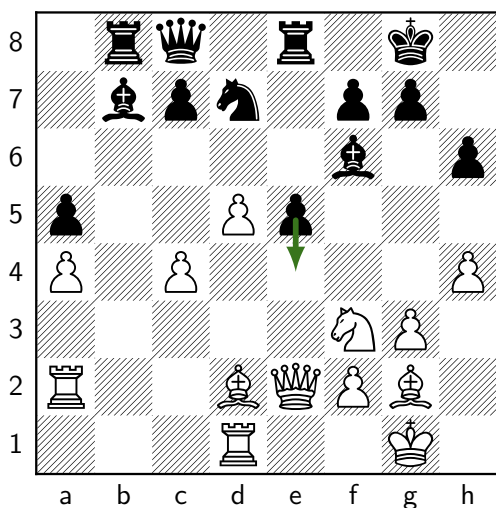
Figure 20: **Move 26, Black: Ba6.** (a) Our position; Melanie plays Ba6 (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1621` (ply 51) [structural distance $d=98$, QAP obj=32]: the master played Be4; under μ our Bc1 plays the role of that game’s Bd4; cross-type roles: our bg2 as their Rc8, our ne1 as their Qb7; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1003` (ply 34) [structural distance $d=101$, QAP obj=27]: the master played b4; under μ our Bc1 plays the role of that game’s Pb2 (a **cross-type** correspondence); cross-type roles: our Pf5 as their Ra1, our Qd8 as their Re1; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3303` (ply 36) [structural distance $d=113$, QAP obj=27]: the master played a3; under μ our Bc1 plays the role of that game’s Pa2 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rb1, our pf4 as their Bd2; transferred move c8a6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Ba6 (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



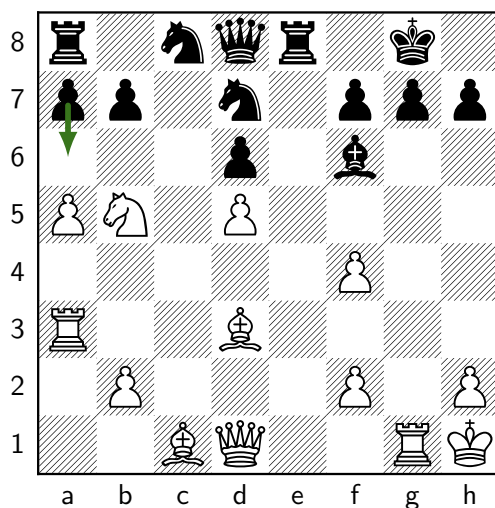
(a)



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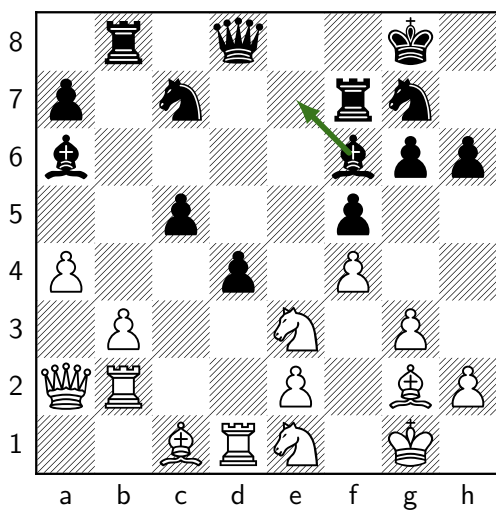


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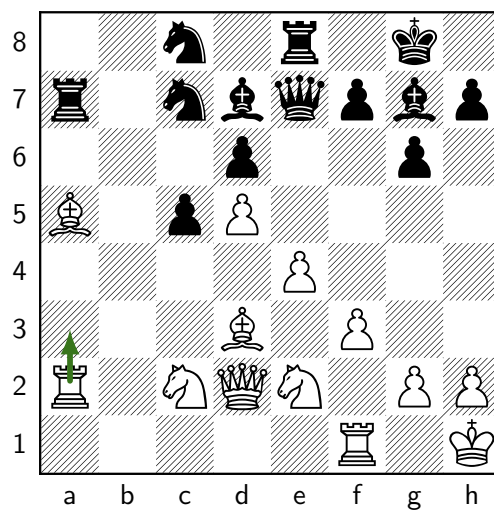


(d)

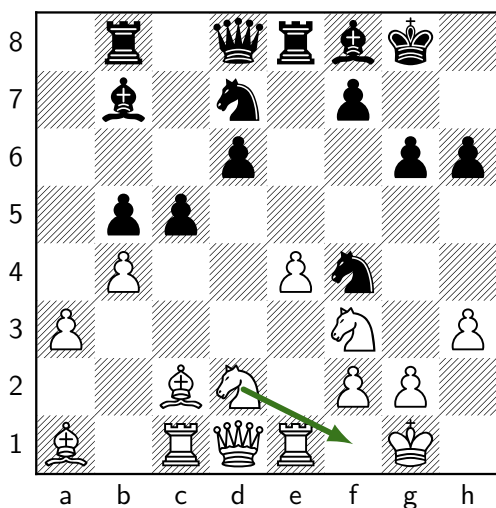
Figure 21: **Move 27, Black: d4**. (a) Our position; Melanie plays d4 (d5d4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1381` (ply 39) [structural distance $d=118$, QAP obj=191]: the master played c3; under μ our $Pd4$ plays the role of that game’s $Pc5$; cross-type roles: our $bc1$ as their $Qd8$, our $Ng7$ as their $Bg7$; transferred move d5d4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1752` (ply 39) [structural distance $d=115$, QAP obj=100]: the master played e4; under μ our $Pd4$ plays the role of that game’s $Pe4$; cross-type roles: our $pg3$ as their $Rb8$, our $Ba6$ as their $Qc8$; transferred move d5d4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2733` (ply 39) [structural distance $d=113$, QAP obj=92]: the master played a6; under μ our $Pd4$ plays the role of that game’s $Pa2$; cross-type roles: our $Ng7$ as their $Ra8$, our $Pg6$ as their $Nc8$; transferred move d5d4 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to d4 (summed QAP objective 500, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



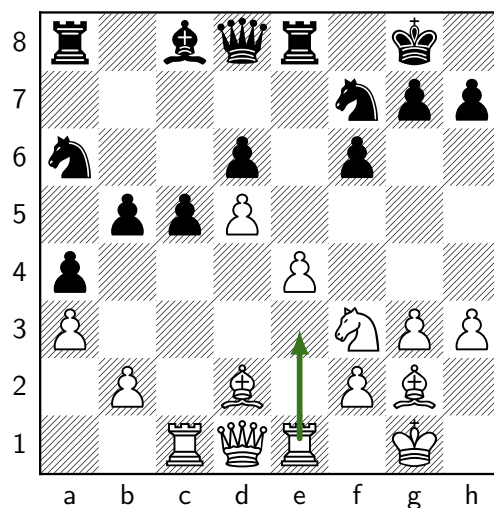
(a)



(b)

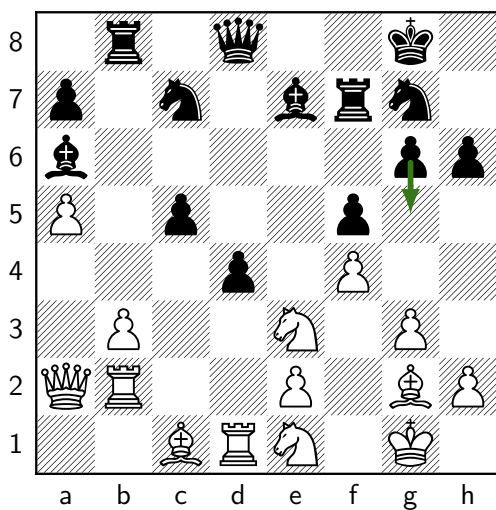


(c)

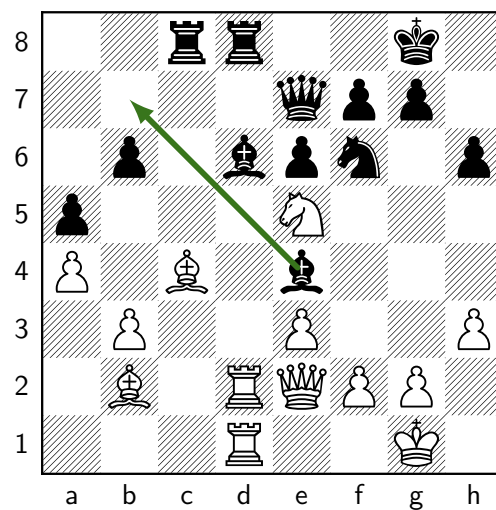


(d)

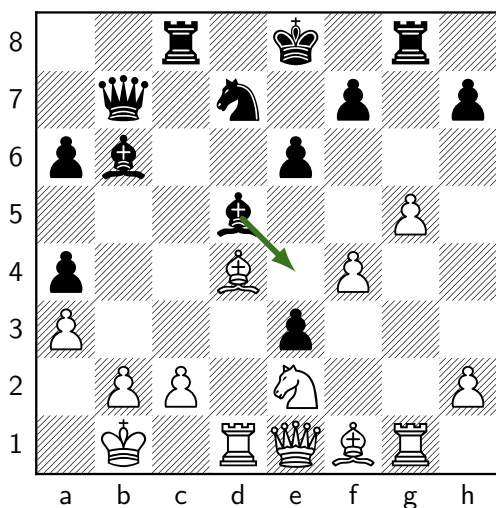
Figure 22: **Move 28, Black: Be7**. (a) Our position; Melanie plays **Be7** (f6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1802` (ply 62) [structural distance $d=94$, QAP obj=61]: the master played **Ra3**; under μ our *Bf3* plays the role of that game’s *Ra2* (a **cross-type** correspondence); cross-type roles: our *Qd8* as their *Rf1*, our *Rb8* as their *Nc2*; transferred move **f6e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#224` (ply 42) [structural distance $d=92$, QAP obj=46]: the master played **Nf1**; under μ our *Bf3* plays the role of that game’s *Nd2* (a **cross-type** correspondence); cross-type roles: our *Rb8* as their *Ba1*, our *Pc5* as their *Rc1*; transferred move **f6e7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2755` (ply 44) [structural distance $d=96$, QAP obj=39]: the master played **Re3**; under μ our *Bf3* plays the role of that game’s *Re1* (a **cross-type** correspondence); cross-type roles: our *Rf7* as their *Pf2*, our *Ph6* as their *Bg2*; transferred move **f6e7** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Be7** (summed QAP objective 146, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



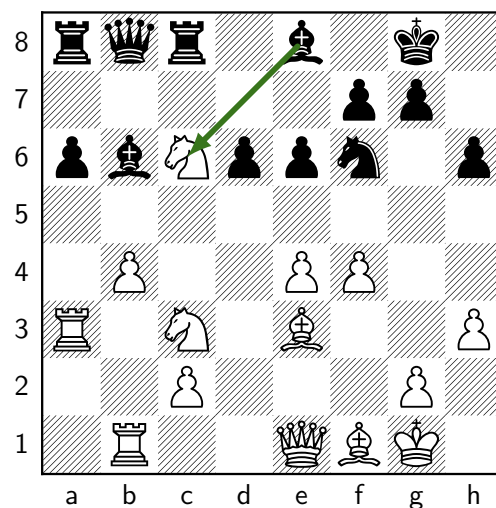
(a)



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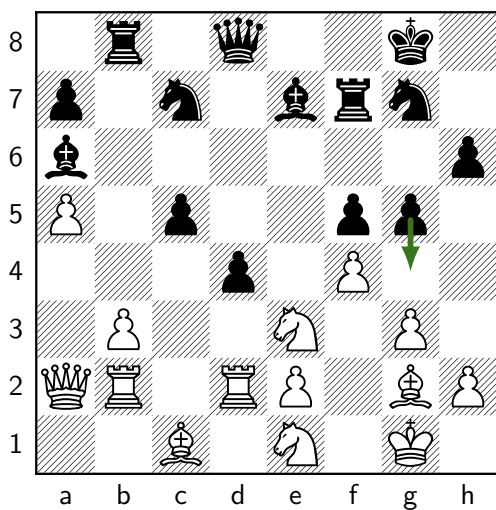


(c)

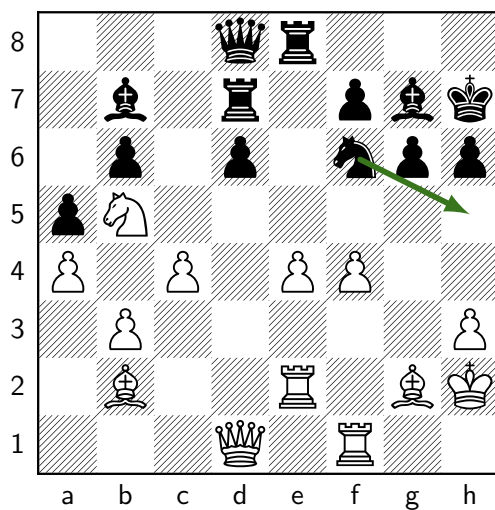


(d)

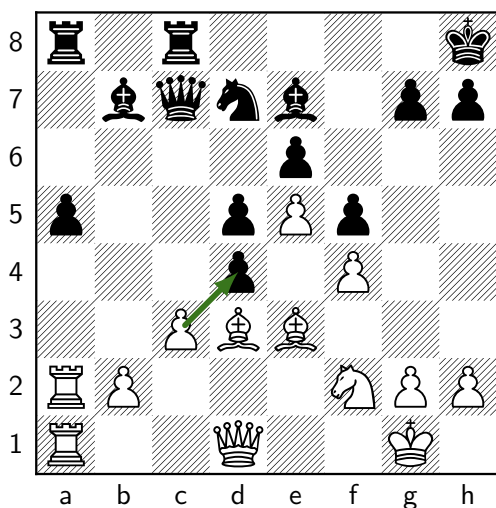
Figure 23: **Move 29, Black: g5.** (a) Our position; Melanie plays g5 (g6g5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1.top_p0002#1585` (ply 57) [structural distance $d=87$, QAP obj=71]: the master played Bb7; under μ our Pg3 plays the role of that game’s Be5 (a **cross-type** correspondence); cross-type roles: our Ba6 as their Rc8, our Qd8 as their Rd8; transferred move g6g5 (**endorses this move**). (c) Retrieved from `ccr1.top_p0003#1621` (ply 51) [structural distance $d=71$, QAP obj=32]: the master played Be4; under μ our Pg3 plays the role of that game’s Bd4 (a **cross-type** correspondence); cross-type roles: our Qd8 as their Rc8, our bc1 as their Qb7; transferred move g6g5 (**endorses this move**). (d) Retrieved from `ccr1.top_p0003#1545` (ply 45) [structural distance $d=79$, QAP obj=27]: the master played Bxc6; under μ our Pg3 plays the role of that game’s Be1 (a **cross-type** correspondence); cross-type roles: our qa2 as their Ra8, our ne3 as their Qb8; transferred move g6g5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to g5 (summed QAP objective 130, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



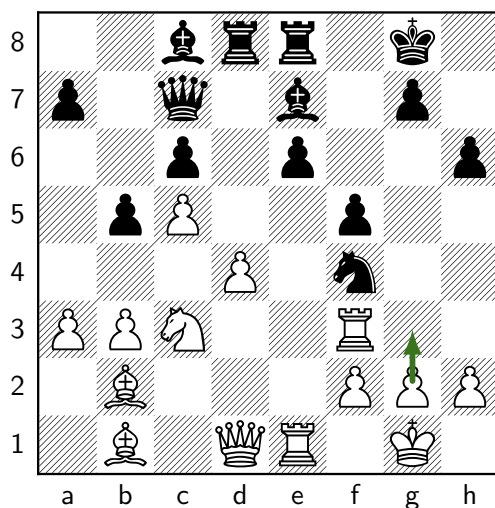
(a)



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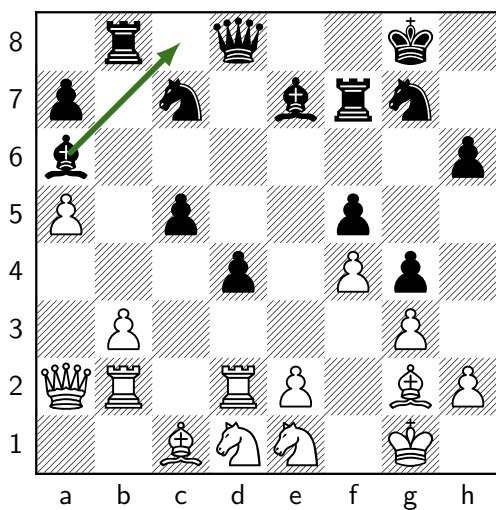


(c)

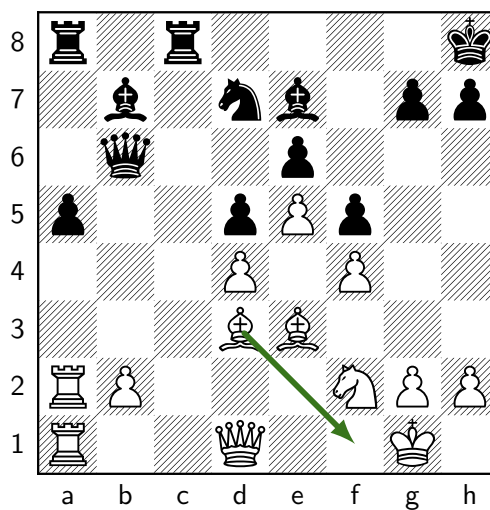


(d)

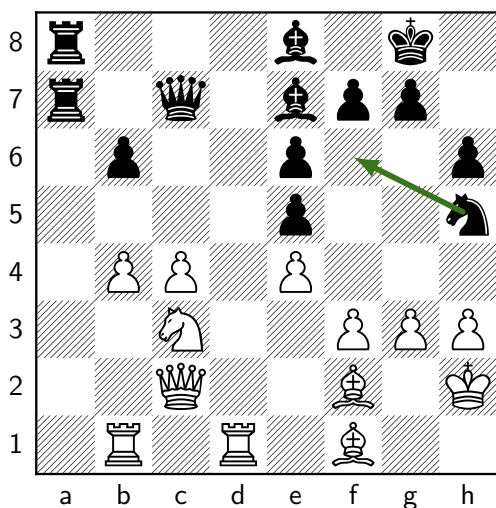
Figure 24: **Move 30, Black: g4**. (a) Our position; Melanie plays g4 (g5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3568` (ply 57) [structural distance $d=89$, QAP obj=48]: the master played Nh5; under μ our Pg4 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our rb2 as their Qd8, our rd2 as their Bb7; transferred move g5g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2358` (ply 50) [structural distance $d=85$, QAP obj=46]: the master played cxd4; under μ our Pg4 plays the role of that game’s Pc3; cross-type roles: our ne1 as their Qd1, our Rf7 as their Nf2; transferred move g5g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3841` (ply 54) [structural distance $d=69$, QAP obj=36]: the master played g3; under μ our Pg4 plays the role of that game’s Pg2; cross-type roles: our pe2 as their Bb1, our rb2 as their Qd1; transferred move g5g4 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to g4 (summed QAP objective 190, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



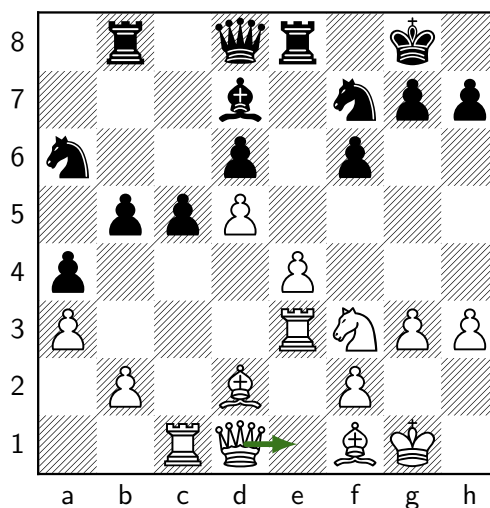
(a)



(b)

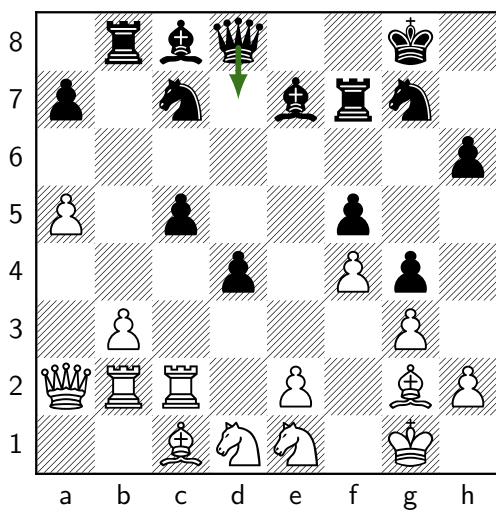


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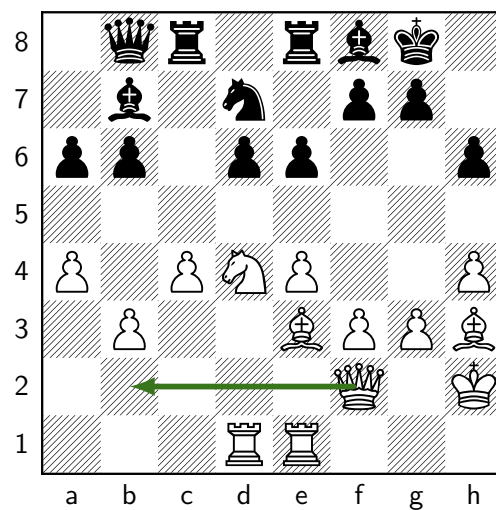


(d)

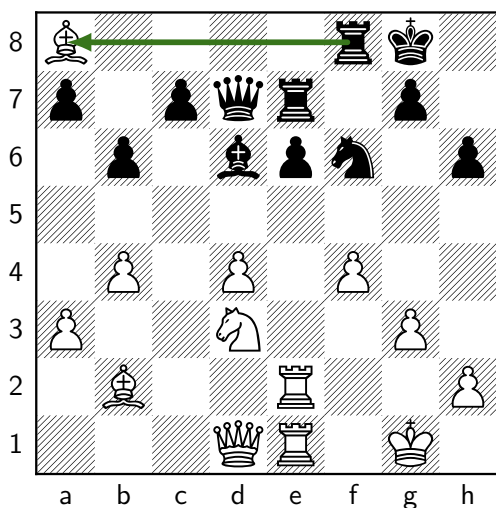
Figure 25: **Move 31, Black: Bc8.** (a) Our position; Melanie plays Bc8 (a6c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2358` (ply 52) [structural distance $d=87$, QAP obj=83]: the master played Bf1; under μ our Ba3 plays the role of that game’s Bd3; cross-type roles: our pb3 as their Ra1, our Rf7 as their Pg2; transferred move a6c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3446` (ply 65) [structural distance $d=101$, QAP obj=63]: the master played Nf6; under μ our Ba3 plays the role of that game’s Nh4 (a **cross-type** correspondence); cross-type roles: our Ng7 as their Ra8, our pf4 as their Be8; transferred move a6c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2755` (ply 48) [structural distance $d=93$, QAP obj=53]: the master played Qe1; under μ our Ba3 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Bf1, our nd1 as their Bd2; transferred move a6c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Bc8 (summed QAP objective 219, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



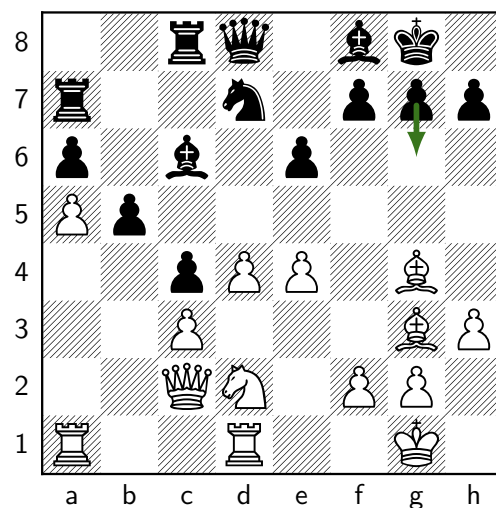
(a)



(b)

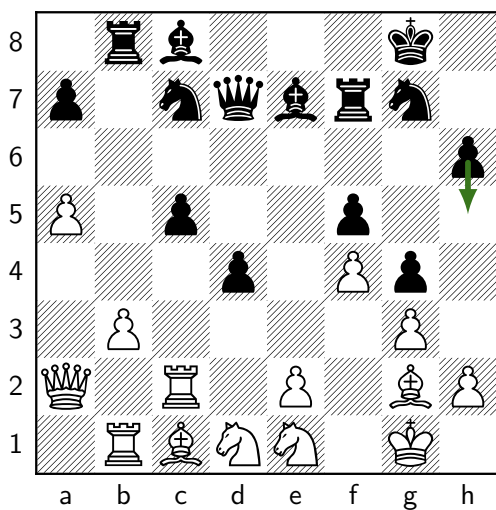


(c)

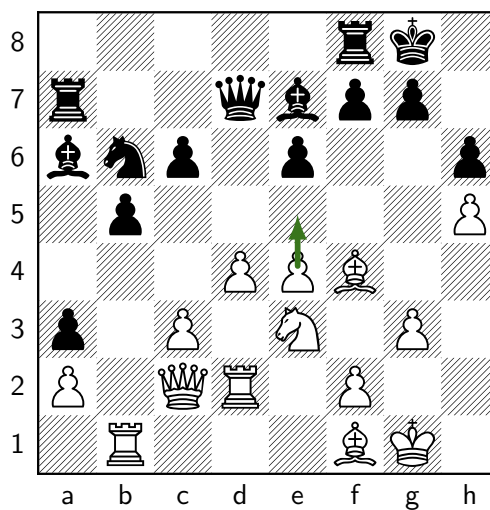


(d)

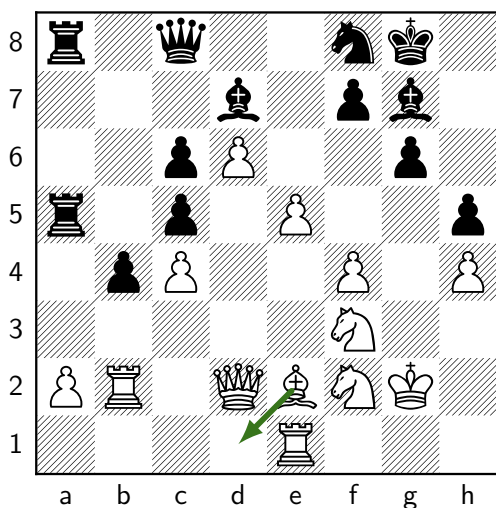
Figure 26: **Move 32, Black: Qd7.** (a) Our position; Melanie plays Qd7 (d8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#575` (ply 50) [structural distance $d=97$, QAP obj=58]: the master played Qb2; under μ our Qd1 plays the role of that game’s Qf2; cross-type roles: our ne1 as their Rd1, our nd1 as their Re1; transferred move d8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#982` (ply 51) [structural distance $d=108$, QAP obj=49]: the master played Rxa8; under μ our Qd1 plays the role of that game’s Rf1 (a **cross-type** correspondence); cross-type roles: our rb2 as their Qd7, our qa2 as their Re7; transferred move d8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2529` (ply 47) [structural distance $d=104$, QAP obj=44]: the master played g6; under μ our Qd1 plays the role of that game’s Pg2 (a **cross-type** correspondence); cross-type roles: our ne1 as their Rc8, our Pd4 as their Qd8; transferred move d8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd7 (summed QAP objective 187, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



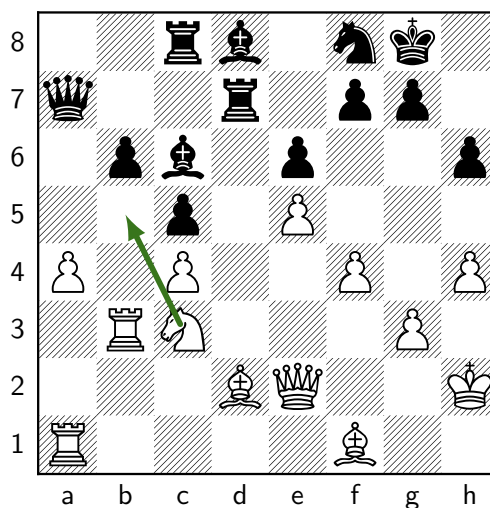
(a)



(b)

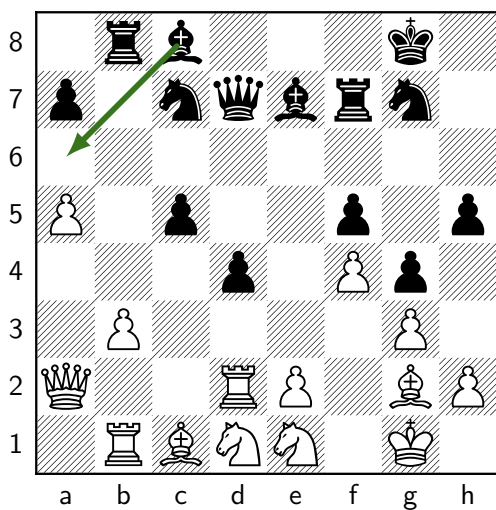


(c)

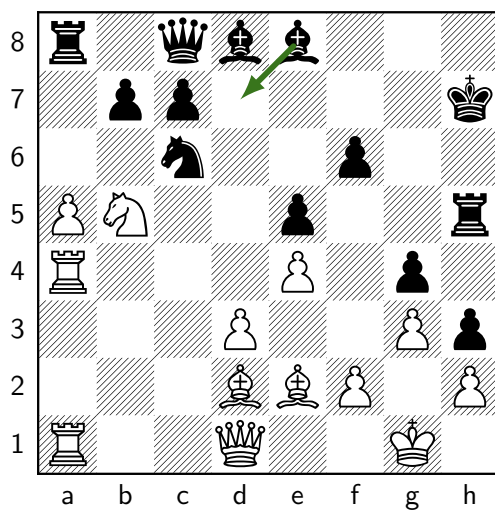


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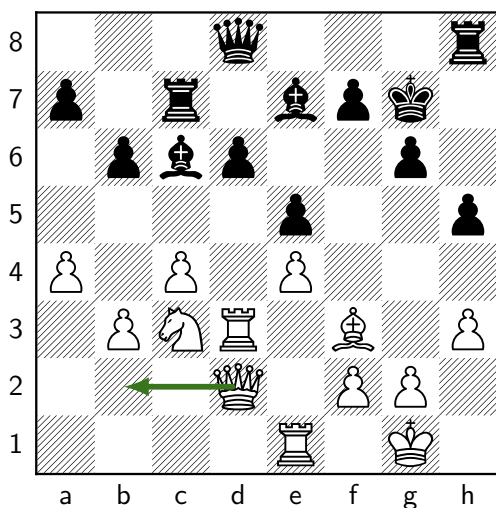
Figure 27: **Move 33, Black: h5**. (a) Our position; Melanie plays h5 (h6h5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3497` (ply 52) [structural distance $d=105$, QAP obj=83]: the master played e5; under μ our *Ph3* plays the role of that game’s *Pe4*; cross-type roles: our *Rb8* as their *Bf1*, our *Pd4* as their *Rd2*; transferred move h6h5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#640` (ply 78) [structural distance $d=105$, QAP obj=66]: the master played Bd1; under μ our *Ph3* plays the role of that game’s *Be2* (a **cross-type** correspondence); cross-type roles: our *qa2* as their *Re1*, our *rc2* as their *Qd2*; transferred move h6h5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#291` (ply 74) [structural distance $d=83$, QAP obj=55]: the master played Nb5; under μ our *Ph3* plays the role of that game’s *Nc3* (a **cross-type** correspondence); cross-type roles: our *Be7* as their *Ra1*, our *pe2* as their *Bd2*; transferred move h6h5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to h5 (summed QAP objective 225, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



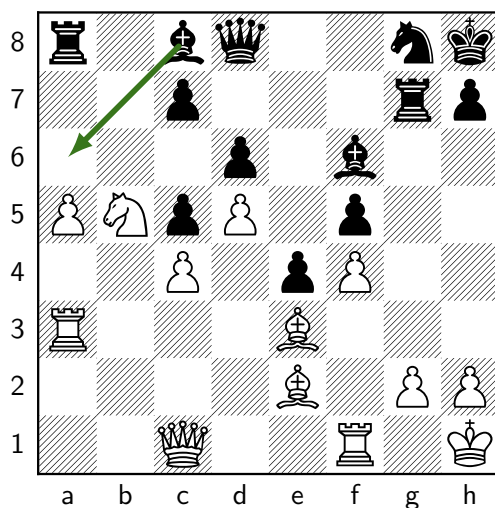
(a)



(b)

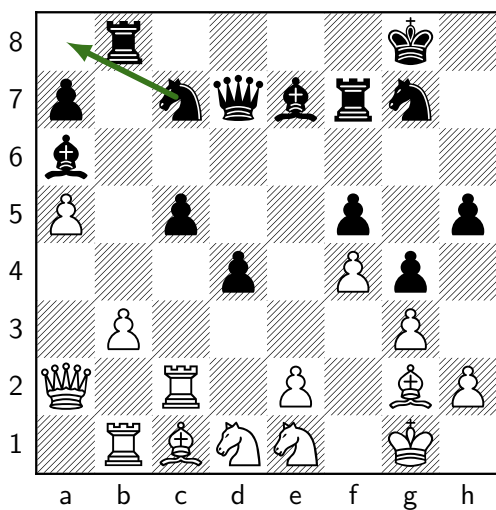


(c)

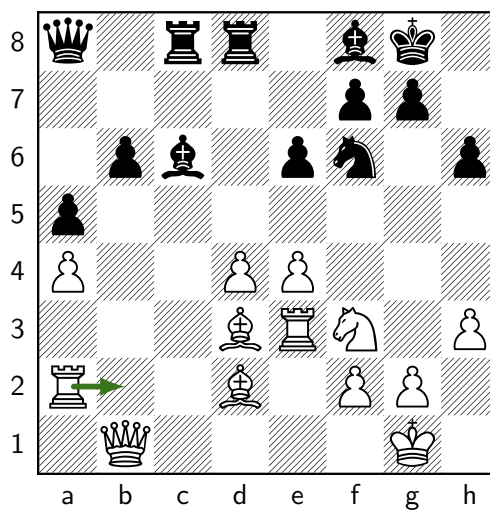


(d)

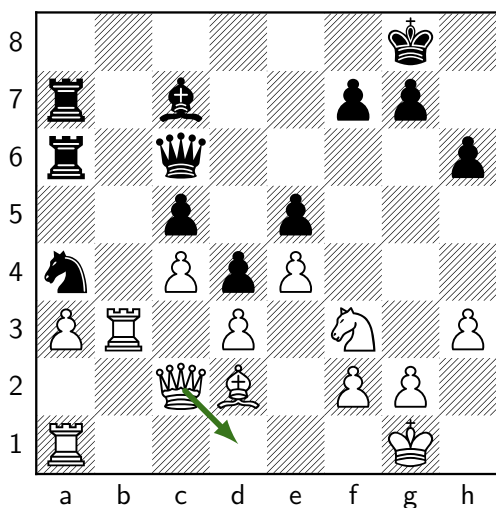
Figure 28: **Move 34, Black: Ba6**. (a) Our position; Melanie plays Ba6 (c8a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#859` (ply 73) [structural distance $d=92$, QAP obj=94]: the master played Bd7; under μ our Bc1 plays the role of that game’s Be1; cross-type roles: our rb1 as their Bd8, our rd2 as their Nc6; transferred move c8a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3490` (ply 82) [structural distance $d=88$, QAP obj=88]: the master played Qb2; under μ our Bc1 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Ng7 as their Re1, our Rf7 as their Pf2; transferred move c8a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1006` (ply 51) [structural distance $d=84$, QAP obj=60]: the master played Ba6; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Rf7 as their Ng8, our bg2 as their Rg7; transferred move c8a6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ba6 (summed QAP objective 330, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



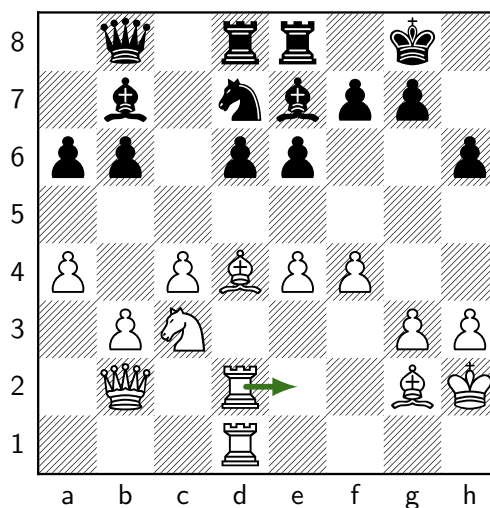
(a)



(b)



(c)



(d)

Figure 29: **Move 35, Black: Na8**. (a) Our position; Melanie plays Na8 (c7a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2719` (ply 82) [structural distance $d=100$, QAP obj=91]: the master played Rb2; under μ our Nc2 plays the role of that game’s Ra2 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Qb1, our Qd7 as their Pg2; transferred move c7a8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3976` (ply 84) [structural distance $d=94$, QAP obj=58]: the master played Qd1; under μ our Nc2 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our rb1 as their Pf2, our pg3 as their Rb3; transferred move c7a8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#970` (ply 60) [structural distance $d=92$, QAP obj=58]: the master played Re2; under μ our Nc2 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our Pf5 as their Rd1, our Be7 as their Qb2; transferred move c7a8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Na8 (summed QAP objective 223, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.

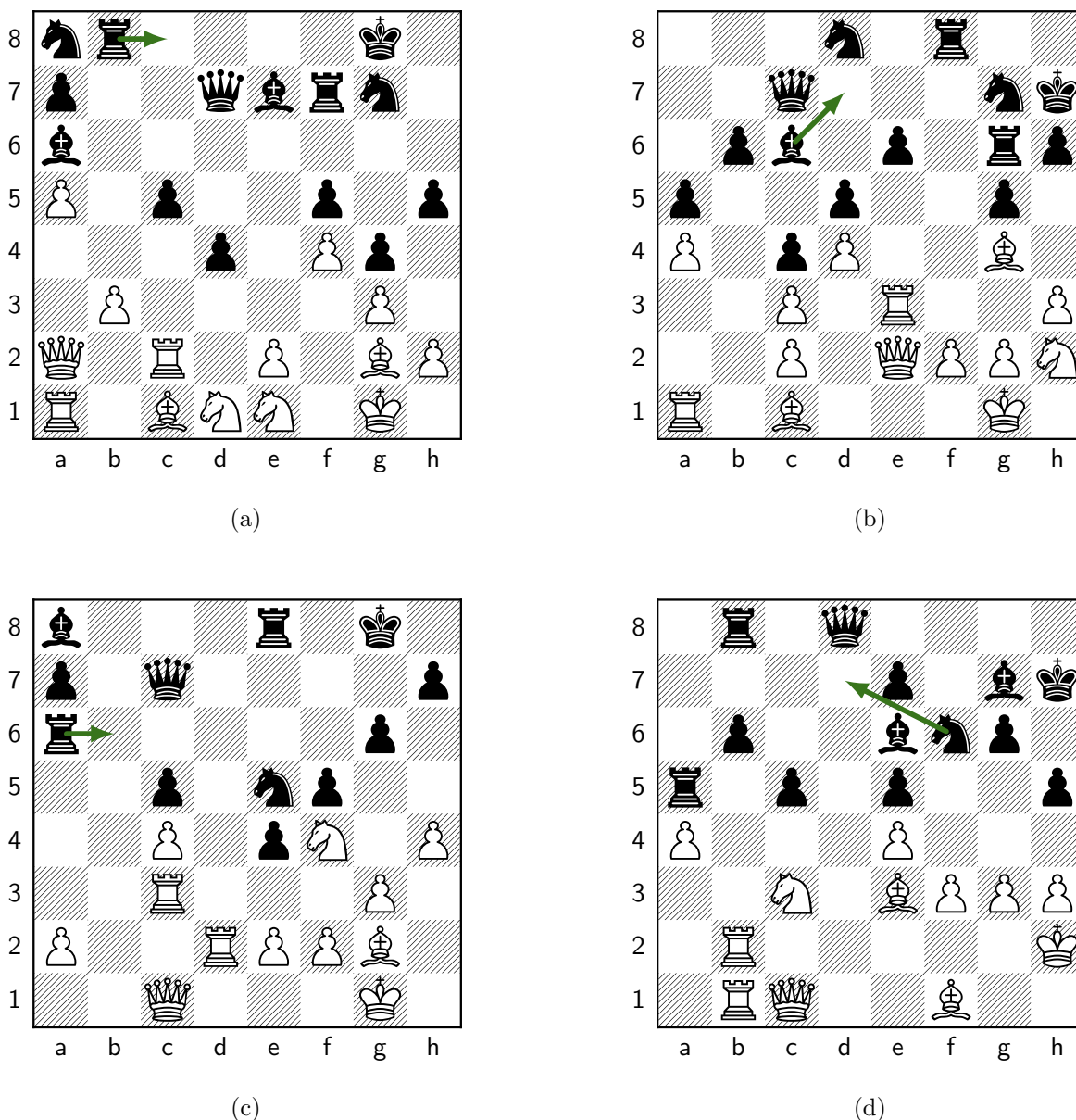
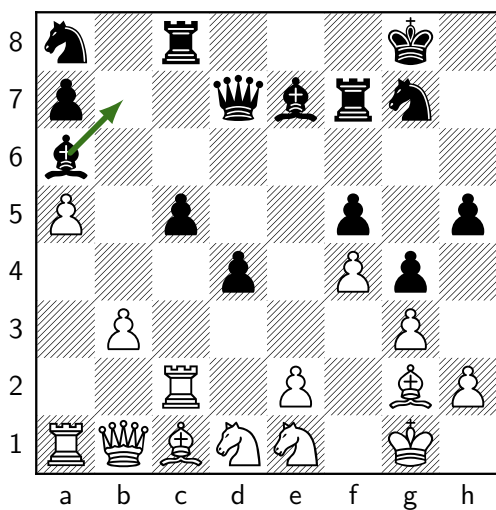
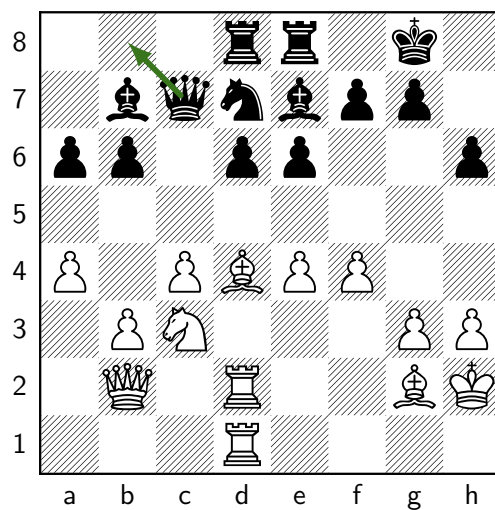


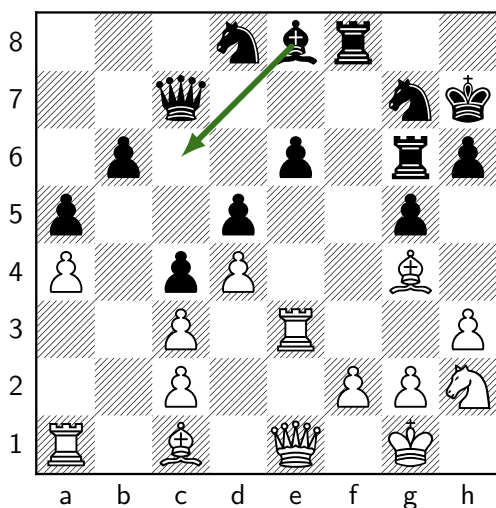
Figure 30: **Move 36, Black: Rc8**. (a) Our position; Melanie plays Rc8 (b8c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1769` (ply 65) [structural distance $d=97$, QAP obj=53]: the master played Bd7; under μ our Rb1 plays the role of that game’s Bc3 (a **cross-type** correspondence); cross-type roles: our Pf5 as their Nd8, our Ba6 as their Rf8; transferred move b8c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#438` (ply 59) [structural distance $d=97$, QAP obj=51]: the master played Rb6; under μ our Rb1 plays the role of that game’s Ra3; cross-type roles: our Pg4 as their Ba8, our Pa7 as their Re8; transferred move b8c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#680` (ply 71) [structural distance $d=83$, QAP obj=25]: the master played Nd7; under μ our Rb1 plays the role of that game’s Nf3 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Rb8, our Pf5 as their Qd8; transferred move b8c8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rc8 (summed QAP objective 147, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



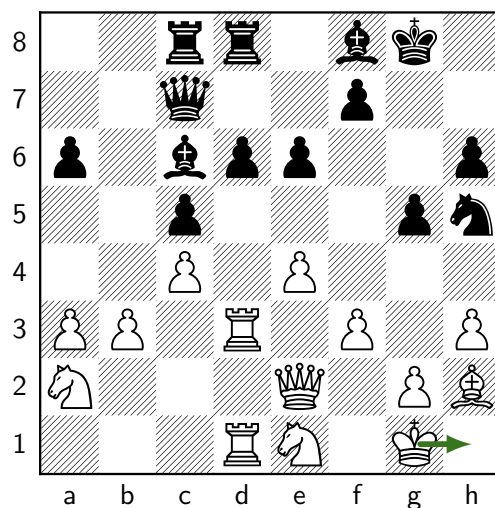
(a)



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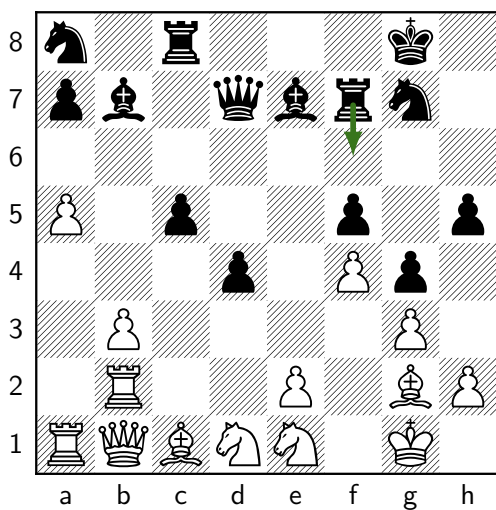


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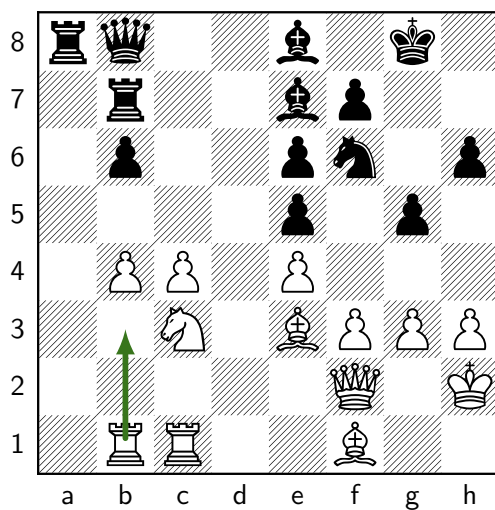


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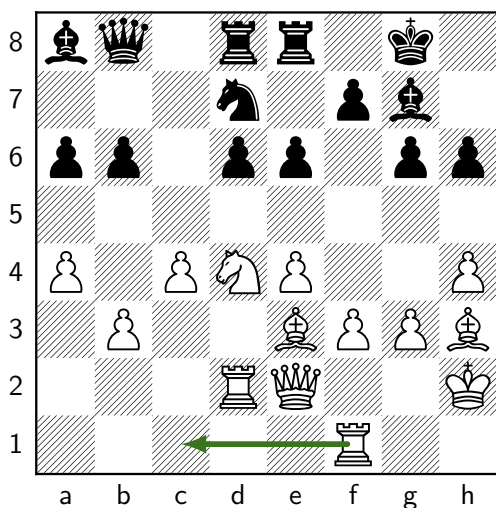
Figure 31: **Move 37, Black: Bb7**. (a) Our position; Melanie plays Bb7 (a6b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#970` (ply 59) [structural distance $d=90$, QAP obj=77]: the master played Qb8; under μ our *Ba3* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *bc1* as their *Re8*, our *rc2* as their *Bb7*; transferred move a6b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1769` (ply 63) [structural distance $d=96$, QAP obj=72]: the master played Bc6; under μ our *Ba3* plays the role of that game’s *Be1*; cross-type roles: our *Be7* as their *Nd8*, our *Pg4* as their *Rf8*; transferred move a6b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#622` (ply 82) [structural distance $d=83$, QAP obj=88]: the master played Kh1; under μ our *Kg1* plays the role of that game’s *Kg1*; cross-type roles: our *ne1* as their *Rd1*, our *qb1* as their *Ne1*; transferred move g8h8. Synthesis: 3 retrieved analogs transferred to Bb7 (summed QAP objective 206, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



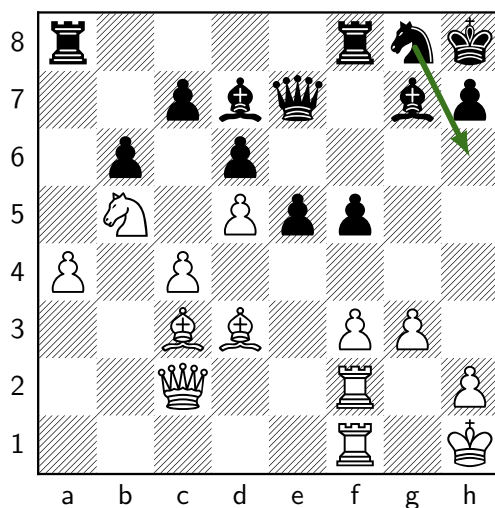
(a)



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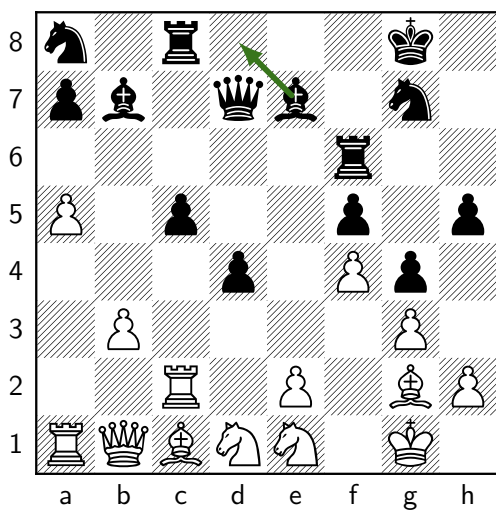


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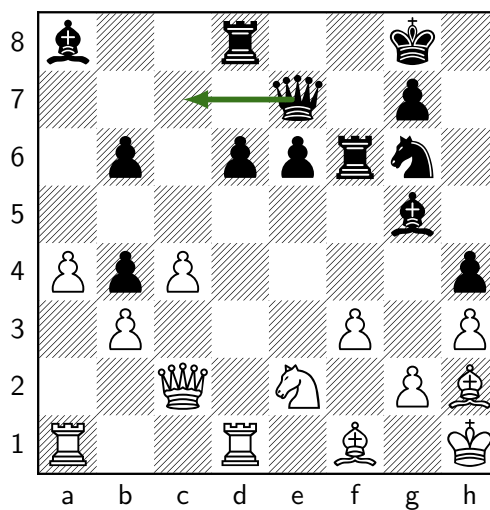


(d)

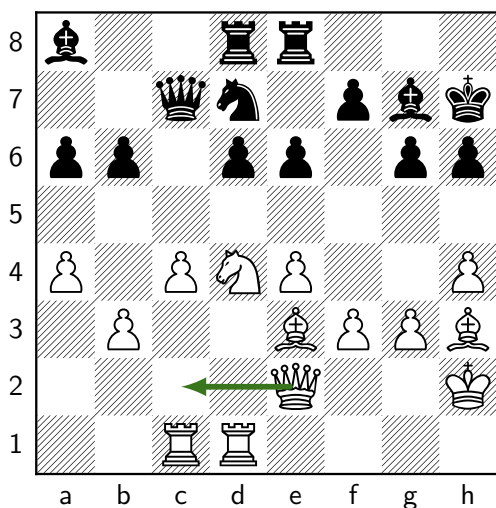
Figure 32: **Move 38, Black: Rf6**. (a) Our position; Melanie plays Rf6 (f7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3446` (ply 72) [structural distance $d=87$, QAP obj=43]: the master played Rb3; under μ our Rf2 plays the role of that game’s Rb1; cross-type roles: our Pc5 as their Rc1, our Rc8 as their Qf2; transferred move f7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#575` (ply 66) [structural distance $d=102$, QAP obj=33]: the master played Rc1; under μ our Rf2 plays the role of that game’s Rf1; cross-type roles: our Qd7 as their Rd2, our Rc8 as their Qe2; transferred move f7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#399` (ply 75) [structural distance $d=94$, QAP obj=22]: the master played Nh6; under μ our Rf2 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Pa7 as their Ra8, our Pc5 as their Rf8; transferred move f7f6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rf6 (summed QAP objective 114, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



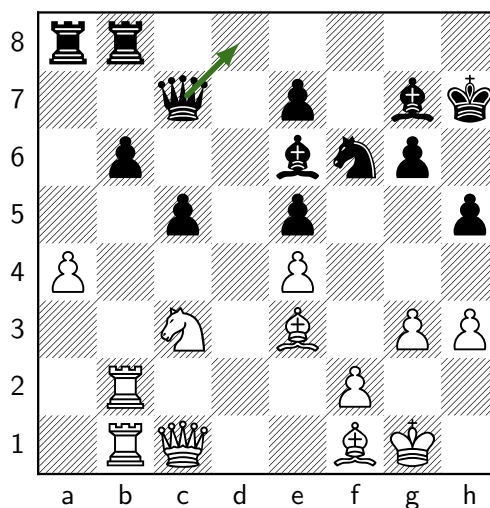
(a)



(b)

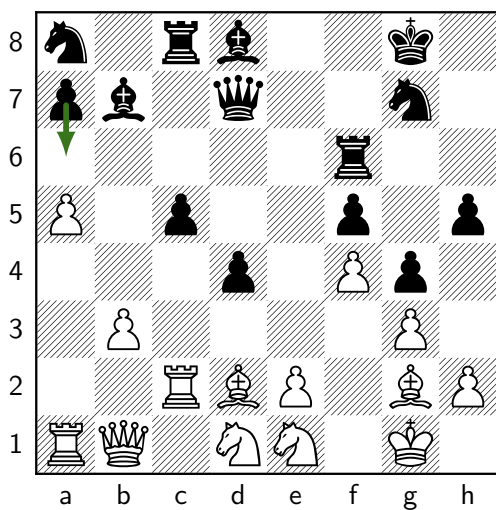


(c)

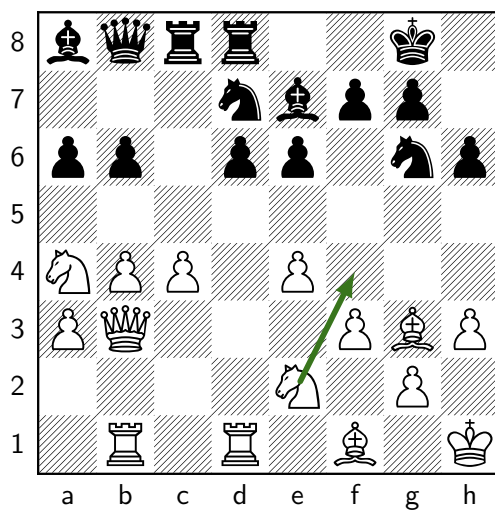


(d)

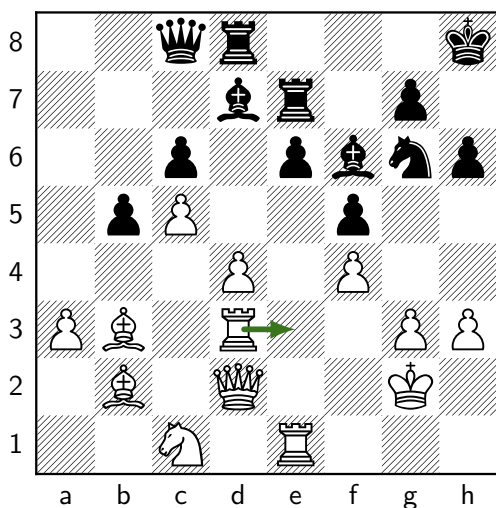
Figure 33: **Move 39, Black: Bd8.** (a) Our position; Melanie plays Bd8 (e7d8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2056` (ply 81) [structural distance $d=99$, QAP obj=67]: the master played Qc7; under μ our *Be2* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *rc2* as their *Ba8*, our *qb1* as their *Pg7*; transferred move e7d8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#575` (ply 70) [structural distance $d=102$, QAP obj=60]: the master played Qc2; under μ our *Be2* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *qb1* as their *Rd1*, our *Na8* as their *Pg3*; transferred move e7d8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#680` (ply 67) [structural distance $d=100$, QAP obj=51]: the master played Qd8; under μ our *Be2* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qd7* as their *Ra8*, our *Pd4* as their *Bg7*; transferred move e7d8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Bd8 (summed QAP objective 251, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



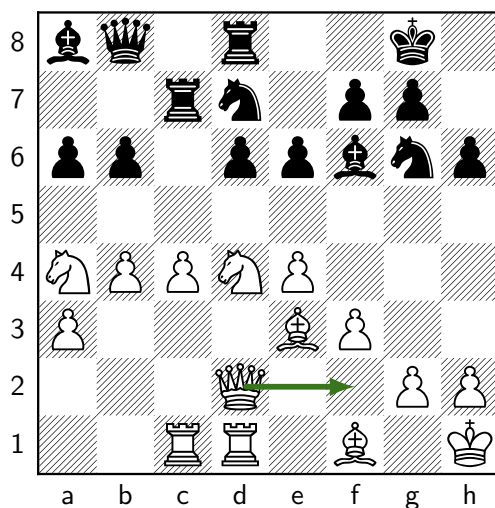
(a)



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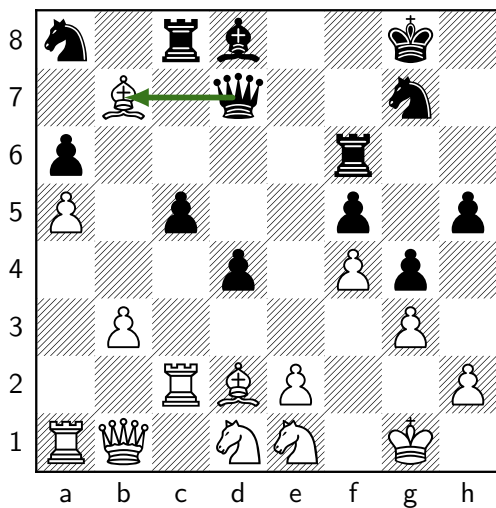


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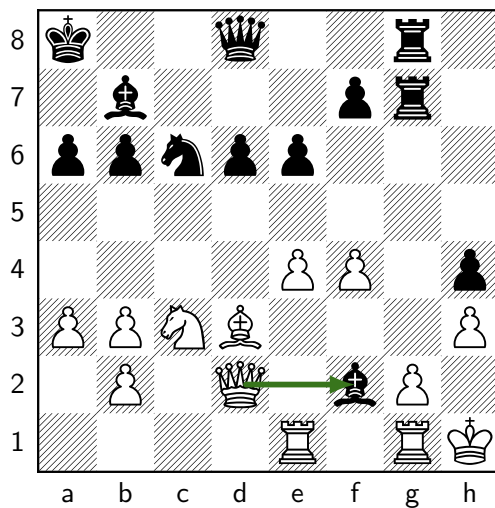


(d)

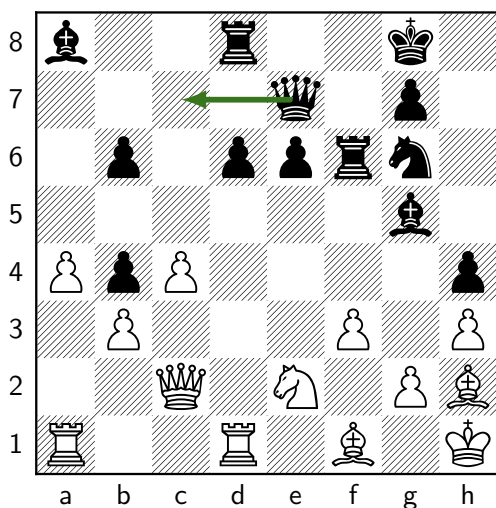
Figure 34: **Move 40, Black: a6.** (a) Our position; Melanie plays a6 (a7a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3405` (ply 62) [structural distance $d=91$, QAP obj=74]: the master played Nf4; under μ our *Pa2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *Ng7* as their *Rb1*, our *Pf5* as their *Bf1*; transferred move a7a6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3841` (ply 80) [structural distance $d=99$, QAP obj=58]: the master played Rde3; under μ our *Pa2* plays the role of that game’s *Rd3* (a **cross-type** correspondence); cross-type roles: our *Rc8* as their *Nc1*, our *Bd8* as their *Re1*; transferred move a7a6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#561` (ply 72) [structural distance $d=105$, QAP obj=41]: the master played Qf2; under μ our *Pa2* plays the role of that game’s *Qd2* (a **cross-type** correspondence); cross-type roles: our *Ng7* as their *Ph2*, our *Rf6* as their *Be3*; transferred move a7a6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to a6 (summed QAP objective 173, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



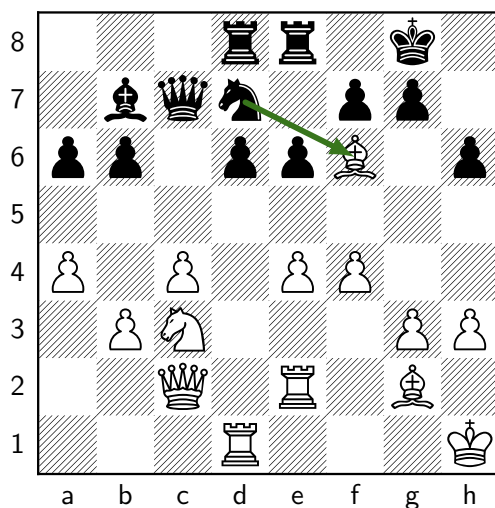
(a)



(b)

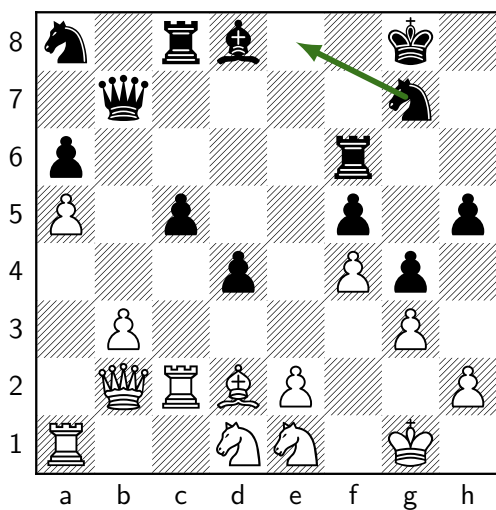


(c)

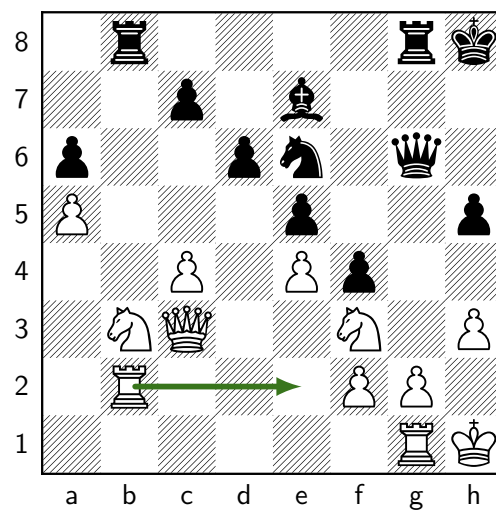


(d)

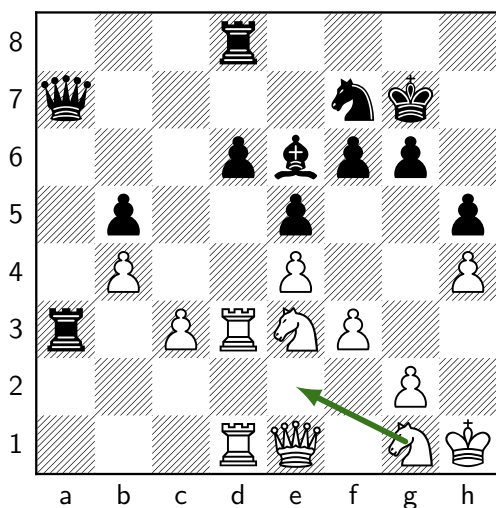
Figure 35: **Move 41, Black: Qxb7.** (a) Our position; Melanie plays Qxb7 (d7b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2802` (ply 64) [structural distance $d=81$, QAP obj=86]: the master played Qxf2; under μ our Qd2 plays the role of that game’s Qd2; cross-type roles: our ra1 as their Pa3, our Pg4 as their Nc3; transferred move d7b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2056` (ply 81) [structural distance $d=87$, QAP obj=31]: the master played Qc7; under μ our Qd2 plays the role of that game’s Qe2; cross-type roles: our Rc8 as their Ba8, our pg3 as their Rd8; transferred move d7b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#970` (ply 73) [structural distance $d=88$, QAP obj=67]: the master played Nxf6; under μ our Qd2 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our nd1 as their Bb7, our rc2 as their Qc7; transferred move d7d6. Synthesis: 2 retrieved analogs transferred to Qxb7 (summed QAP objective 117, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



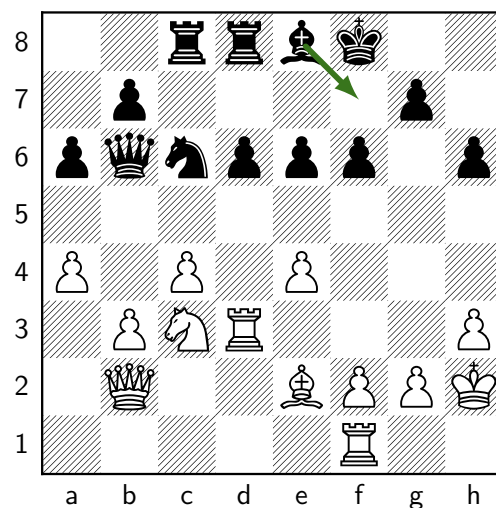
(a)



(b)

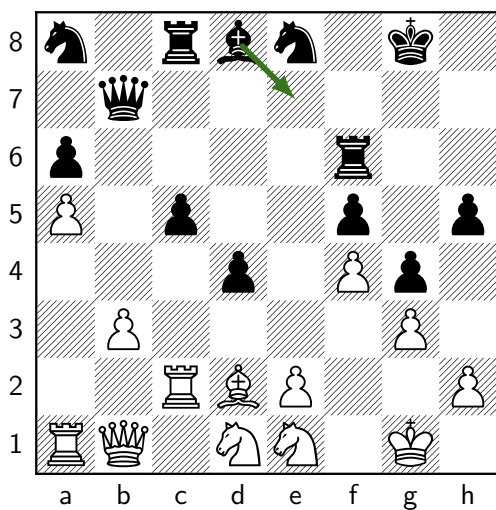


(c)

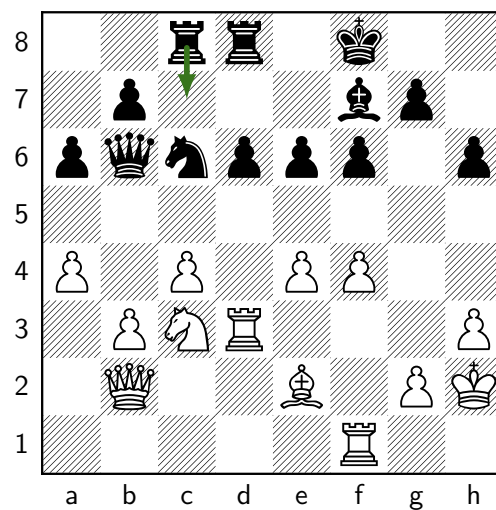


(d)

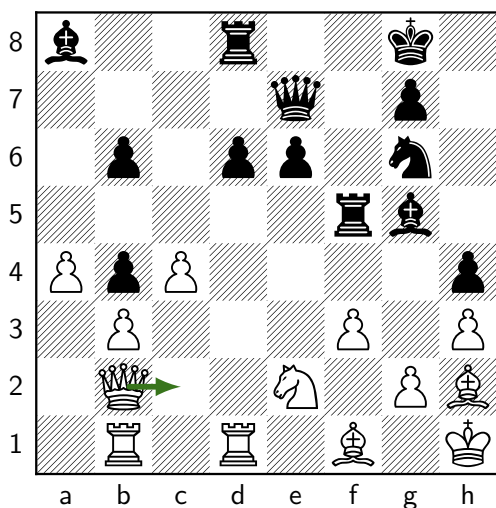
Figure 36: **Move 42, Black: Ne8 (g7e8)**. (a) Our position; Melanie plays Ne8 (g7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#925` (ply 66) [structural distance $d=80$, QAP obj=95]: the master played Re2; under μ our Ng2 plays the role of that game’s Rb2 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Nb3, our qb2 as their Nf3; transferred move g7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#915` (ply 96) [structural distance $d=93$, QAP obj=69]: the master played Ne2; under μ our Ng2 plays the role of that game’s Ng1; cross-type roles: our Bd8 as their Qe1, our qb2 as their ra3; transferred move g7e8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1753` (ply 67) [structural distance $d=88$, QAP obj=67]: the master played Bf7; under μ our Ng2 plays the role of that game’s Be1 (a **cross-type** correspondence); cross-type roles: our qb2 as their Rd8, our ne1 as their Pb7; transferred move g7e8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Ne8 (summed QAP objective 300, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



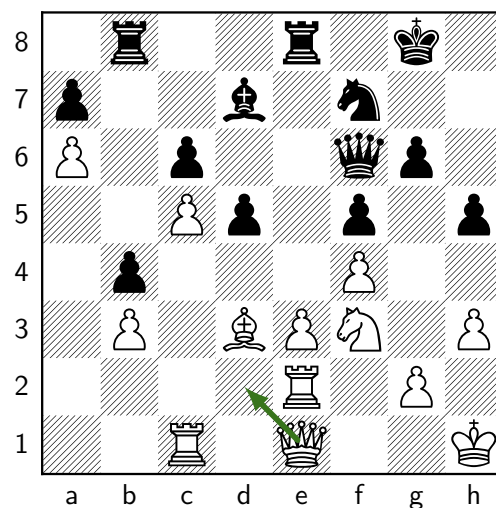
(a)



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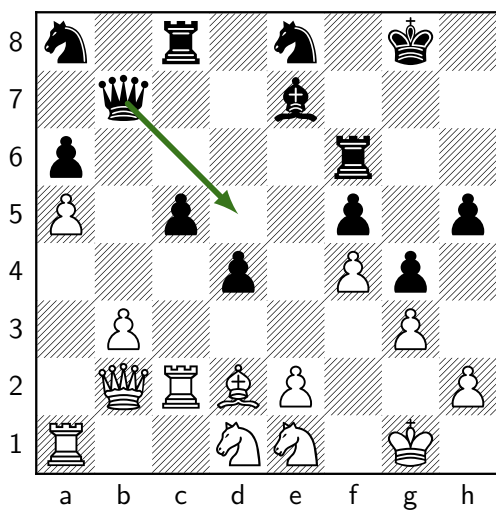


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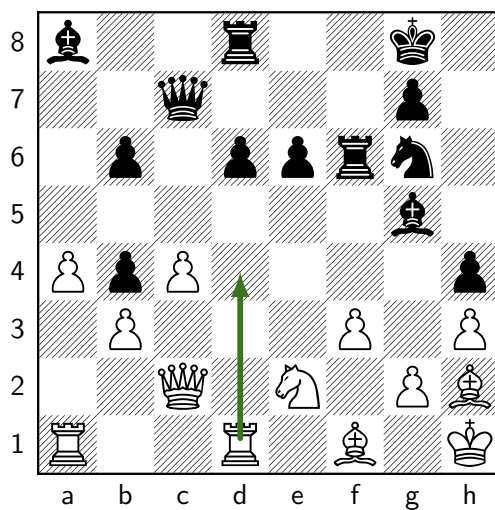


(d)

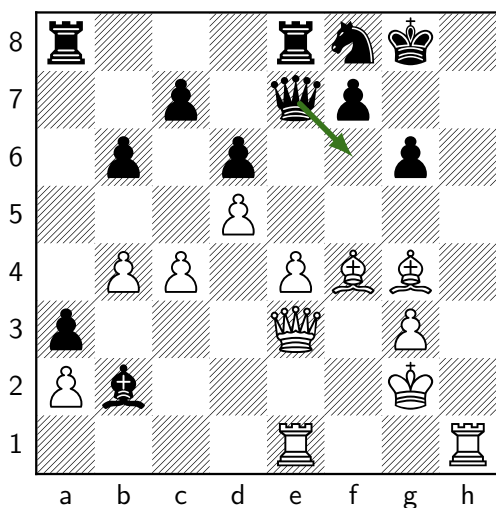
Figure 37: **Move 43, Black: Be7.** (a) Our position; Melanie plays **Be7** (**d8e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1753` (ply 69) [structural distance $d=74$, QAP obj=83]: the master played **Rc7**; under μ our *Bd1* plays the role of that game’s *Rc1* (a **cross-type** correspondence); cross-type roles: our *Pg4* as their *Bf7*, our *ph2* as their *Nc6*; transferred move **d8e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#2056` (ply 78) [structural distance $d=89$, QAP obj=68]: the master played **Qc2**; under μ our *Bd1* plays the role of that game’s *Qb2* (a **cross-type** correspondence); cross-type roles: our *Qb7* as their *Rd1*, our *Pa6* as their *Bf1*; transferred move **d8e7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1797` (ply 94) [structural distance $d=94$, QAP obj=54]: the master played **Qd2**; under μ our *Bd1* plays the role of that game’s *Qe1* (a **cross-type** correspondence); cross-type roles: our *Qb7* as their *Bd3*, our *Pd4* as their *Nf3*; transferred move **d8e7** (**endorses this move**). Synthesis: 5 retrieved analogs transferred to **Be7** (summed QAP objective 282, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



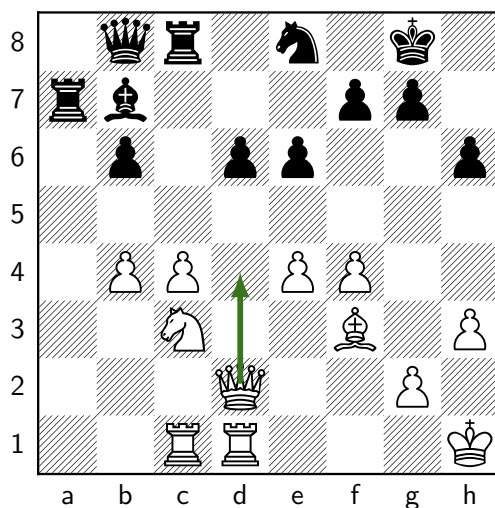
(a)



(b)

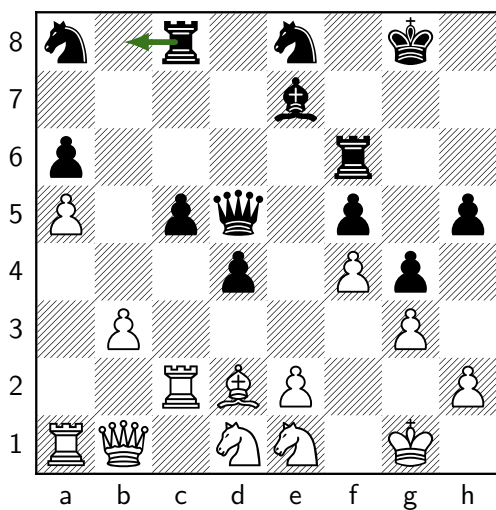


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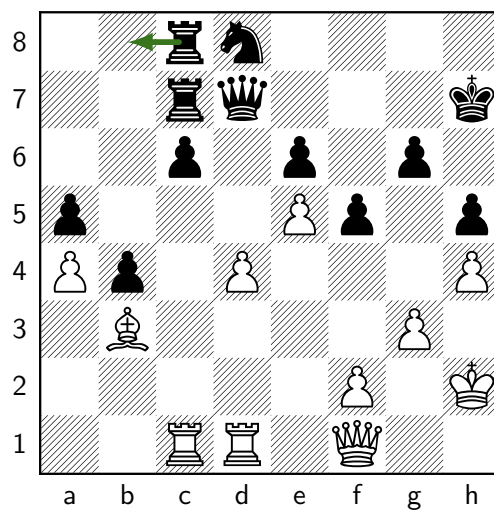


(d)

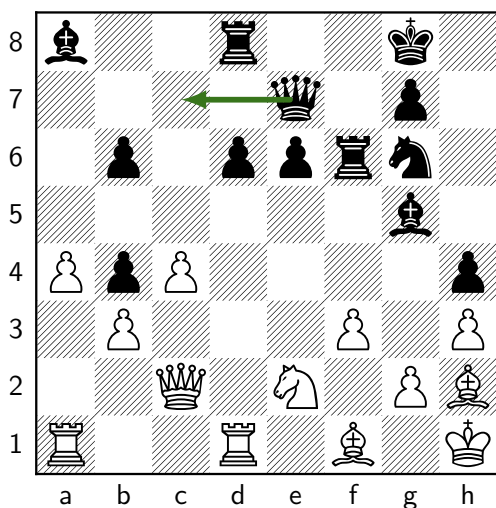
Figure 38: **Move 44, Black: Qd5.** (a) Our position; Melanie plays Qd5 (b7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2056` (ply 82) [structural distance $d=95$, QAP obj=85]: the master played Rd4; under μ our Qb2 plays the role of that game’s Rd1 (a **cross-type** correspondence); cross-type roles: our Be7 as their Ra1, our Na8 as their Bf1; transferred move b7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#812` (ply 95) [structural distance $d=98$, QAP obj=75]: the master played Qf6; under μ our Qb2 plays the role of that game’s Qe2; cross-type roles: our Pg4 as their Ra8, our Be7 as their Pc7; transferred move b7d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1250` (ply 100) [structural distance $d=93$, QAP obj=70]: the master played Qd4; under μ our Qb2 plays the role of that game’s Qd2; cross-type roles: our Be7 as their Rd1, our Rf6 as their Bf3; transferred move b7d5 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qd5 (summed QAP objective 229, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



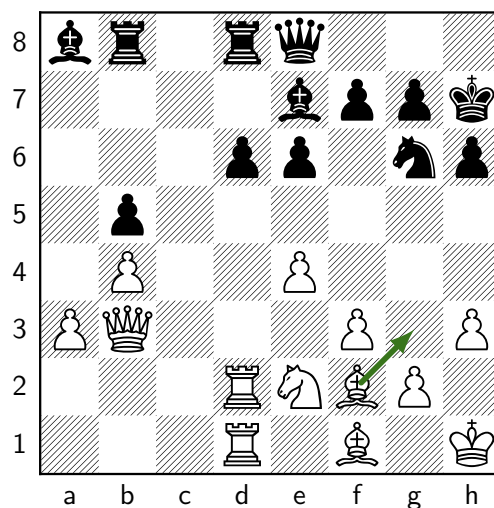
(a)



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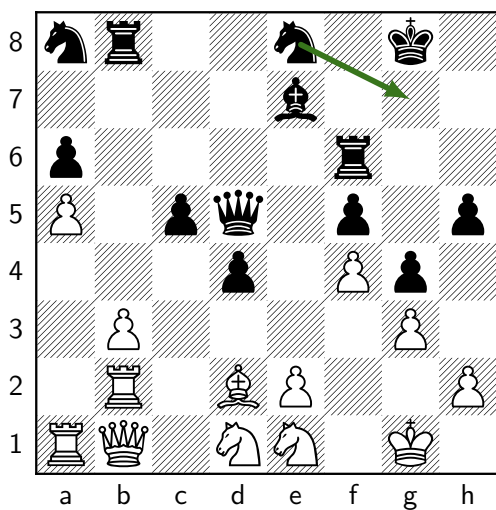


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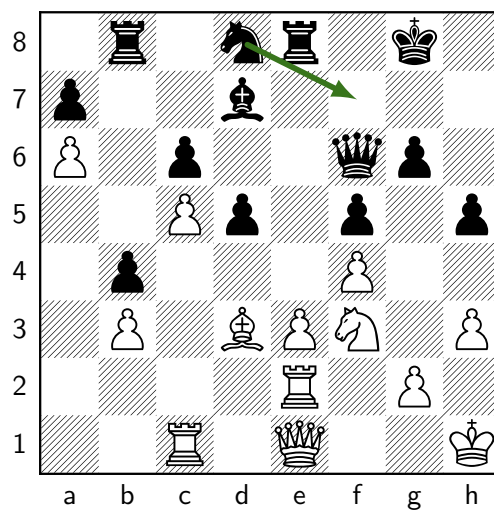


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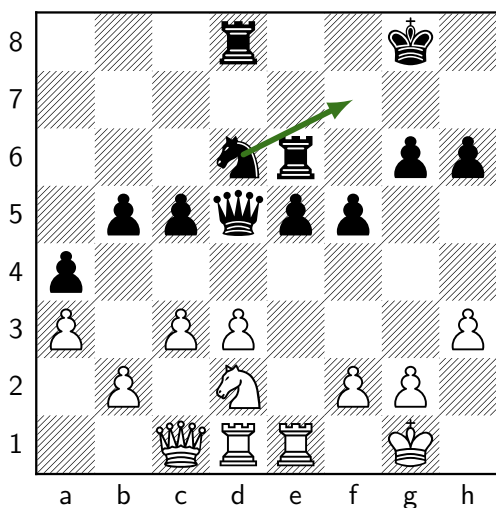
Figure 39: **Move 45, Black: Rb8.** (a) Our position; Melanie plays Rb8 (c8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#2734` (ply 87) [structural distance $d=94$, QAP obj=66]: the master played Rb8; under μ our *Rc1* plays the role of that game’s *Rc1*; cross-type roles: our *Be7* as their *Nd8*, our *ne1* as their *Rc7*; transferred move c8b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#2056` (ply 81) [structural distance $d=98$, QAP obj=59]: the master played Qc7; under μ our *Rc1* plays the role of that game’s *Qe2* (a **cross-type** correspondence); cross-type roles: our *rc2* as their *Ba8*, our *qb1* as their *Rd8*; transferred move c8b8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0002#3541` (ply 90) [structural distance $d=101$, QAP obj=45]: the master played Bg3; under μ our *Rc1* plays the role of that game’s *Bf2* (a **cross-type** correspondence); cross-type roles: our *ne1* as their *Rd1*, our *Na8* as their *Bf1*; transferred move c8b8 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Rb8 (summed QAP objective 222, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



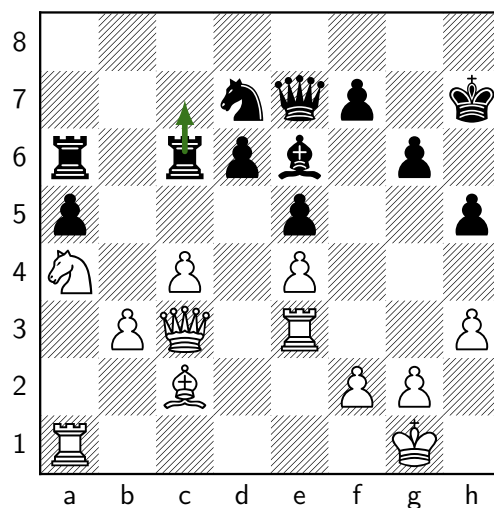
(a)



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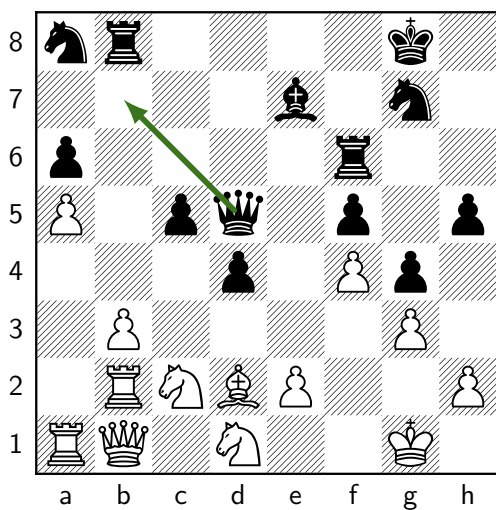


(c)

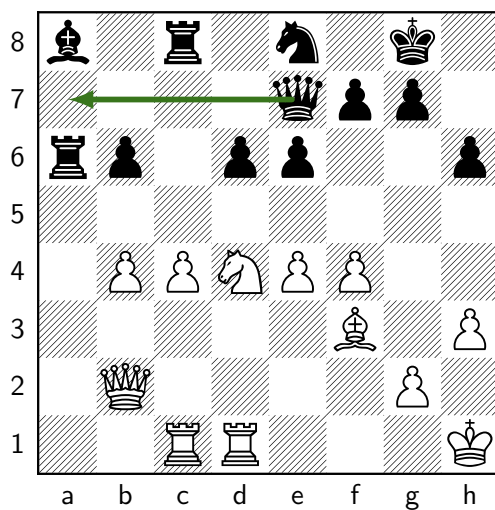


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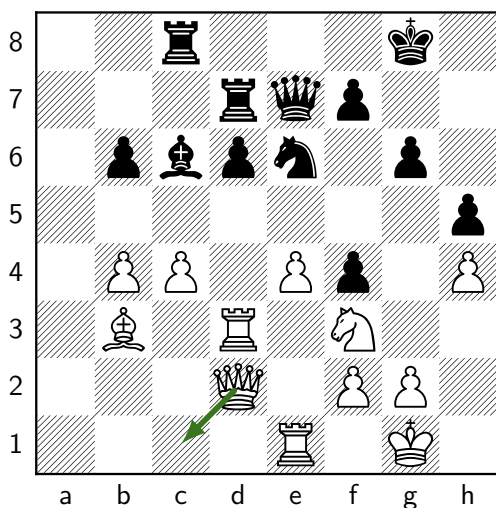
Figure 40: **Move 46, Black: Ng7**. (a) Our position; Melanie plays Ng7 (e8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1797` (ply 93) [structural distance $d=46$, QAP obj=64]: the master played Nf7; under μ our Ne1 plays the role of that game’s Nd1; cross-type roles: our pf4 as their Bd7, our bd2 as their Qf6; transferred move e8g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1300` (ply 101) [structural distance $d=94$, QAP obj=45]: the master played Nf7; under μ our Ne1 plays the role of that game’s Nd3; cross-type roles: our Na8 as their Rd8, our Rb8 as their Ph6; transferred move e8g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3975` (ply 89) [structural distance $d=102$, QAP obj=79]: the master played Rc7; under μ our Rb1 plays the role of that game’s Rc3; cross-type roles: our Pd4 as their Nd7, our bd2 as their Qe7; transferred move b8c8. Synthesis: 4 retrieved analogs transferred to Ng7 (summed QAP objective 215, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



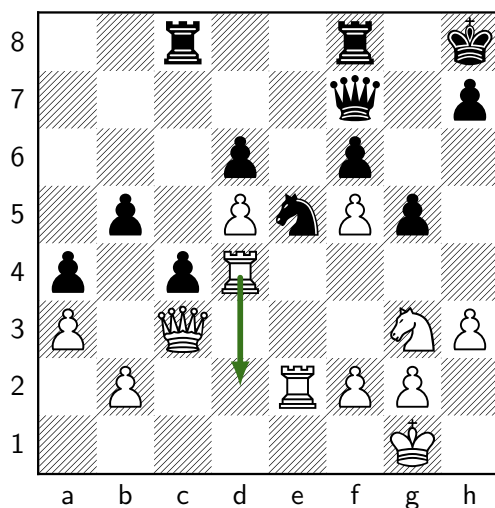
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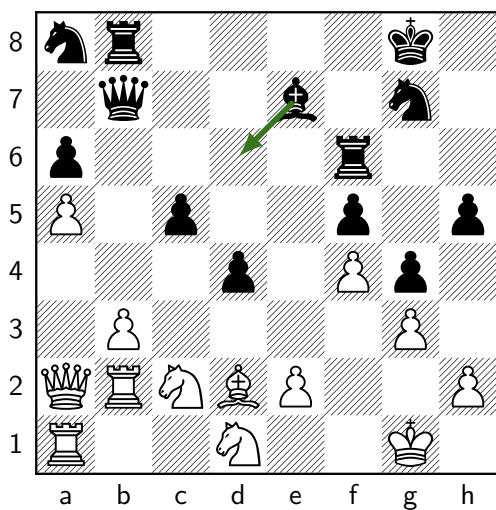


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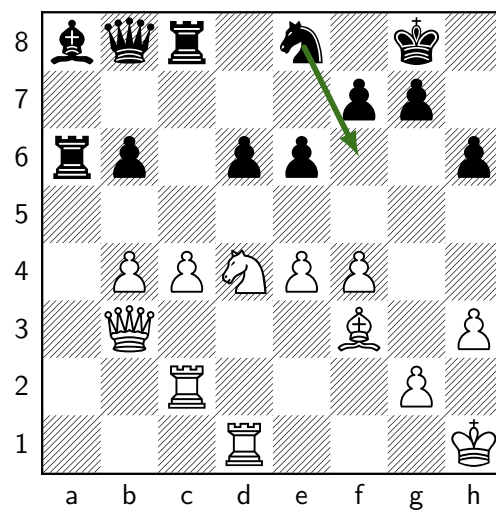


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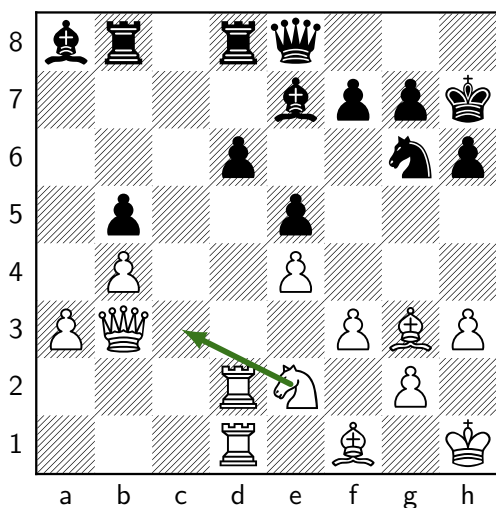
Figure 41: **Move 47, Black: Qb7**. (a) Our position; Melanie plays Qb7 (d5b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1250` (ply 81) [structural distance $d=85$, QAP obj=63]: the master played Qa7; under μ our Qd4 plays the role of that game's Qe2; cross-type roles: our Rb8 as their Ba8, our Pf5 as their Ne8; transferred move d5b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3664` (ply 106) [structural distance $d=107$, QAP obj=54]: the master played Qc1; under μ our Qd4 plays the role of that game's Qd2; cross-type roles: our Pg4 as their Re1, our Pf5 as their Bb3; transferred move d5b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3325` (ply 90) [structural distance $d=100$, QAP obj=31]: the master played Rdd2; under μ our Qd4 plays the role of that game's Rd4 (a **cross-type** correspondence); cross-type roles: our bd2 as their Re2, our Rb8 as their Pf2; transferred move d5b7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qb7 (summed QAP objective 201, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



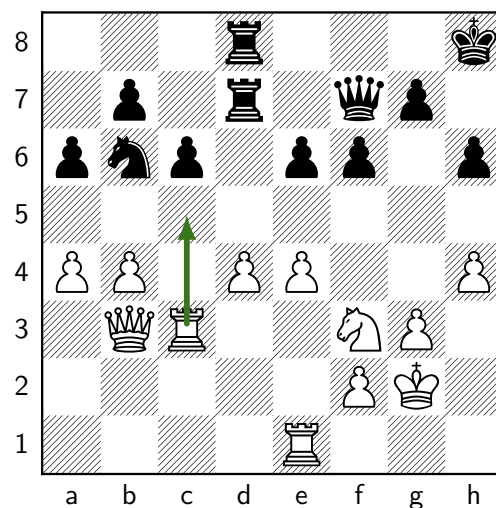
(a)



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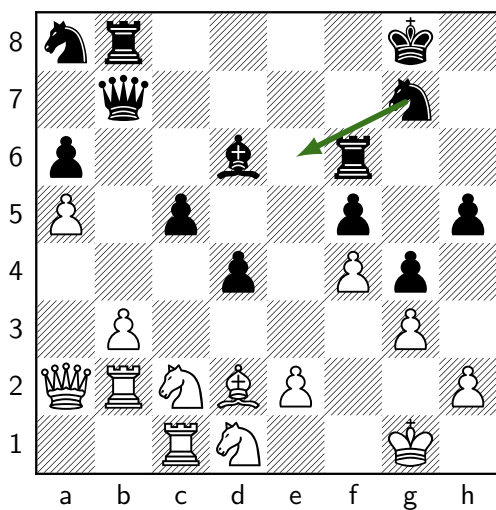


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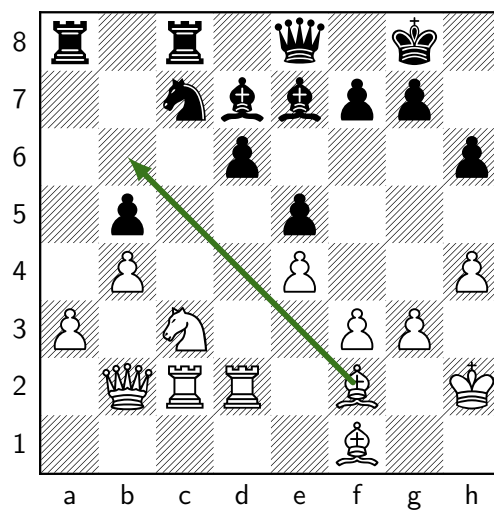


(d)

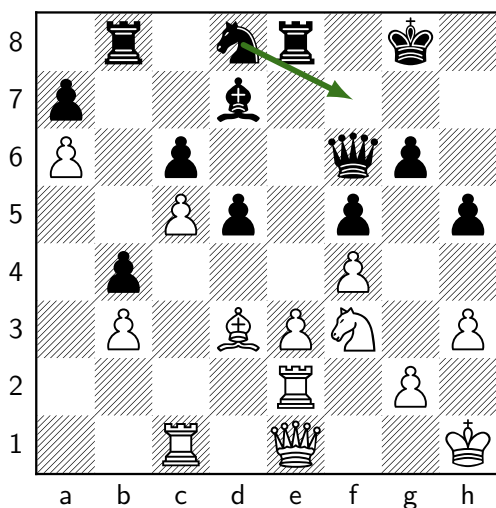
Figure 42: **Move 48, Black: Bd6.** (a) Our position; Melanie plays Bd6 (e7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1250` (ply 85) [structural distance $d=95$, QAP obj=83]: the master played Nf6; under μ our *Be2* plays the role of that game’s *Ne1* (a **cross-type** correspondence); cross-type roles: our *Qb7* as their *Ba8*, our *Rb8* as their *Qb8*; transferred move e7d6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#3541` (ply 92) [structural distance $d=97$, QAP obj=61]: the master played Nc3; under μ our *Be2* plays the role of that game’s *Ne2* (a **cross-type** correspondence); cross-type roles: our *rb2* as their *Bf1*, our *Qb7* as their *Rd2*; transferred move e7d6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3080` (ply 88) [structural distance $d=99$, QAP obj=59]: the master played Rc5; under μ our *Be2* plays the role of that game’s *Rc3* (a **cross-type** correspondence); cross-type roles: our *pg3* as their *Nf3*, our *nd1* as their *pa6*; transferred move e7d6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Bd6 (summed QAP objective 204, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



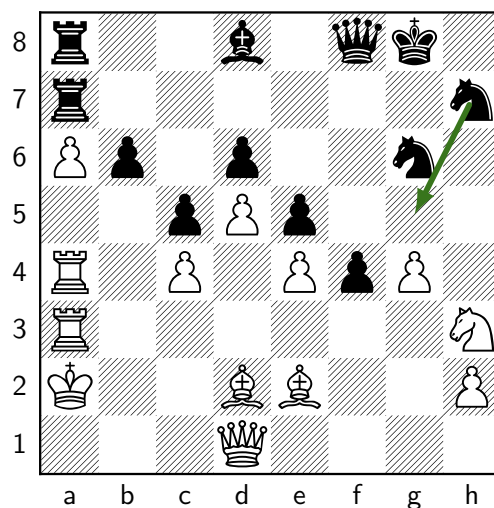
(a)



(b)

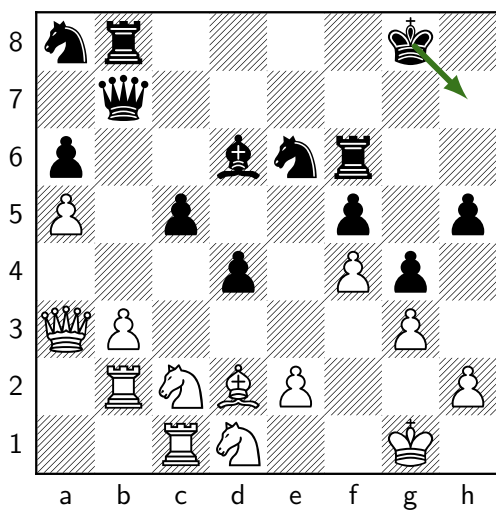


(c)

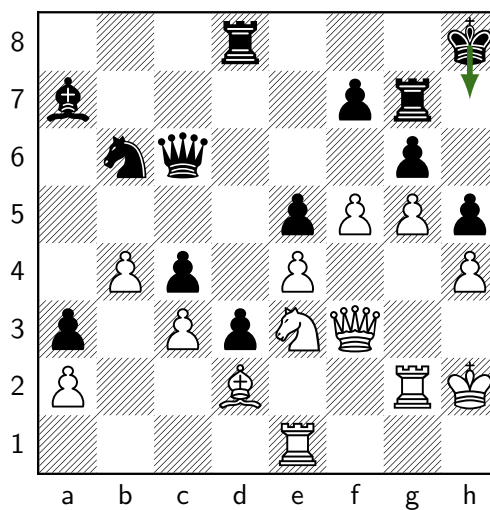


(d)

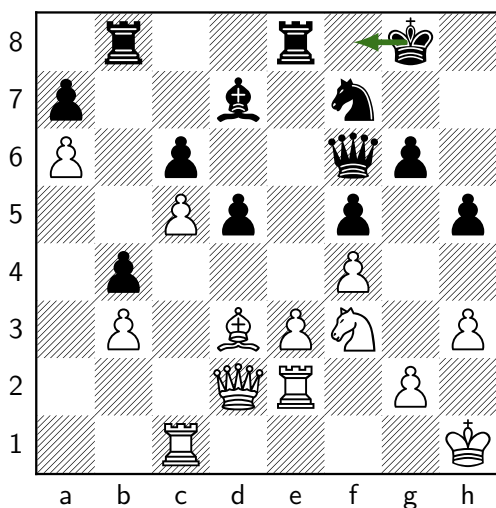
Figure 43: **Move 49, Black: Ne6 (g7e6)**. (a) Our position; Melanie plays Ne6 (g7e6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3541` (ply 108) [structural distance $d=107$, QAP obj=78]: the master played Bb6; under μ our Ng2 plays the role of that game’s Bf2 (a **cross-type** correspondence); cross-type roles: our Rf6 as their Qb2, our Qb7 as their Rd2; transferred move g7e6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1797` (ply 93) [structural distance $d=89$, QAP obj=54]: the master played Nf7; under μ our Ng2 plays the role of that game’s Nd1; cross-type roles: our Pg4 as their Rb8, our Qb7 as their Re8; transferred move g7e6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#2984` (ply 99) [structural distance $d=99$, QAP obj=37]: the master played Ng5; under μ our Ng2 plays the role of that game’s Nh2; cross-type roles: our Pf5 as their Bd8, our Pc5 as their Qf8; transferred move g7e6 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Ne6 (summed QAP objective 240, the strongest correspondence among the candidate moves); Melanie committed it via the search selector.



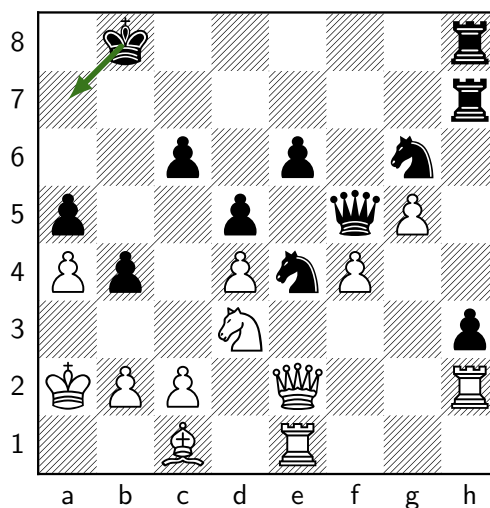
(a)



(b)

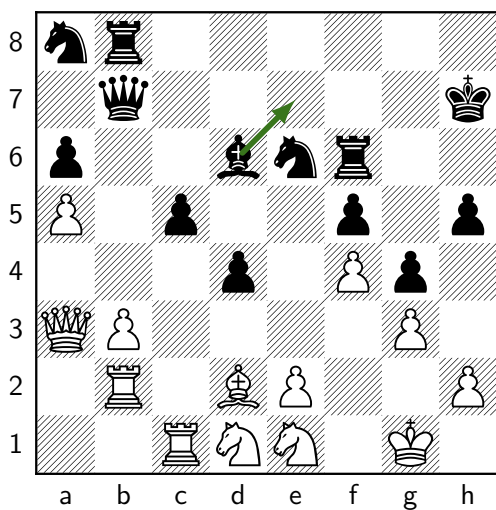


(c)

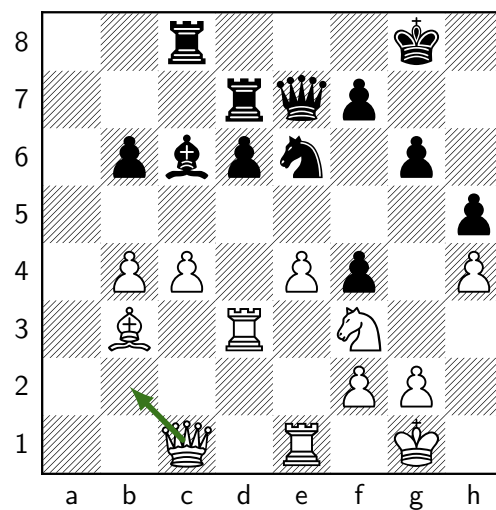


(d)

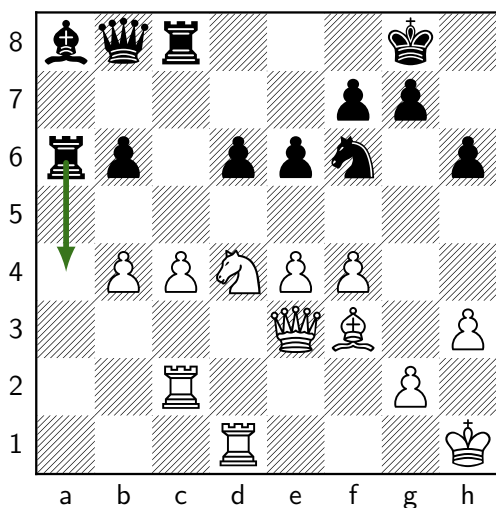
Figure 44: **Move 50, Black: Kh7**. (a) Our position; Melanie plays Kh7 (**g8h7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2432` (ply 107) [structural distance $d=76$, QAP obj=63]: the master played Kh7; under μ our *Kg1* plays the role of that game's *Kh1*; cross-type roles: our *pa5* as their *Rd8*, our *pg3* as their *Ba7*; transferred move **g8h7 (endorses this move)**. (c) Retrieved from `ccr1_top_p0002#1797` (ply 95) [structural distance $d=94$, QAP obj=60]: the master played Kf8; under μ our *Kg1* plays the role of that game's *Kg1*; cross-type roles: our *Qb7* as their *Re8*, our *pg3* as their *Bd7*; transferred move **g8h7 (endorses this move)**. (d) Retrieved from `ccr1_top_p0002#3315` (ply 87) [structural distance $d=102$, QAP obj=58]: the master played Ka7; under μ our *Kg1* plays the role of that game's *Kb1*; cross-type roles: our *Qb7* as their *Rh7*, our *Rf6* as their *Pe6*; transferred move **g8h7 (endorses this move)**. Synthesis: 5 retrieved analogs transferred to Kh7 (summed QAP objective 254, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



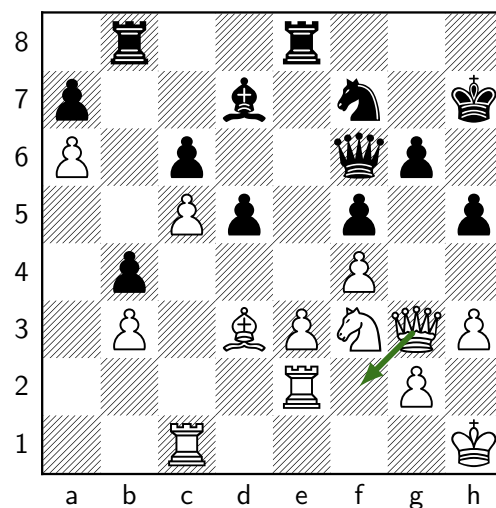
(a)



(b)

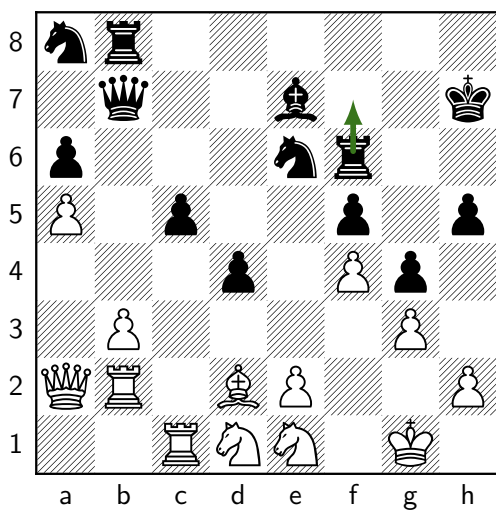


(c)

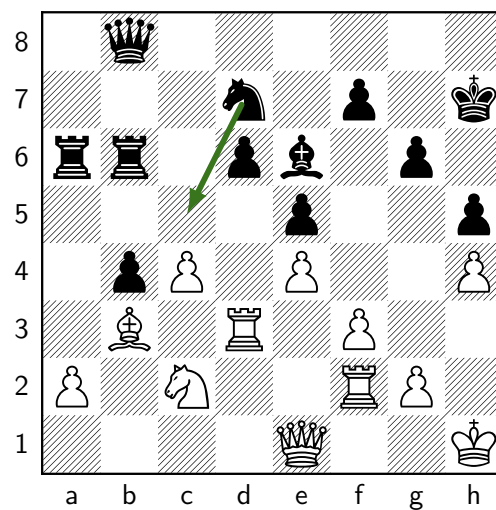


(d)

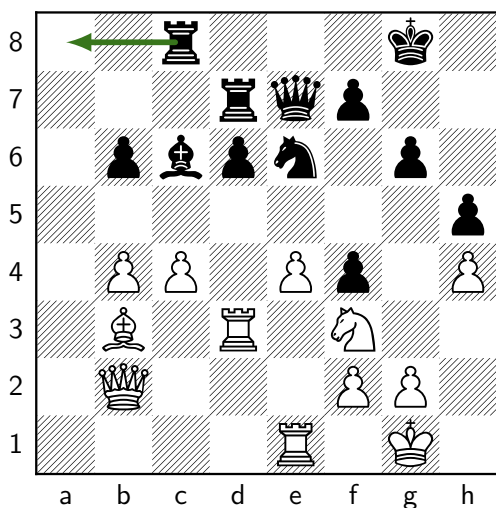
Figure 45: **Move 51, Black: Be7**. (a) Our position; Melanie plays **Be7** (d6e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#3664` (ply 102) [structural distance $d=104$, QAP obj=87]: the master played **Qb2**; under μ our *Bd3* plays the role of that game’s *Qc1* (a **cross-type** correspondence); cross-type roles: our *Ne6* as their *Pg2*, our *Pa6* as their *Bb3*; transferred move **d6e7** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1250` (ply 87) [structural distance $d=99$, QAP obj=82]: the master played **Ra4**; under μ our *Bd3* plays the role of that game’s *Ra3* (a **cross-type** correspondence); cross-type roles: our *qa3* as their *Ba8*, our *Rb8* as their *Qb8*; transferred move **d6e7** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1797` (ply 86) [structural distance $d=97$, QAP obj=55]: the master played **Qf2**; under μ our *Bd3* plays the role of that game’s *Qg3* (a **cross-type** correspondence); cross-type roles: our *Ne6* as their *Rc1*, our *bd2* as their *Pg2*; transferred move **d6e7** (**endorses this move**). Synthesis: 4 retrieved analogs transferred to **Be7** (summed QAP objective 270, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



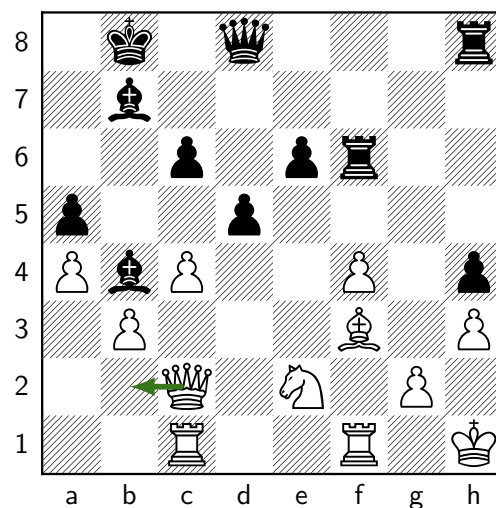
(a)



(b)

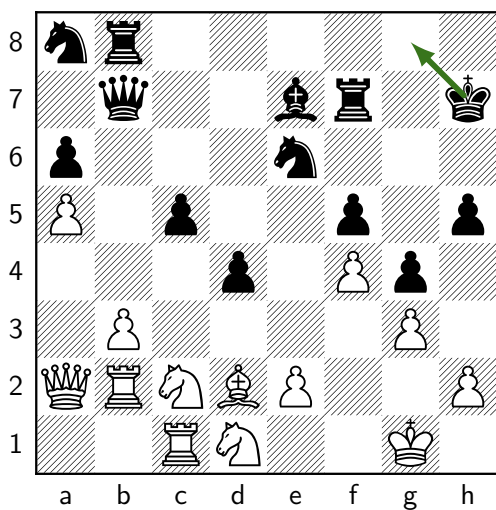


(c)

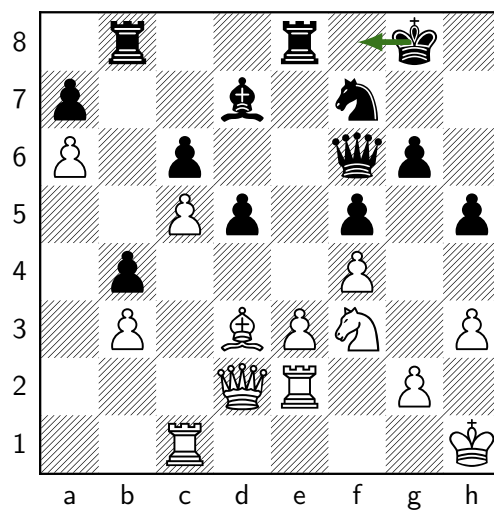


(d)

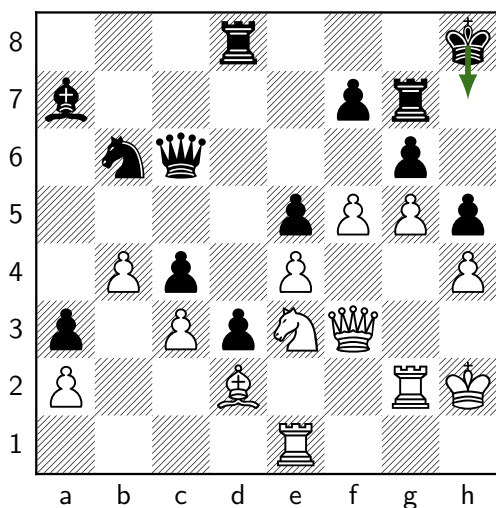
Figure 46: **Move 52, Black: Rf7 (f6f7)**. (a) Our position; Melanie plays Rf7 (f6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2892` (ply 107) [structural distance $d=104$, QAP obj=74]: the master played Nc5; under μ our Rf3 plays the role of that game’s Nd2 (a **cross-type** correspondence); cross-type roles: our Ne6 as their Be6, our bd2 as their Pg6; transferred move f6f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3664` (ply 103) [structural distance $d=99$, QAP obj=58]: the master played Ra8; under μ our Rf3 plays the role of that game’s Rc1; cross-type roles: our qa2 as their Rd7, our rc1 as their Qe7; transferred move f6f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#598` (ply 112) [structural distance $d=98$, QAP obj=40]: the master played Qb2; under μ our Rf3 plays the role of that game’s Qc2 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Rc1, our qa2 as their Ne2; transferred move f6f7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Rf7 (summed QAP objective 172, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



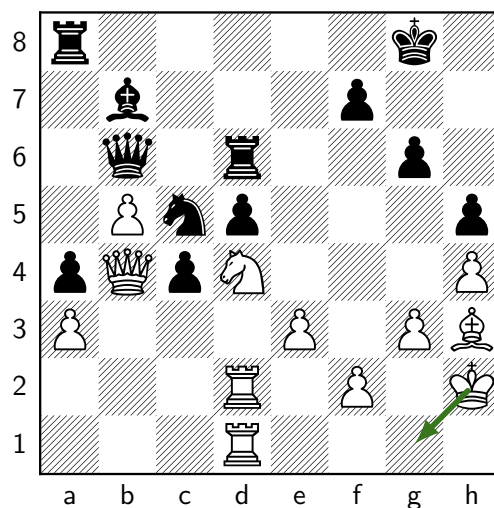
(a)



(b)

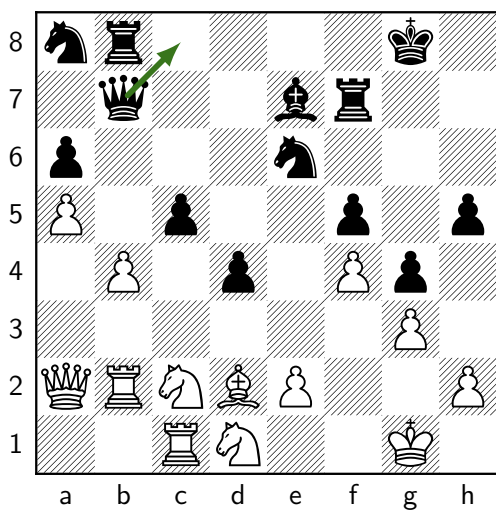


(c)

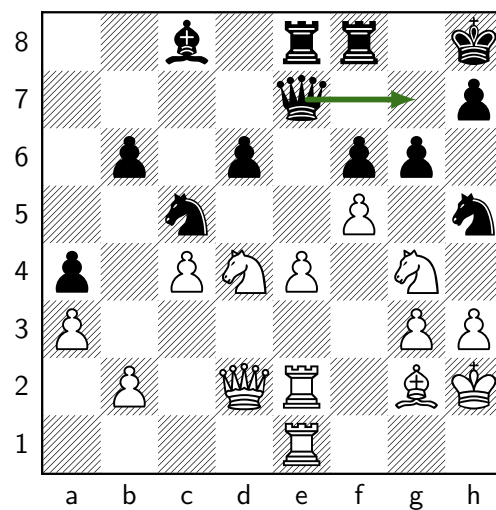


(d)

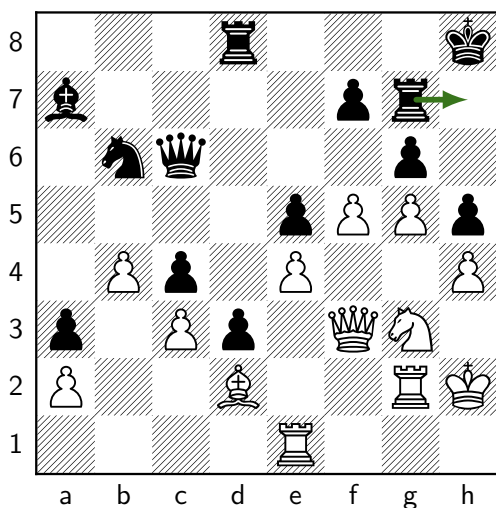
Figure 47: **Move 53, Black: Kg8.** (a) Our position; Melanie plays Kg8 (h7g8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1797` (ply 95) [structural distance $d=99$, QAP obj=63]: the master played Kf8; under μ our Kh2 plays the role of that game’s Kg1; cross-type roles: our Pg4 as their Rb8, our Qb7 as their Re8; transferred move h7g8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2432` (ply 107) [structural distance $d=93$, QAP obj=44]: the master played Kh7; under μ our Kh2 plays the role of that game’s Kh1; cross-type roles: our Na8 as their Ba7, our ph2 as their Rg7; transferred move h7g8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2023` (ply 116) [structural distance $d=99$, QAP obj=7]: the master played Kg1; under μ our Kh2 plays the role of that game’s Kh2; cross-type roles: our pa5 as their Rd1, our pe2 as their Rd2; transferred move h7g8 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kg8 (summed QAP objective 171, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



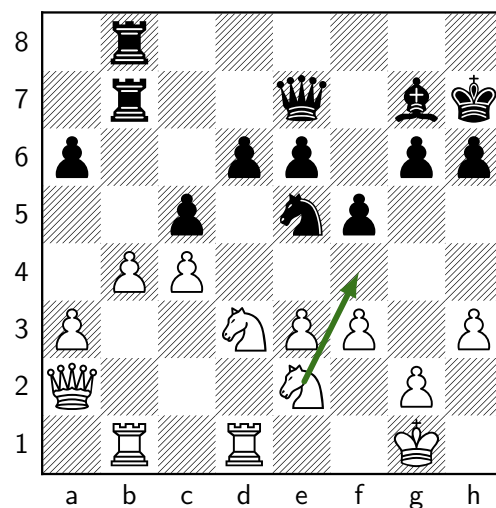
(a)



(b)

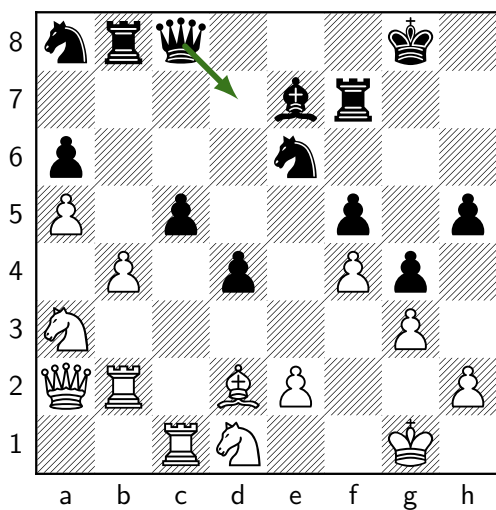


(c)

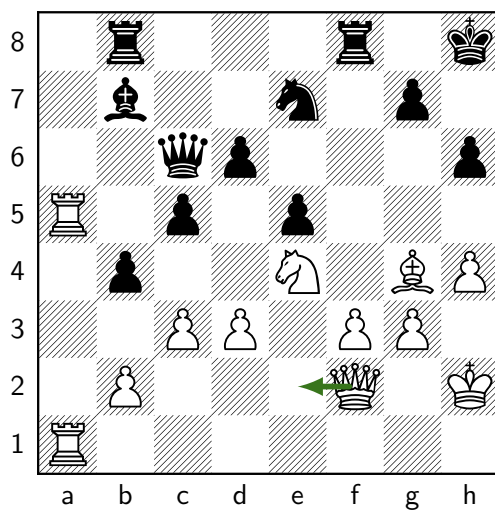


(d)

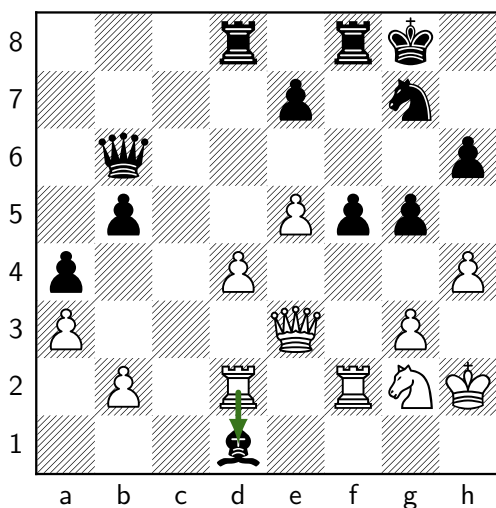
Figure 48: **Move 54, Black: Qc8.** (a) Our position; Melanie plays Qc8 (b7c8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#3814` (ply 107) [structural distance $d=95$, QAP obj=115]: the master played Qg7; under μ our Qb2 plays the role of that game’s Qe2; cross-type roles: our Qb7 as their Bc8, our Be7 as their Ph7; transferred move b7c8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2432` (ply 111) [structural distance $d=82$, QAP obj=90]: the master played Rh7; under μ our Qb2 plays the role of that game’s Rg2 (a **cross-type** correspondence); cross-type roles: our Ph5 as their Rd8, our rb2 as their Nb6; transferred move b7c8 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2133` (ply 94) [structural distance $d=98$, QAP obj=97]: the master played Nef4; under μ our Pd5 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our bd2 as their Rd1, our Ne6 as their Pg2; transferred move d4d3. Synthesis: 2 retrieved analogs transferred to Qc8 (summed QAP objective 205, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



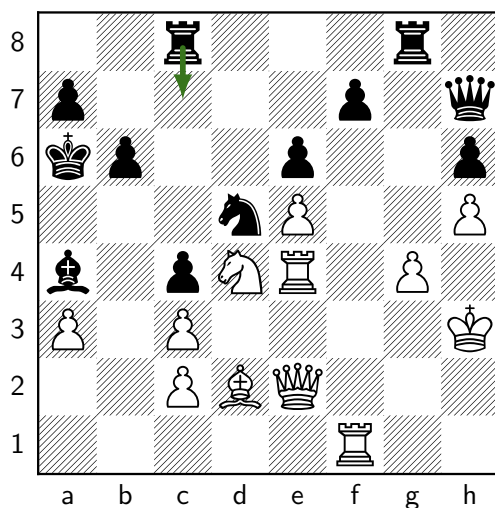
(a)



(b)

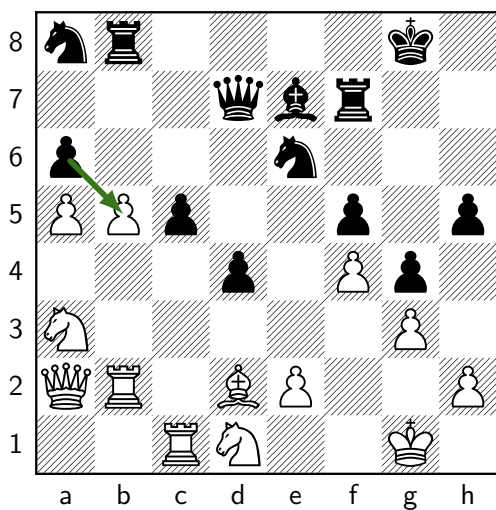


(c)

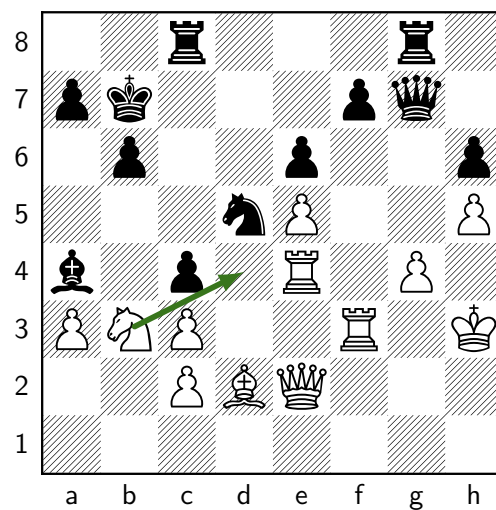


(d)

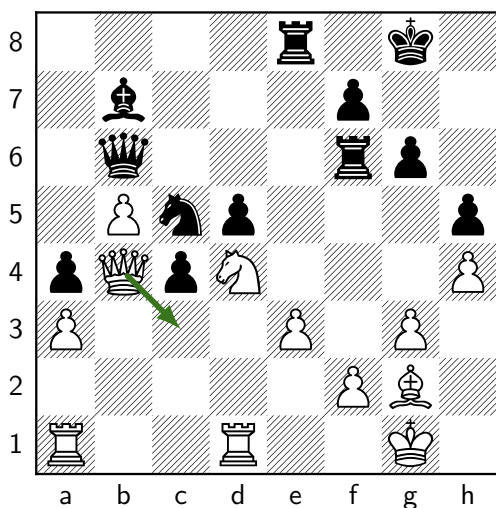
Figure 49: **Move 55, Black: Qd7.** (a) Our position; Melanie plays Qd7 (c8d7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#958` (ply 98) [structural distance $d=84$, QAP obj=59]: the master played Qe2; under μ our Qc1 plays the role of that game’s Qf2; cross-type roles: our pa5 as their Ra1, our Ne6 as their Ra5; transferred move c8d7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2576` (ply 94) [structural distance $d=88$, QAP obj=51]: the master played Rxd1; under μ our Qc1 plays the role of that game’s Rd2 (a **cross-type** correspondence); cross-type roles: our pa5 as their bd1, our rc1 as their Qe3; transferred move c8d7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#91` (ply 101) [structural distance $d=92$, QAP obj=40]: the master played Rc7; under μ our Qc1 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Na8 as their Rg8, our Pc5 as their Qh7; transferred move c8d7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd7 (summed QAP objective 179, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



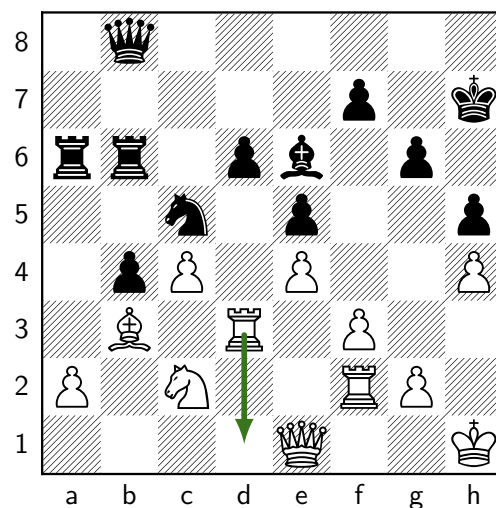
(a)



(b)

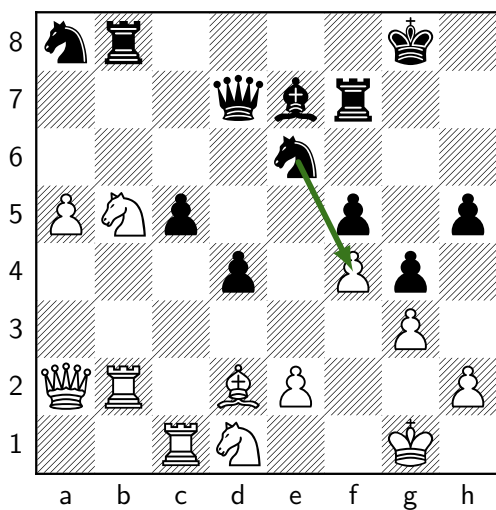


(c)

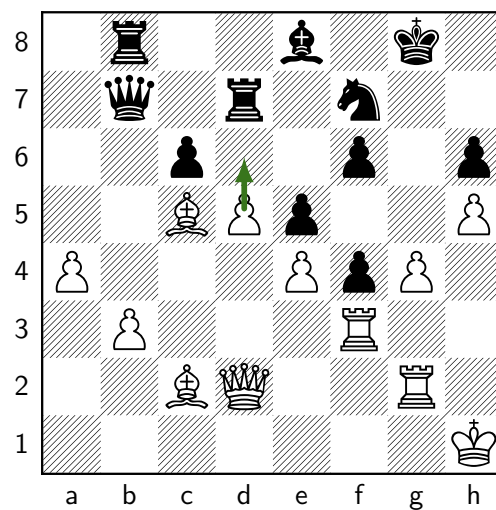


(d)

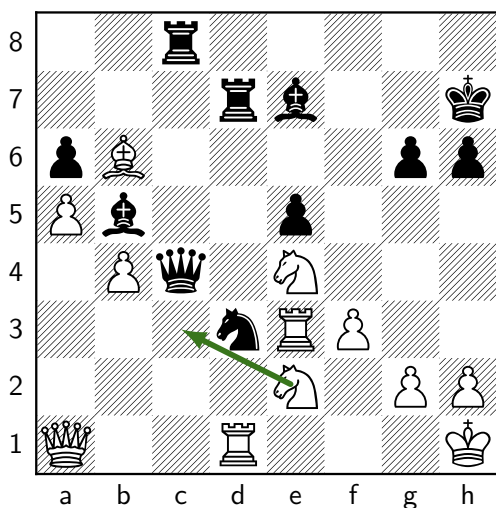
Figure 50: **Move 56, Black: axb5.** (a) Our position; Melanie plays axb5 (a6b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#91 (ply 124) [structural distance $d=97$, QAP obj=60]: the master played Nd4; under μ our $Pa3$ plays the role of that game’s Nb3 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Bd2, our pb5 as their ba4; transferred move a6b5 (**endorses this move**). (c) Retrieved from ccr1_top_p0002#2023 (ply 100) [structural distance $d=89$, QAP obj=36]: the master played Qc3; under μ our $Pa3$ plays the role of that game’s Qb4 (a **cross-type** correspondence); cross-type roles: our Be7 as their Ra1, our Na8 as their Rd1; transferred move a6b5 (**endorses this move**). (d) Retrieved from ccr1_top_p0003#2892 (ply 108) [structural distance $d=86$, QAP obj=35]: the master played Rd1; under μ our $Pa3$ plays the role of that game’s Rd3 (a **cross-type** correspondence); cross-type roles: our Rb8 as their Nc2, our na3 as their Rf2; transferred move a6b5 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to axb5 (summed QAP objective 170, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



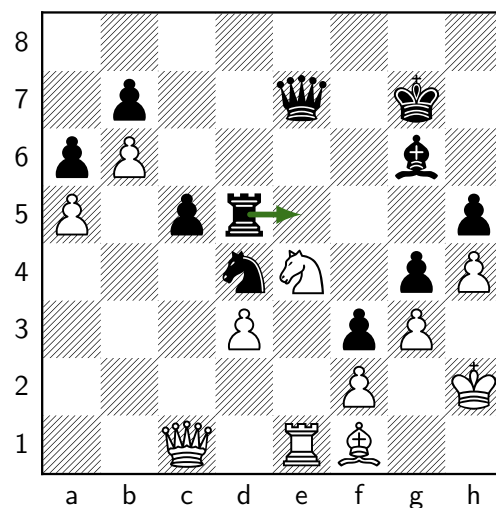
(a)



(b)

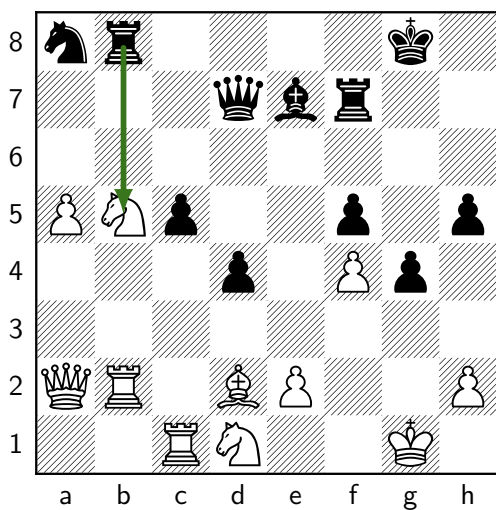


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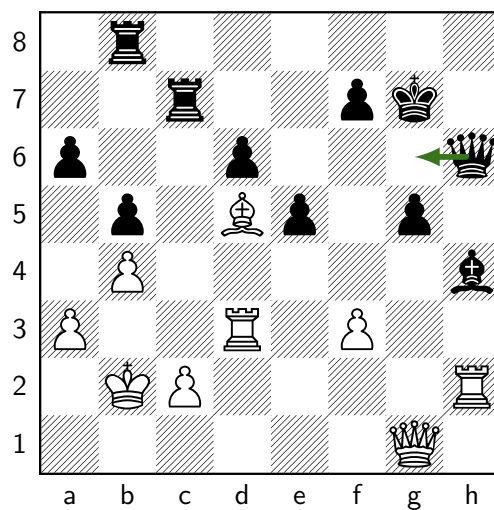


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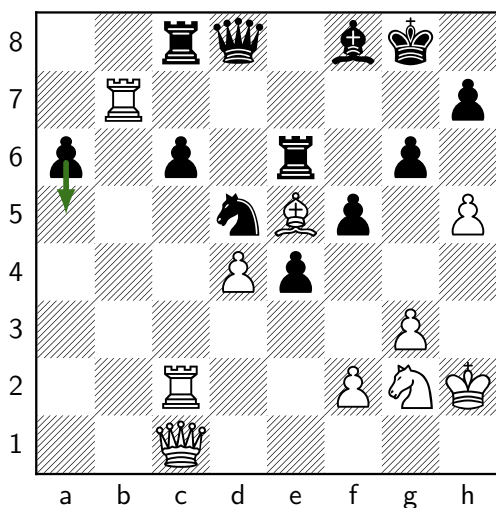
Figure 51: **Move 57, Black: Nxf4.** (a) Our position; Melanie plays Nxf4 (e6f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2089 (ply 110) [structural distance $d=97$, QAP obj=57]: the master played d6; under μ our Ne3 plays the role of that game’s Pd5 (a **cross-type** correspondence); cross-type roles: our Na8 as their Pe4, our nb5 as their pf4; transferred move e6f4 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#2804 (ply 98) [structural distance $d=93$, QAP obj=100]: the master played N2c3; under μ our Be2 plays the role of that game’s Ne2 (a **cross-type** correspondence); cross-type roles: our Na8 as their Qa1, our Rf7 as their Ph2; transferred move e7d6. (d) Retrieved from ccr1_top_p0002#316 (ply 99) [structural distance $d=90$, QAP obj=96]: the master played Re5; under μ our Ne3 plays the role of that game’s Rd4 (a **cross-type** correspondence); cross-type roles: our Na8 as their Pb7, our Rb8 as their Qe7; transferred move e6g5. Synthesis: 1 retrieved analog transferred to Nxf4 (summed QAP objective 57, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



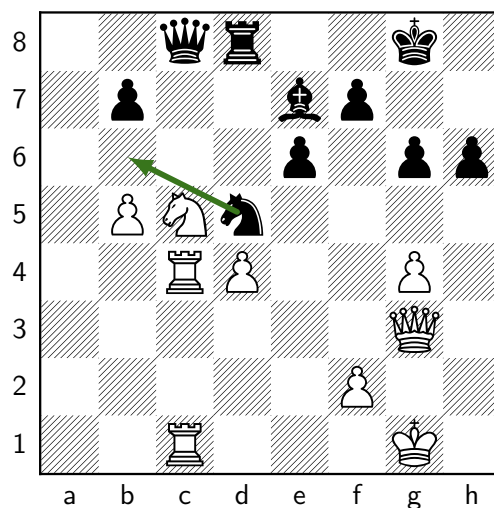
(a)



(b)

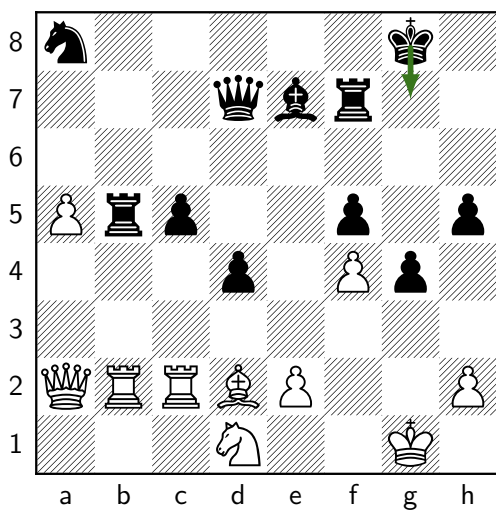


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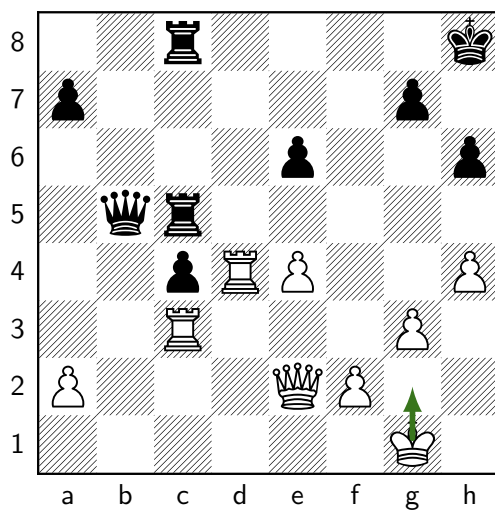


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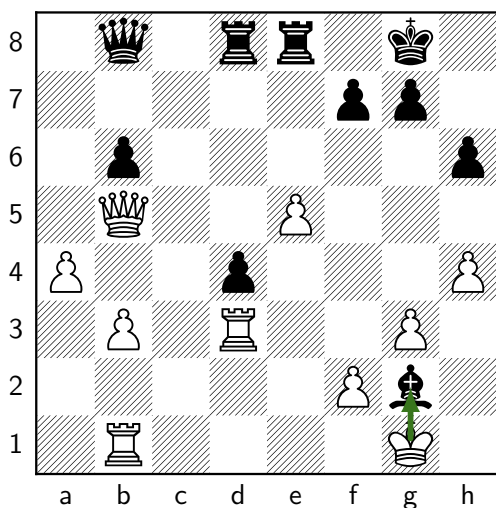
Figure 52: **Move 58, Black: Rxb5.** (a) Our position; Melanie plays Rxb5 (b8b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1051` (ply 101) [structural distance $d=108$, QAP obj=42]: the master played Qg6; under μ our *Rb1* plays the role of that game’s *Qh3* (a **cross-type** correspondence); cross-type roles: our *pf4* as their *Rb8*, our *nb5* as their *Pf7*; transferred move b8b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#3989` (ply 105) [structural distance $d=92$, QAP obj=20]: the master played a5; under μ our *Rb1* plays the role of that game’s *Pa3* (a **cross-type** correspondence); cross-type roles: our *Pd4* as their *Rc8*, our *Pf5* as their *Qd8*; transferred move b8b5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1885` (ply 113) [structural distance $d=106$, QAP obj=99]: the master played Nb6; under μ our *Pg5* plays the role of that game’s *Nd4* (a **cross-type** correspondence); cross-type roles: our *Rb8* as their *Qc8*, our *Na8* as their *Rd8*; transferred move g4g3. Synthesis: 2 retrieved analogs transferred to Rxb5 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



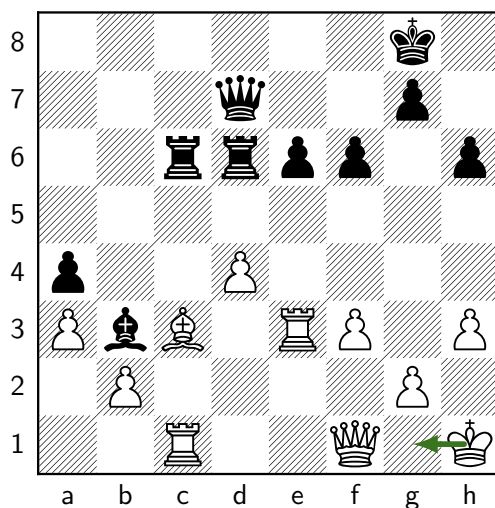
(a)



(b)

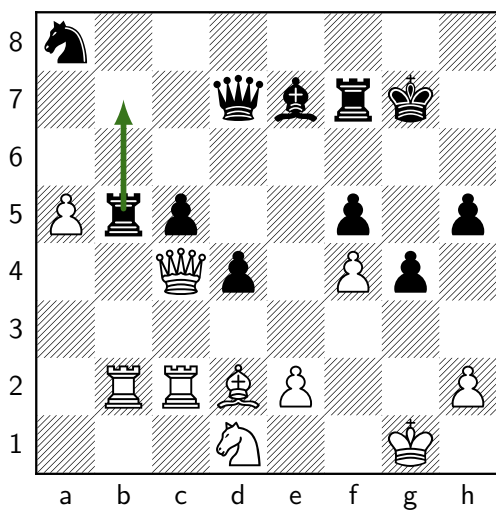


(c)

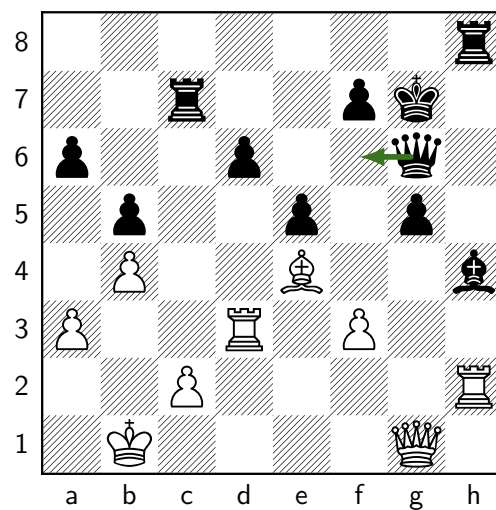


(d)

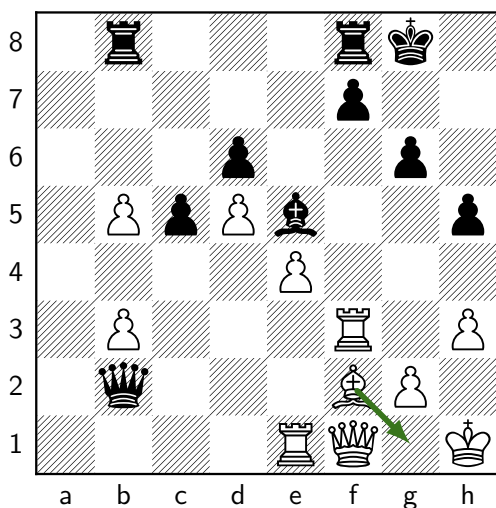
Figure 53: **Move 59, Black: Kg7.** (a) Our position; Melanie plays Kg7 (g8g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#1783 (ply 116) [structural distance $d=96$, QAP obj=51]: the master played Kg2; under μ our Kg1 plays the role of that game's Kg1; cross-type roles: our Na8 as their Pf2, our qa2 as their Rc3; transferred move g8g7 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#2371 (ply 102) [structural distance $d=87$, QAP obj=48]: the master played Kxg2; under μ our Kg1 plays the role of that game's Kg1; cross-type roles: our Pf5 as their Rb1, our Pg4 as their bg2; transferred move g8g7 (**endorses this move**). (d) Retrieved from ccr1_top_p0001#1734 (ply 106) [structural distance $d=79$, QAP obj=47]: the master played Kg1; under μ our Kg1 plays the role of that game's Kh1; cross-type roles: our Be7 as their Rc1, our Ph5 as their Qf1; transferred move g8g7 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kg7 (summed QAP objective 384, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



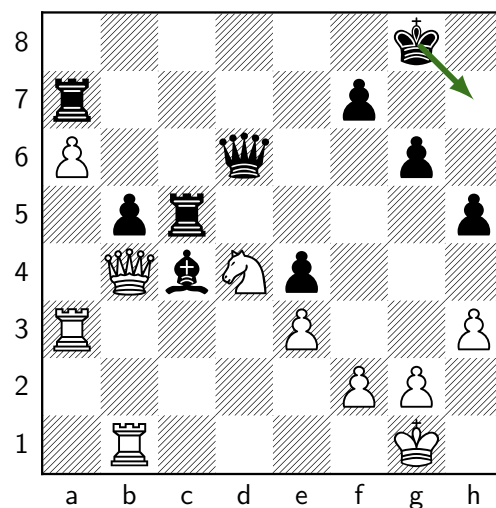
(a)



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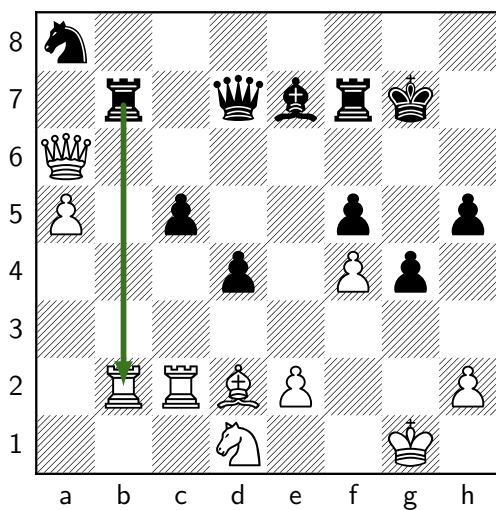


(c)

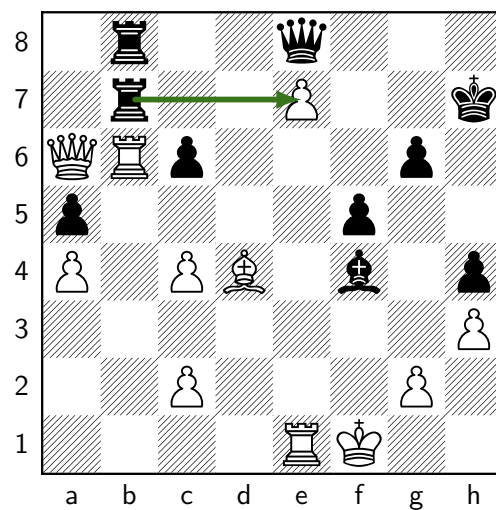


(d)

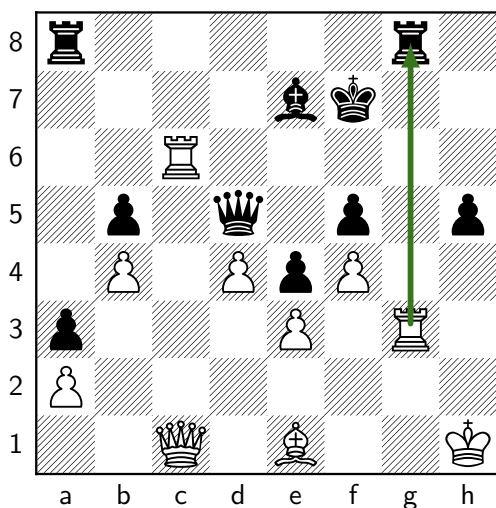
Figure 54: **Move 60, Black: Rb7.** (a) Our position; Melanie plays Rb7 (b5b7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1051` (ply 105) [structural distance $d=94$, QAP obj=23]: the master played Qf6; under μ our Rb4 plays the role of that game’s Qg3 (a **cross-type** correspondence); cross-type roles: our pf4 as their Rc7, our Qd7 as their Pf7; transferred move b5b7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1080` (ply 134) [structural distance $d=90$, QAP obj=5]: the master played Bg1; under μ our Rb4 plays the role of that game’s Bf2 (a **cross-type** correspondence); cross-type roles: our qc4 as their Re1, our Rf7 as their Qf1; transferred move b5b7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#703` (ply 111) [structural distance $d=99$, QAP obj=55]: the master played Kh7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our Pd4 as their Ra7, our nd1 as their pa6; transferred move g7g8. Synthesis: 2 retrieved analogs transferred to Rb7 (summed QAP objective 29, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



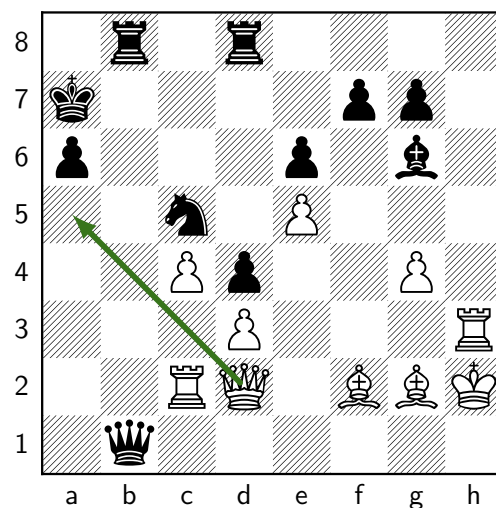
(a)



(b)

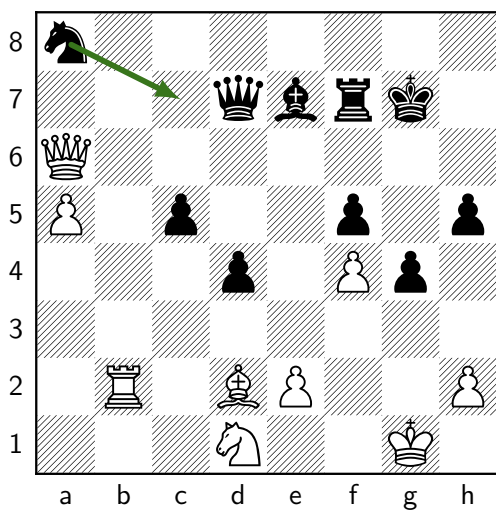


(c)

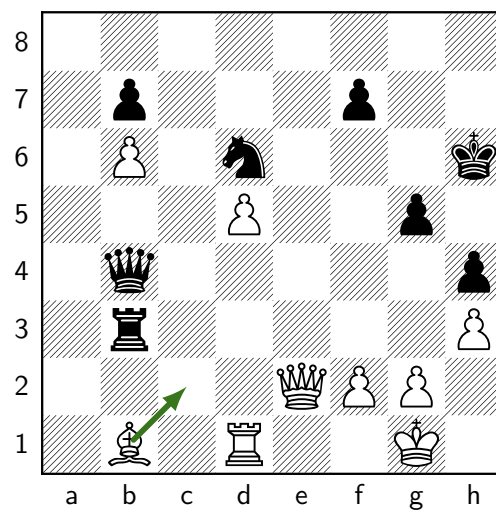


(d)

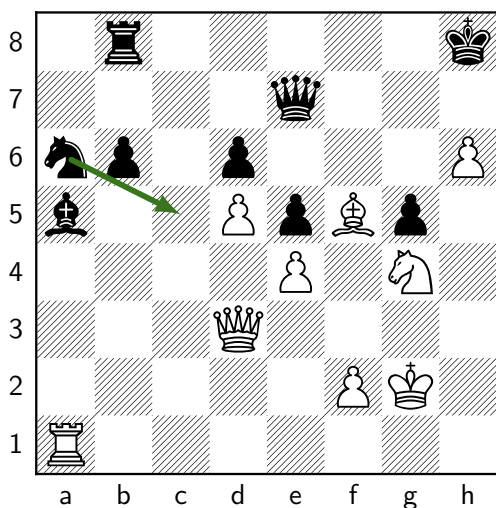
Figure 55: **Move 61, Black: Rxb2**. (a) Our position; Melanie plays Rxb2 (b7b2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3836` (ply 137) [structural distance $d=83$, QAP obj=47]: the master played Rxe7; under μ our Rb2 plays the role of that game’s Rb2; cross-type roles: our Qd7 as their Rb8, our Pf5 as their Qe8; transferred move b7b5. (c) Retrieved from `ccr1_top_p0002#3866` (ply 114) [structural distance $d=86$, QAP obj=28]: the master played Rxg8; under μ our Pg5 plays the role of that game’s Rg3 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Be1, our Rf7 as their pa3; transferred move g4g3. (d) Retrieved from `ccr1_top_p0002#441` (ply 114) [structural distance $d=88$, QAP obj=27]: the master played Qa5; under μ our Qd2 plays the role of that game’s Qd2; cross-type roles: our Pd4 as their Bf2, our Pc5 as their Bg2; transferred move d7a4. Synthesis: no retrieved analog transferred to Rxb2 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



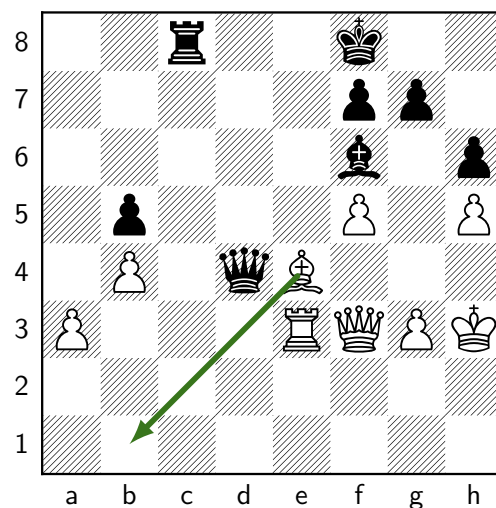
(a)



(b)

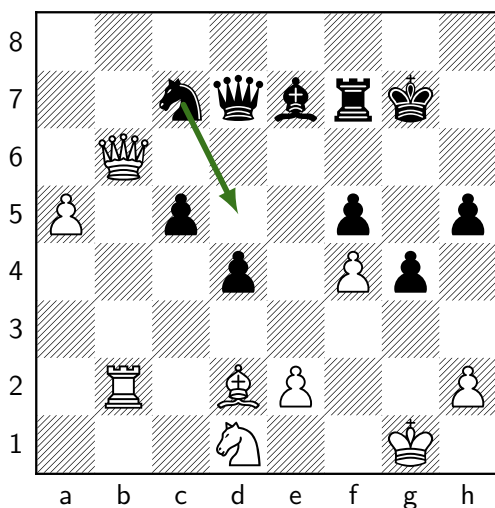


(c)

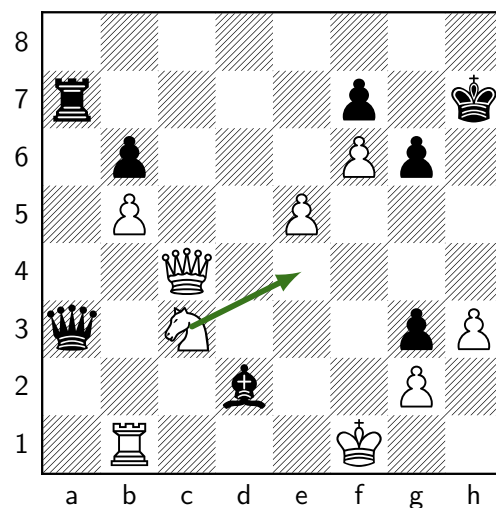


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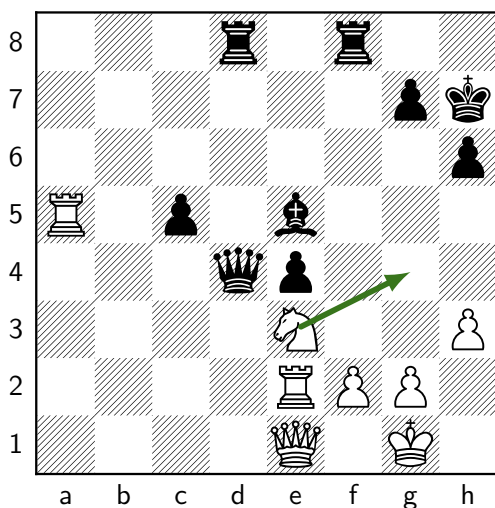
Figure 56: **Move 62, Black: Nc7 (a8c7)**. (a) Our position; Melanie plays Nc7 (a8c7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1686` (ply 106) [structural distance $d=72$, QAP obj=23]: the master played Bc2; under μ our *Na1* plays the role of that game’s *Bb1* (a **cross-type** correspondence); cross-type roles: our *qa6* as their *rb3*, our *pa5* as their *qb4*; transferred move a8c7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2042` (ply 133) [structural distance $d=57$, QAP obj=17]: the master played Nc5; under μ our *Na1* plays the role of that game’s *Na3*; cross-type roles: our *Pg4* as their *Qe7*, our *Qd7* as their *Ba5*; transferred move a8c7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1093` (ply 122) [structural distance $d=77$, QAP obj=14]: the master played Bb1; under μ our *Na1* plays the role of that game’s *Be4* (a **cross-type** correspondence); cross-type roles: our *Na8* as their *Re3*, our *Be7* as their *Pg3*; transferred move a8c7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Nc7 (summed QAP objective 62, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



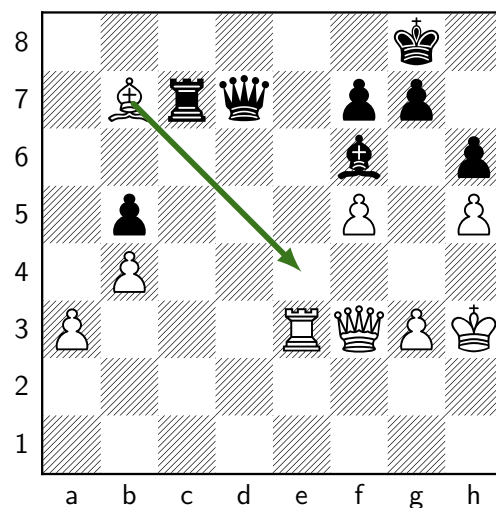
(a)



(b)

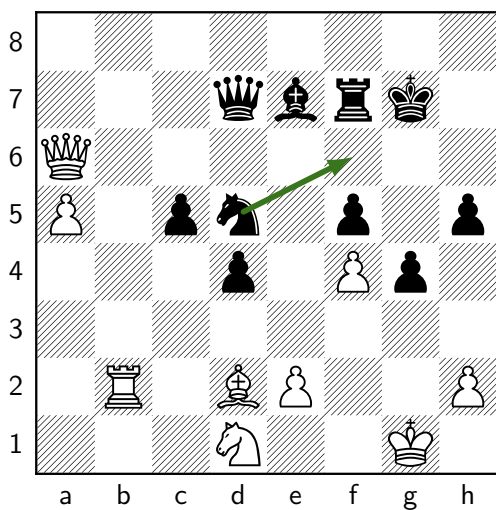


(c)

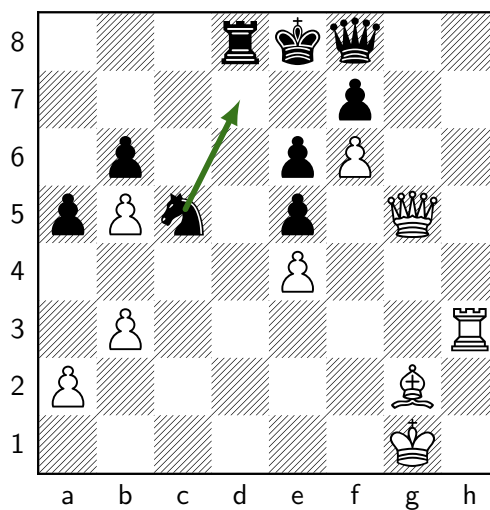


(d)

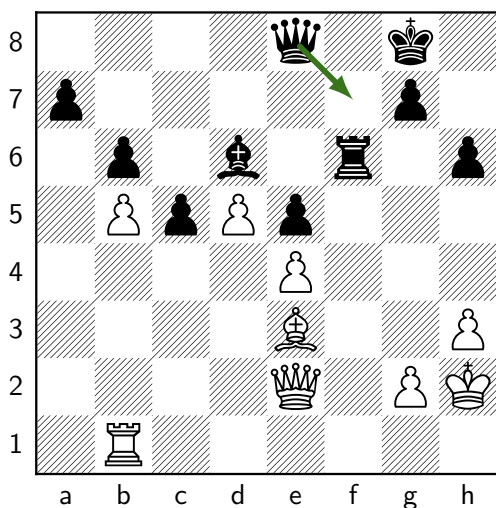
Figure 57: **Move 63, Black: Nd5.** (a) Our position; Melanie plays Nd5 (c7d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#1804` (ply 116) [structural distance $d=70$, QAP obj=32]: the master played Ne4; under μ our Nc2 plays the role of that game’s Nc3; cross-type roles: our Be7 as their Rb1, our qb6 as their bd2; transferred move c7d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2798` (ply 128) [structural distance $d=79$, QAP obj=27]: the master played Ng4; under μ our Nc2 plays the role of that game’s Ne3; cross-type roles: our Rf7 as their Qe1, our Be7 as their Re2; transferred move c7d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#1093` (ply 134) [structural distance $d=68$, QAP obj=25]: the master played Be4; under μ our Nc2 plays the role of that game’s Bb7 (a **cross-type** correspondence); cross-type roles: our Be7 as their Pb4, our Rf7 as their pb5; transferred move c7d5 (**endorses this move**). Synthesis: 6 retrieved analogs transferred to Nd5 (summed QAP objective 138, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



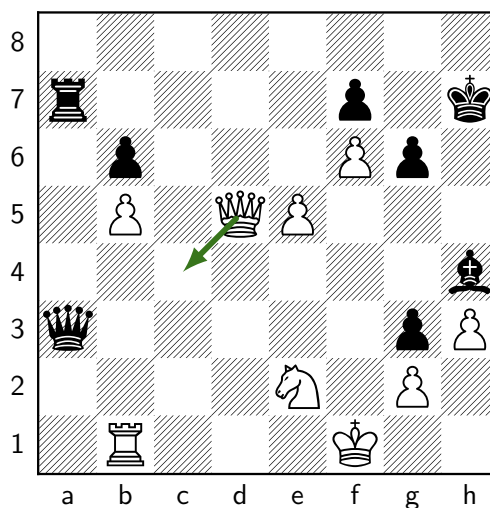
(a)



(b)



(c)



(d)

Figure 58: **Move 64, Black: Nf6.** (a) Our position; Melanie plays Nf6 (d5f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2143` (ply 121) [structural distance $d=83$, QAP obj=48]: the master played Nd7; under μ our Nd4 plays the role of that game's Nc4; cross-type roles: our bd2 as their Pe6, our qa6 as their pf6; transferred move d5f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#126` (ply 133) [structural distance $d=89$, QAP obj=37]: the master played Qf7; under μ our Nd4 plays the role of that game's Qe1 (a **cross-type** correspondence); cross-type roles: our nd1 as their Pg7, our qa6 as their Pb6; transferred move d5f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1804` (ply 112) [structural distance $d=71$, QAP obj=33]: the master played Qc4; under μ our Nd4 plays the role of that game's Qd5 (a **cross-type** correspondence); cross-type roles: our bd2 as their qa3, our qa6 as their Ph3; transferred move d5f4. Synthesis: 2 retrieved analogs transferred to Nf6 (summed QAP objective 85, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.

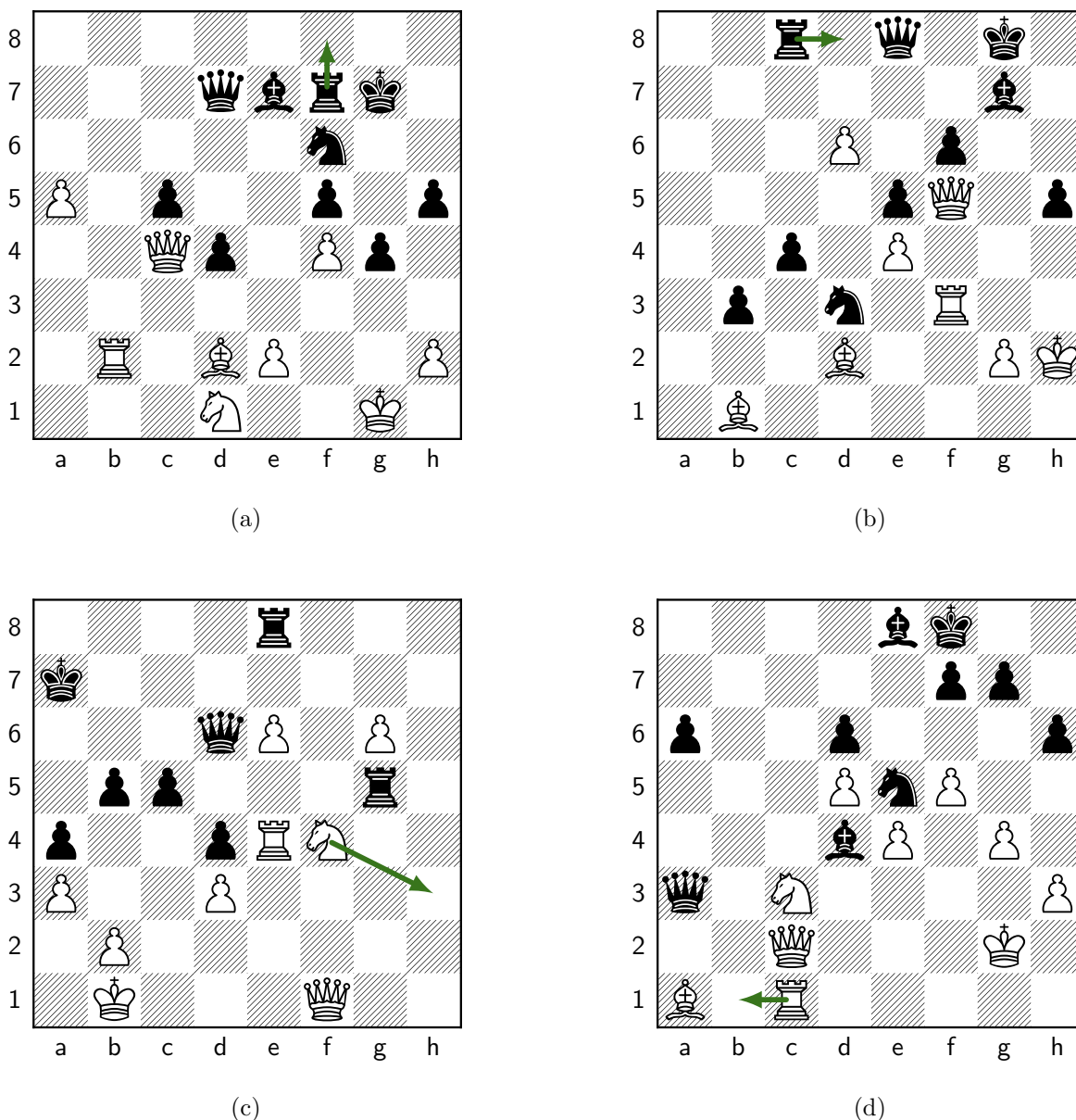
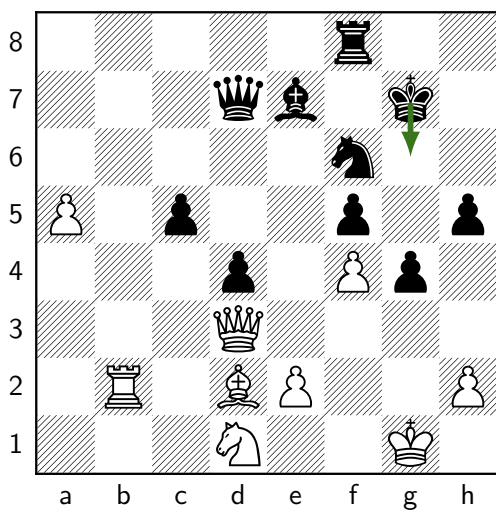
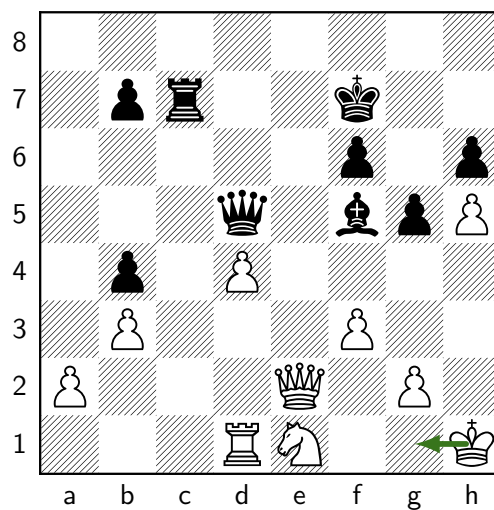


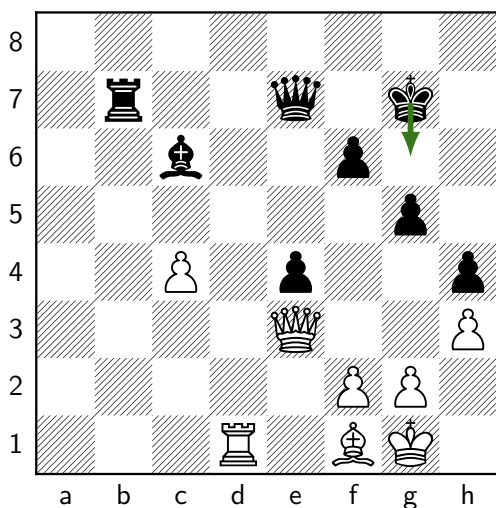
Figure 59: **Move 65, Black: Rf8.** (a) Our position; Melanie plays Rf8 (f7f8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#804 (ply 141) [structural distance $d=100$, QAP obj=73]: the master played Rd8; under μ our Rf2 plays the role of that game's Rc1; cross-type roles: our Pf5 as their Bg7, our Be7 as their Pc4; transferred move f7f8 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#999 (ply 124) [structural distance $d=102$, QAP obj=42]: the master played Nh3; under μ our Rf2 plays the role of that game's Nf4 (a **cross-type** correspondence); cross-type roles: our rb2 as their Pb2, our Be7 as their Pd3; transferred move f7f8 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#686 (ply 118) [structural distance $d=87$, QAP obj=38]: the master played Rb1; under μ our Rf2 plays the role of that game's Rc1; cross-type roles: our Pc5 as their Ba1, our Rf7 as their Nc3; transferred move f7f8 (**endorses this move**). Synthesis: 13 retrieved analogs transferred to Rf8 (summed QAP objective 292, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



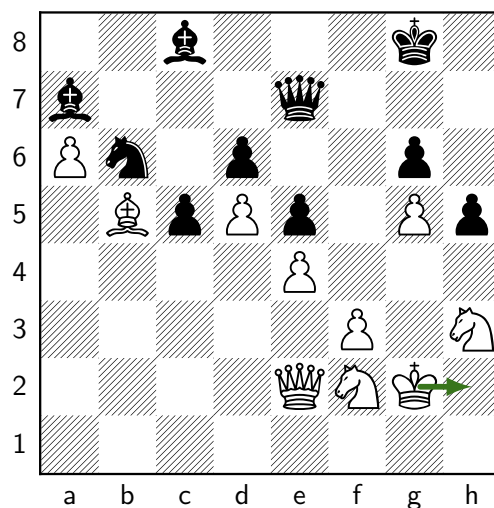
(a)



(b)

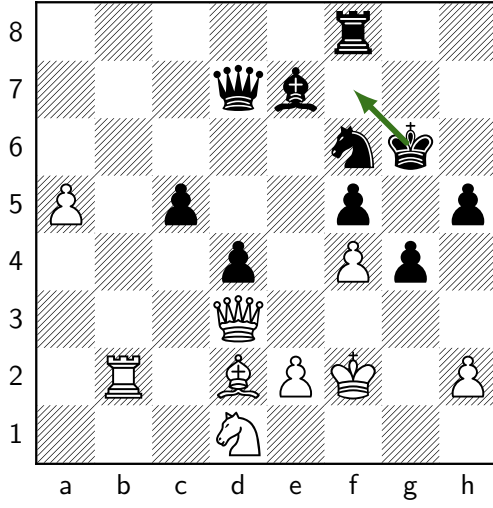


(c)

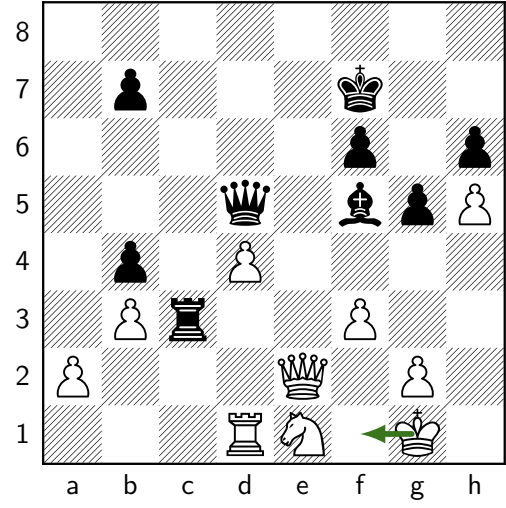


(d)

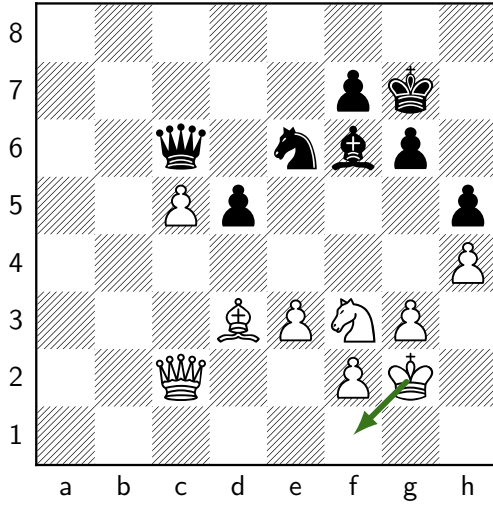
Figure 60: **Move 66, Black: Kg6 (g7g6)**. (a) Our position; Melanie plays Kg6 (g7g6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2302` (ply 126) [structural distance $d=93$, QAP obj=95]: the master played Kg1; under μ our Kg2 plays the role of that game’s Kh1; cross-type roles: our Qd7 as their Rd1, our Pc5 as their Ne1; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1742` (ply 115) [structural distance $d=105$, QAP obj=81]: the master played Kg6; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Nf6 as their Rb7, our bd2 as their ph3; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1674` (ply 116) [structural distance $d=106$, QAP obj=26]: the master played Kh2; under μ our Kg2 plays the role of that game’s Kg2; cross-type roles: our Be7 as their Qe2, our Rf8 as their Nf2; transferred move g7g6 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Kg6 (summed QAP objective 201, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



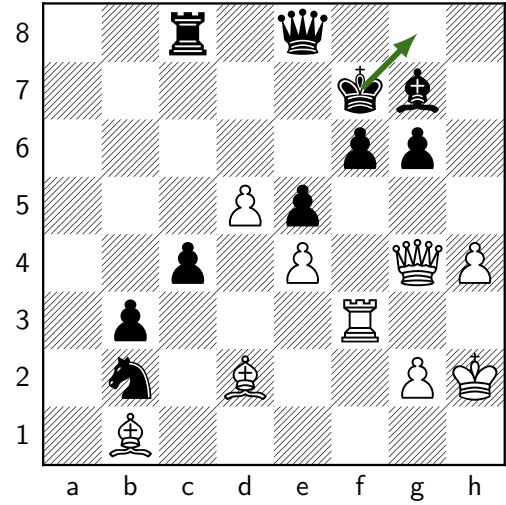
(a)



(b)

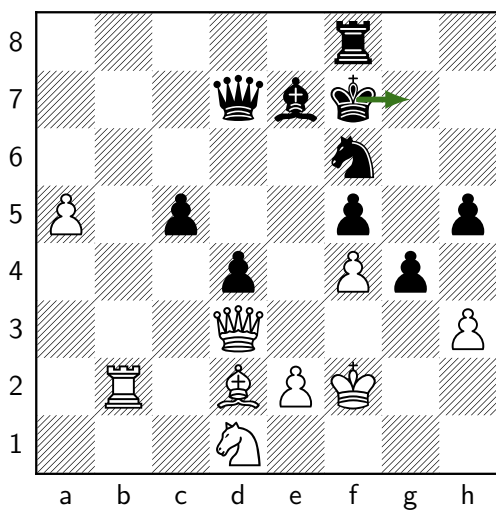


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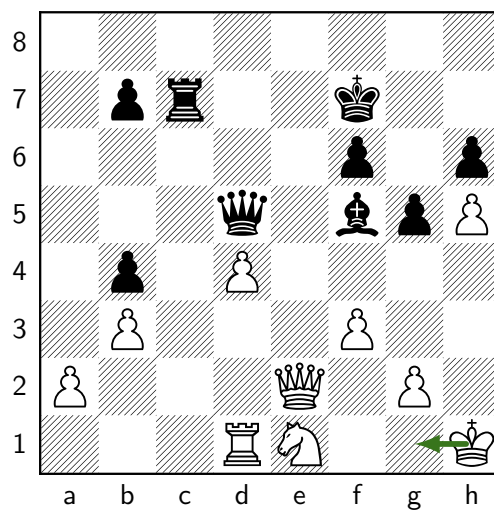


(d)

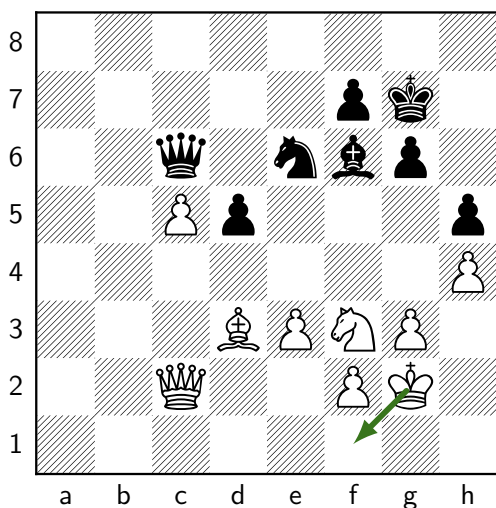
Figure 61: **Move 67, Black: Kf7.** (a) Our position; Melanie plays Kf7 (g6f7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2302` (ply 128) [structural distance $d=122$, QAP obj=105]: the master played Kf1; under μ our Kg3 plays the role of that game’s Kg1; cross-type roles: our Qd7 as their Rd1, our Be7 as their Qe2; transferred move g6f7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1106` (ply 124) [structural distance $d=117$, QAP obj=54]: the master played Kf1; under μ our Kg3 plays the role of that game’s Kg2; cross-type roles: our Nf6 as their Bd3, our ph2 as their Nf3; transferred move g6f7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#804` (ply 135) [structural distance $d=119$, QAP obj=53]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kf2; cross-type roles: our Pd4 as their Rc8, our Nf6 as their Bg7; transferred move g6f7 (**endorses this move**). Synthesis: 5 retrieved analogs transferred to Kf7 (summed QAP objective 284, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



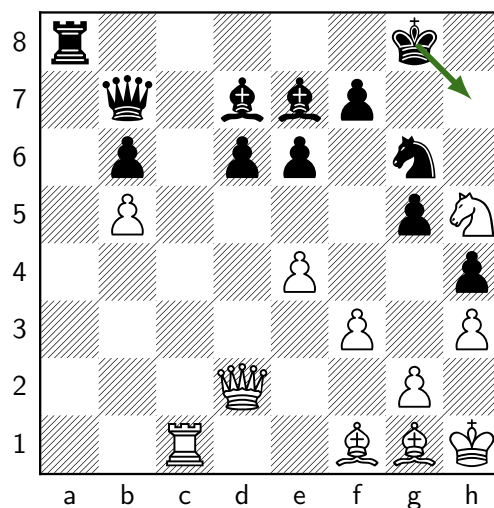
(a)



(b)

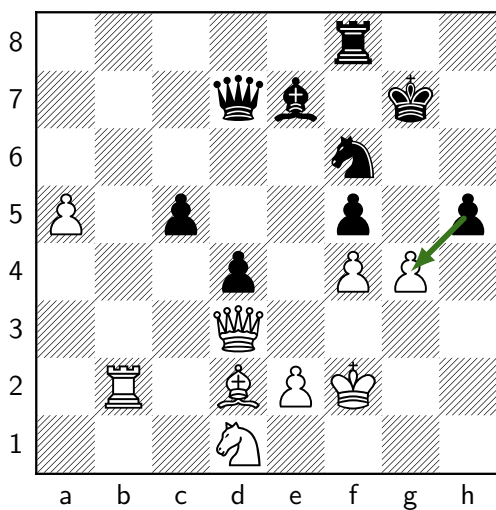


(c)

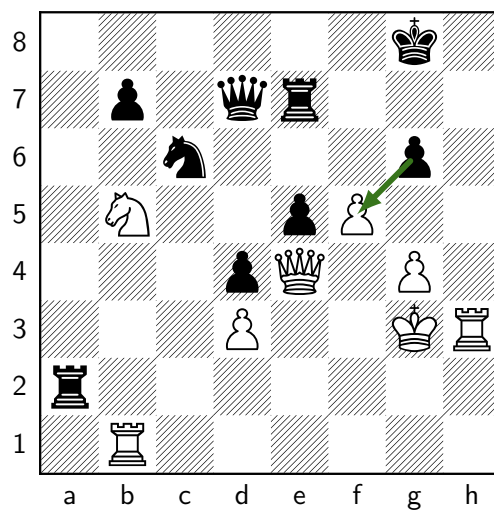


(d)

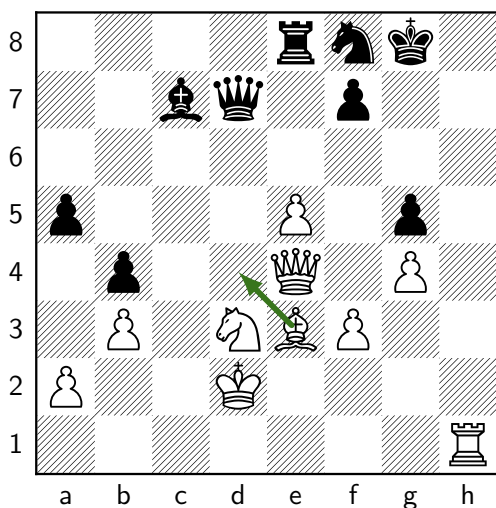
Figure 62: **Move 68, Black: Kg7 (f7g7)**. (a) Our position; Melanie plays Kg7 (f7g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#2302` (ply 126) [structural distance $d=108$, QAP obj=95]: the master played Kg1; under μ our Kf2 plays the role of that game’s Kh1; cross-type roles: our Nf6 as their Rd1, our Be7 as their Ne1; transferred move f7g7 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#1106` (ply 124) [structural distance $d=114$, QAP obj=60]: the master played Kf1; under μ our Kf2 plays the role of that game’s Kg2; cross-type roles: our nd1 as their Pf2, our qd3 as their Bd3; transferred move f7g7 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#3952` (ply 141) [structural distance $d=111$, QAP obj=41]: the master played Kh7; under μ our Kf2 plays the role of that game’s Kg1; cross-type roles: our Nf6 as their Qb7, our Pg4 as their Bd7; transferred move f7g7 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Kg7 (summed QAP objective 213, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



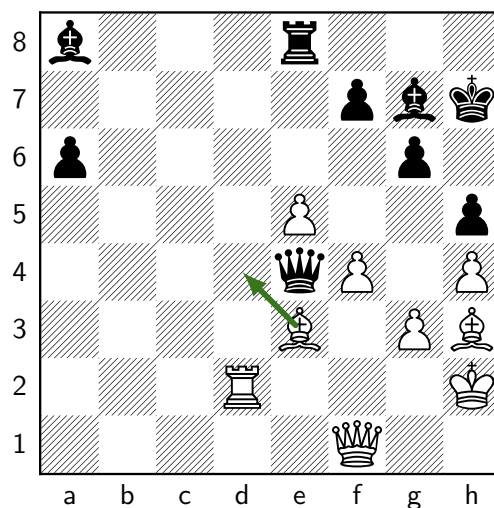
(a)



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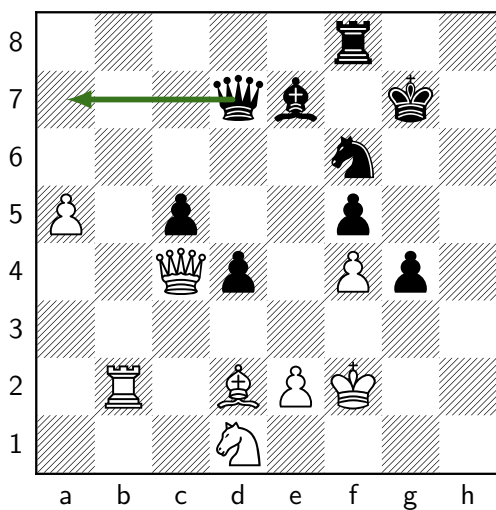


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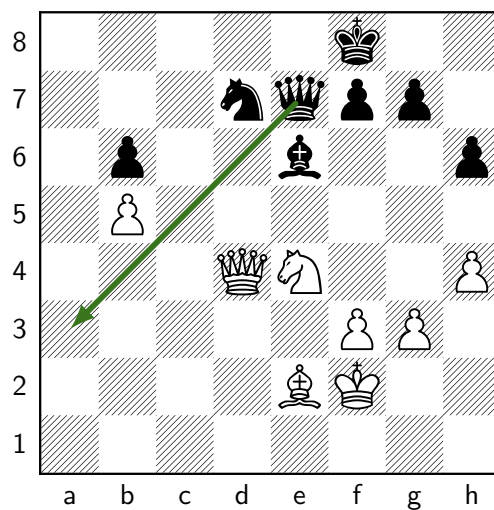


(d)

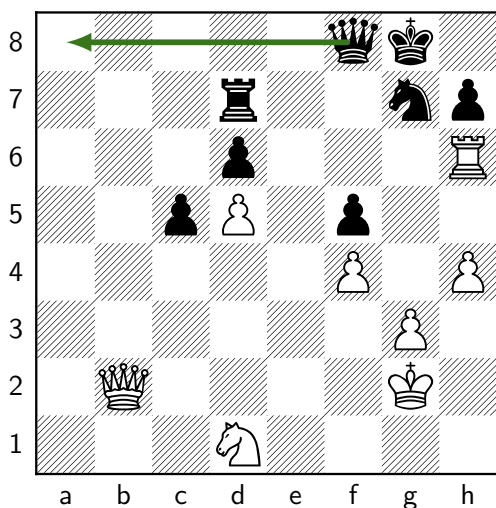
Figure 63: **Move 69, Black: hxg4**. (a) Our position; Melanie plays hxg4 (h5g4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#796` (ply 133) [structural distance $d=103$, QAP obj=109]: the master played gxf5; under μ our $Ph4$ plays the role of that game’s $Pg3$; cross-type roles: our $Nf6$ as their $Qd7$, our $Qd7$ as their $Re7$; transferred move h5g4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1490` (ply 124) [structural distance $d=108$, QAP obj=70]: the master played Bd4; under μ our $Ph4$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $Be7$ as their $Pa2$, our $Pd4$ as their $Nd3$; transferred move h5g4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#575` (ply 130) [structural distance $d=106$, QAP obj=18]: the master played Bd4; under μ our $Ph4$ plays the role of that game’s $Be3$ (a **cross-type** correspondence); cross-type roles: our $Nf6$ as their $Rd2$, our $rb2$ as their $Pg3$; transferred move h5g4 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to hxg4 (summed QAP objective 197, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



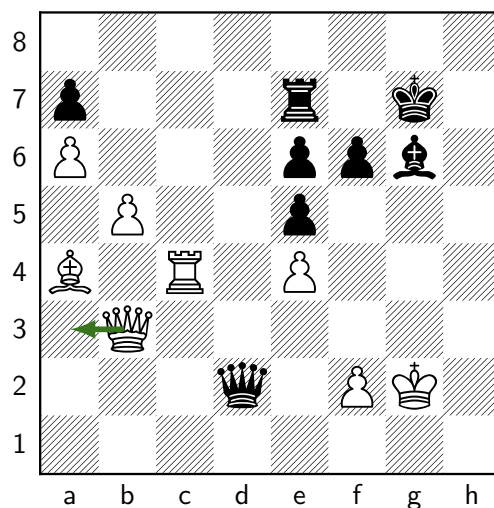
(a)



(b)

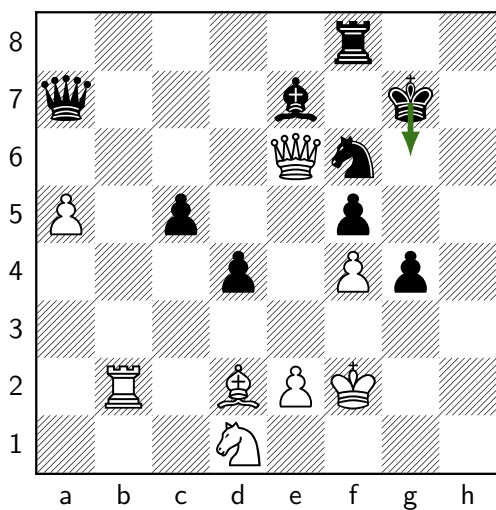


(c)

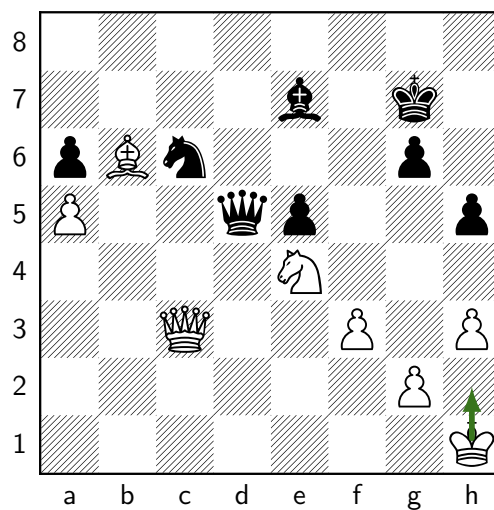


(d)

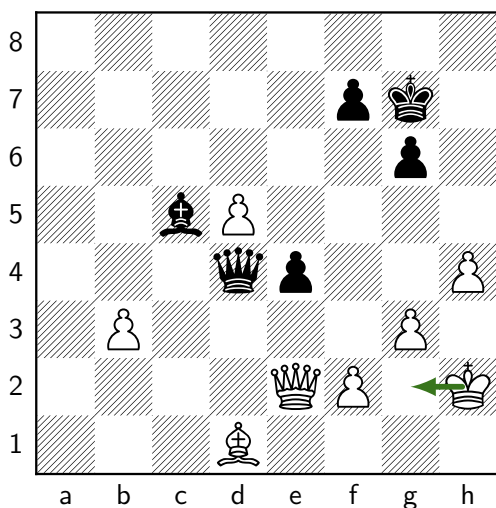
Figure 64: **Move 70, Black: Qa7**. (a) Our position; Melanie plays Qa7 (d7a7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2225` (ply 123) [structural distance $d=90$, QAP obj=39]: the master played Qa3; under μ our Qd2 plays the role of that game's Qe2; cross-type roles: our Pc5 as their Nd7, our Nf6 as their Pf7; transferred move d7a7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1802` (ply 133) [structural distance $d=105$, QAP obj=30]: the master played Qa8; under μ our Qd2 plays the role of that game's Qf1; cross-type roles: our Be7 as their Rd7, our Pd4 as their Ng7; transferred move d7a7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#3876` (ply 132) [structural distance $d=102$, QAP obj=18]: the master played Qa3; under μ our Qd2 plays the role of that game's Qb3; cross-type roles: our Nf6 as their Ba4, our Be7 as their Rc4; transferred move d7a7 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qa7 (summed QAP objective 87, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



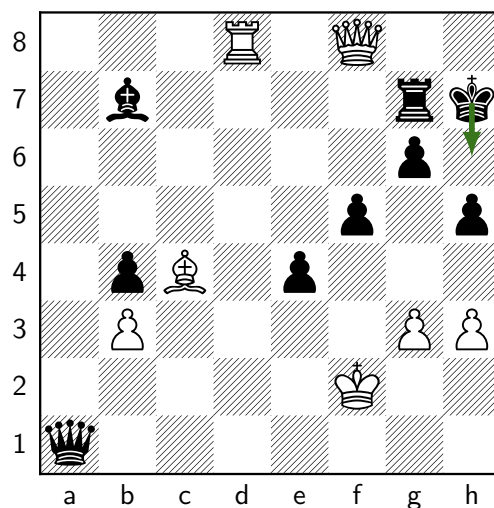
(a)



(b)



(c)



(d)

Figure 65: **Move 71, Black: Kg6**. (a) Our position; Melanie plays Kg6 (**g7g6**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2804` (ply 126) [structural distance $d=93$, QAP obj=25]: the master played Kh2; under μ our Kg2 plays the role of that game's Kh1; cross-type roles: our bd2 as their Pg2, our nd1 as their ph5; transferred move g7g6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0001#3490` (ply 154) [structural distance $d=90$, QAP obj=23]: the master played Kg2; under μ our Kg2 plays the role of that game's Kh2; cross-type roles: our Rf8 as their Qe2, our pe2 as their bc5; transferred move g7g6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#1925` (ply 149) [structural distance $d=81$, QAP obj=20]: the master played Kh6; under μ our Kg2 plays the role of that game's Kh2; cross-type roles: our pe2 as their rd8, our Pd4 as their Bb7; transferred move g7g6 (**endorses this move**). Synthesis: 10 retrieved analogs transferred to Kg6 (summed QAP objective 102, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

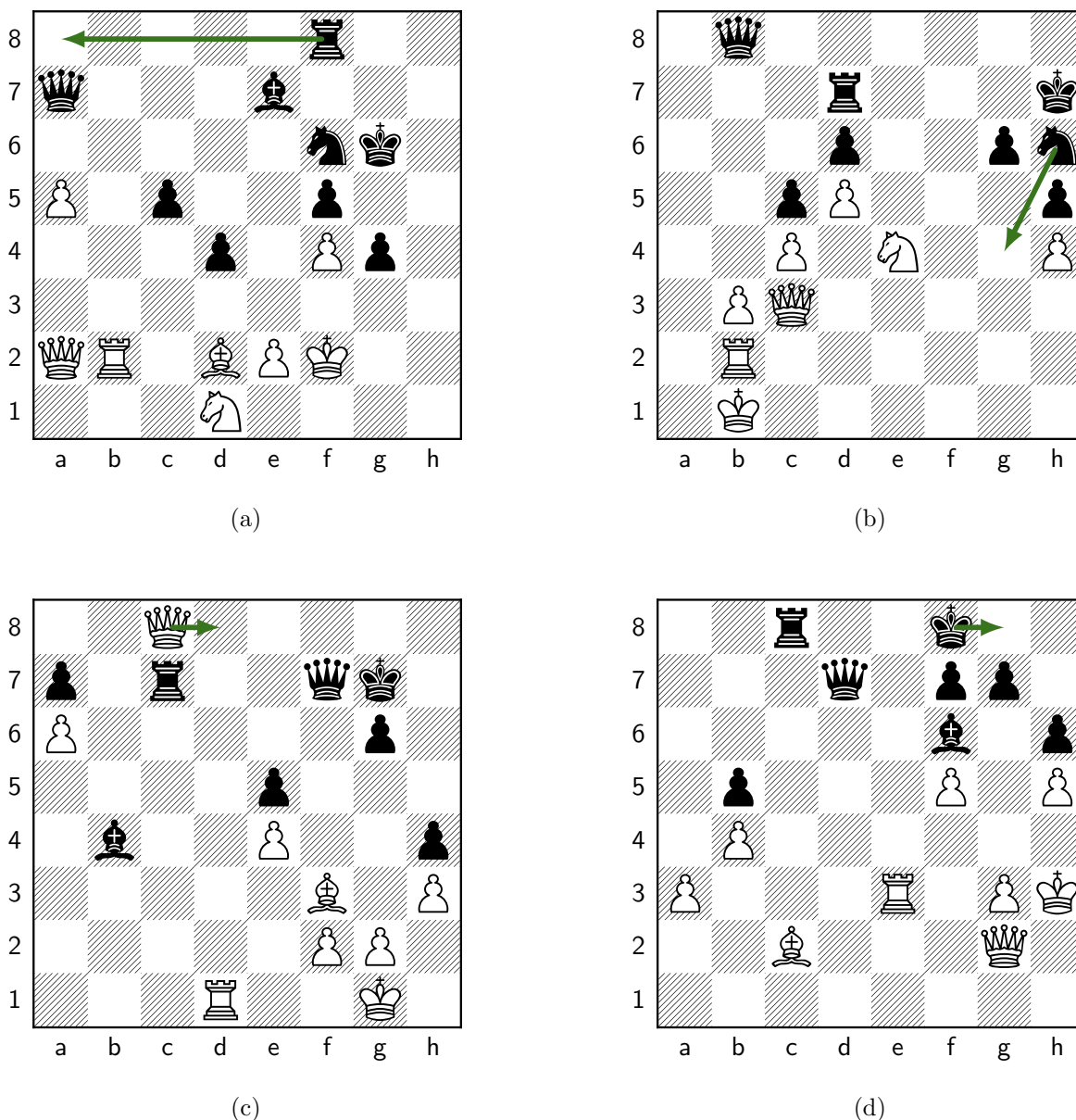
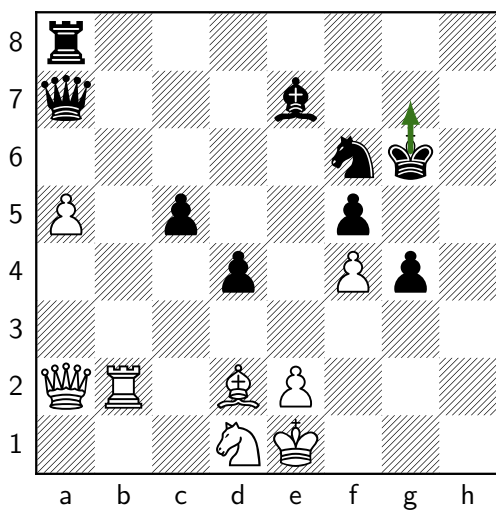
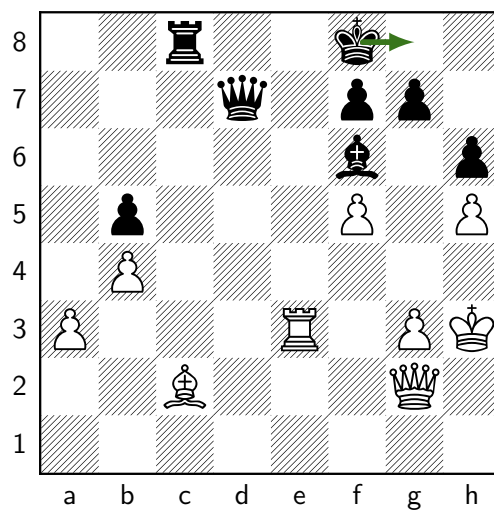


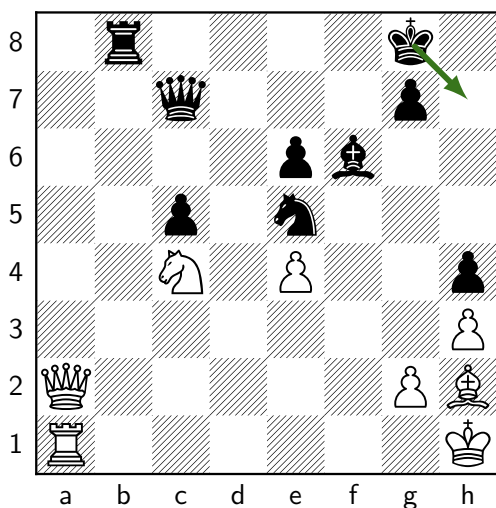
Figure 66: **Move 72, Black: Ra8.** (a) Our position; Melanie plays Ra8 (f8a8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#115` (ply 149) [structural distance $d=90$, QAP obj=44]: the master played Ng4; under μ our Rf1 plays the role of that game’s Nh3 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Rd7, our rb2 as their Pg6; transferred move f8a8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0001#1928` (ply 128) [structural distance $d=90$, QAP obj=2]: the master played Qd8; under μ our Rf1 plays the role of that game’s Qc8 (a **cross-type** correspondence); cross-type roles: our pf4 as their Rd1, our Qa7 as their Pf2; transferred move f8a8 (**endorses this move**). (d) Retrieved from `ccrl_top_p0001#1093` (ply 157) [structural distance $d=96$, QAP obj=53]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kf1; cross-type roles: our Qa7 as their Rc8, our Pd4 as their Bf6; transferred move g6g7. Synthesis: 2 retrieved analogs transferred to Ra8 (summed QAP objective 46, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



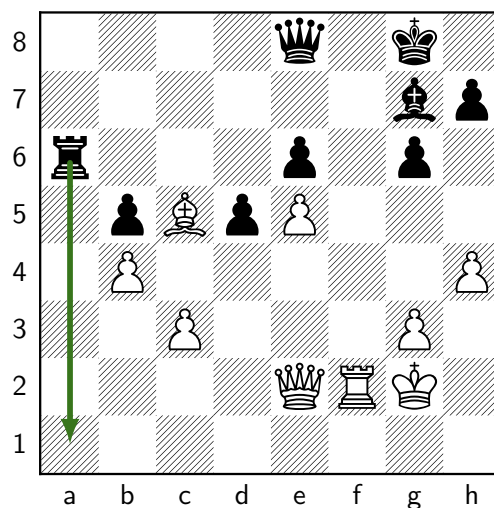
(a)



(b)



(c)



(d)

Figure 67: **Move 73, Black: Kg7**. (a) Our position; Melanie plays Kg7 (g6g7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#1093` (ply 157) [structural distance $d=83$, QAP obj=54]: the master played Kg8; under μ our Kg3 plays the role of that game’s Kf1; cross-type roles: our Qa7 as their Rc8, our Pd4 as their Bf6; transferred move g6g7 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#561` (ply 155) [structural distance $d=81$, QAP obj=23]: the master played Kh7; under μ our Kg3 plays the role of that game’s Kg1; cross-type roles: our pe2 as their Rb8, our nd1 as their Pe6; transferred move g6g7 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#2465` (ply 131) [structural distance $d=85$, QAP obj=43]: the master played Ra1; under μ our Pd5 plays the role of that game’s Ra3 (a **cross-type** correspondence); cross-type roles: our bd2 as their Qe8, our Qa7 as their Bg7; transferred move d4d3. Synthesis: 2 retrieved analogs transferred to Kg7 (summed QAP objective 77, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

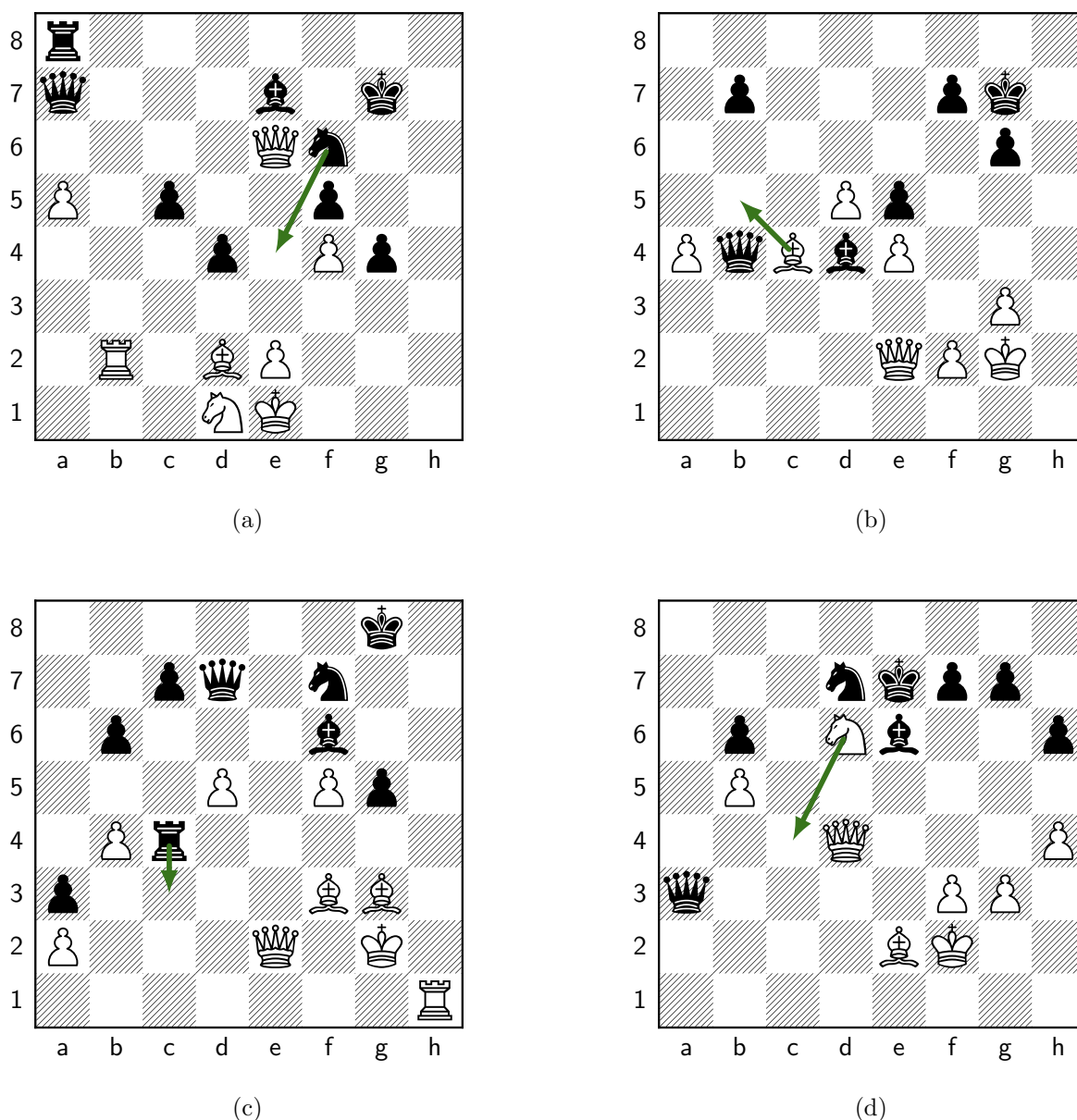
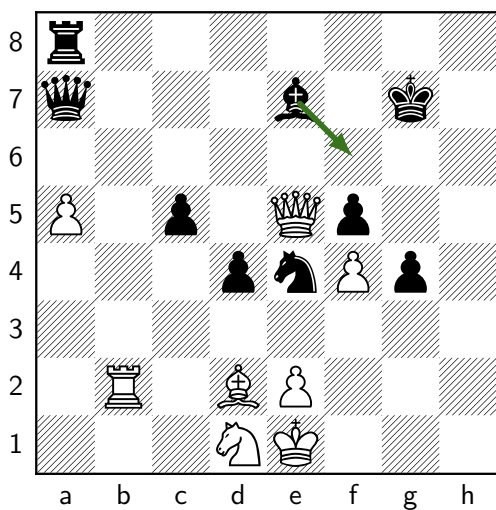
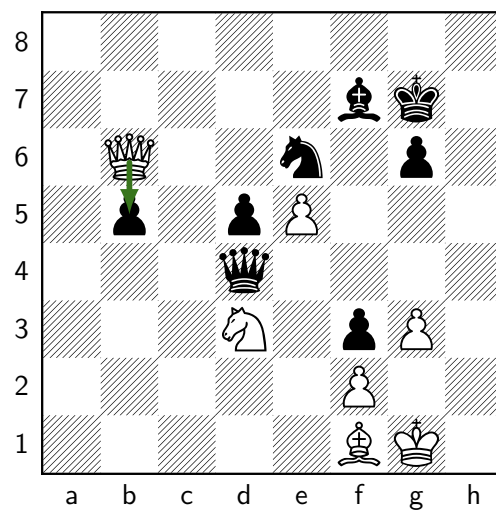


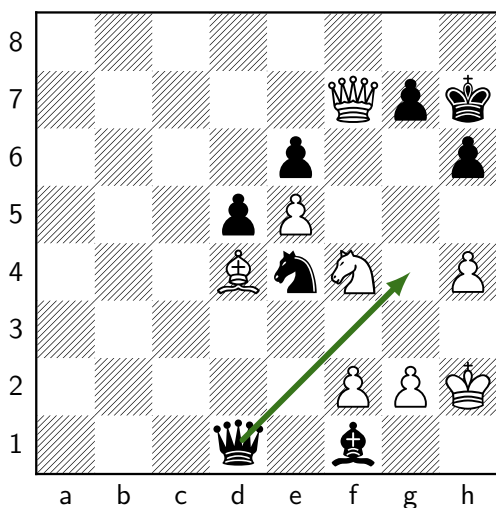
Figure 68: **Move 74, Black: Ne4**. (a) Our position; Melanie plays **Ne4** (f6e4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#1365` (ply 144) [structural distance $d=74$, QAP obj=33]: the master played **Bb5**; under μ our *Nf3* plays the role of that game's *Bc4* (a **cross-type** correspondence); cross-type roles: our *Be7* as their *Qe2*, our *Qa7* as their *bd4*; transferred move **f6e4** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#812` (ply 133) [structural distance $d=76$, QAP obj=19]: the master played **Rc3**; under μ our *Nf3* plays the role of that game's *Rc5* (a **cross-type** correspondence); cross-type roles: our *Pf5* as their *Nf7*, our *rb2* as their *pd5*; transferred move **f6e4** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#2225` (ply 130) [structural distance $d=81$, QAP obj=12]: the master played **Nc4**; under μ our *Nf3* plays the role of that game's *Nd6*; cross-type roles: our *Pc5* as their *Be2*, our *Be7* as their *Pf3*; transferred move **f6e4** (**endorses this move**). Synthesis: 3 retrieved analogs transferred to **Ne4** (summed QAP objective 64, the strongest correspondence among the candidate moves); Melanie committed it via the **search selector**.



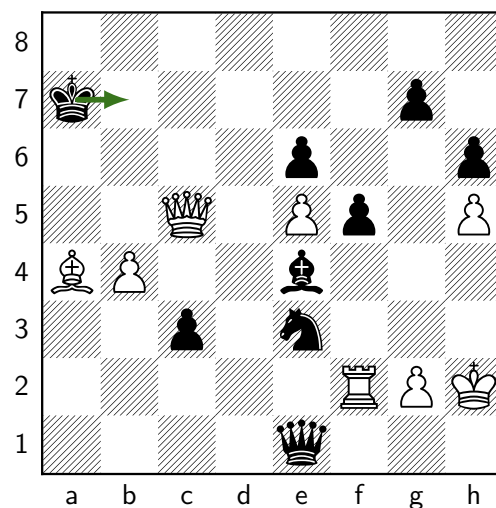
(a)



(b)

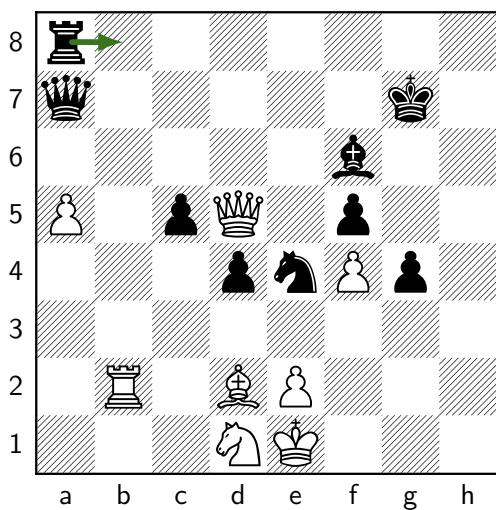


(c)

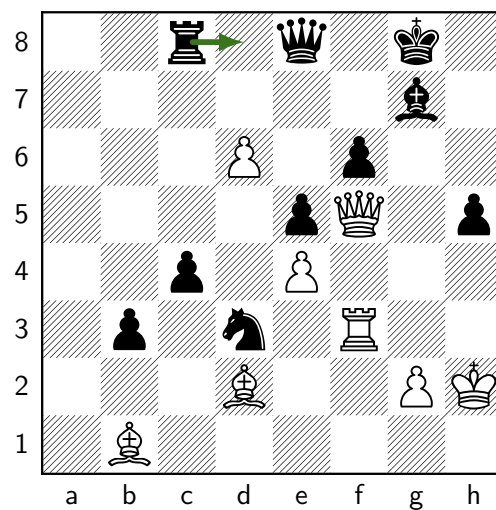


(d)

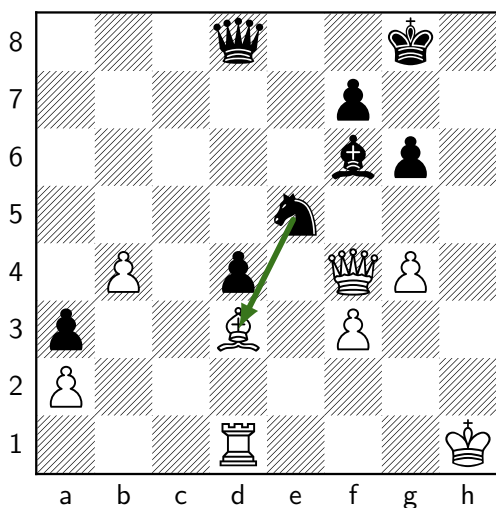
Figure 69: **Move 75, Black: Bf6**. (a) Our position; Melanie plays Bf6 (e7f6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#2792` (ply 132) [structural distance $d=98$, QAP obj=6]: the master played Qxb5; under μ our *Be2* plays the role of that game’s *Qb6* (a **cross-type** correspondence); cross-type roles: our *Qa7* as their *Bf1*, our *nd1* as their *pb5*; transferred move e7f6 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#1643` (ply 137) [structural distance $d=95$, QAP obj=4]: the master played Qg4; under μ our *Be2* plays the role of that game’s *Qd8* (a **cross-type** correspondence); cross-type roles: our *Ne4* as their *Pg7*, our *rb2* as their *Ph6*; transferred move e7f6 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3164` (ply 143) [structural distance $d=85$, QAP obj=72]: the master played Kb7; under μ our *Kg2* plays the role of that game’s *Ka2*; cross-type roles: our *Ra8* as their *pe5*, our *Qa7* as their *ba4*; transferred move g7h7. Synthesis: 2 retrieved analogs transferred to Bf6 (summed QAP objective 10, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



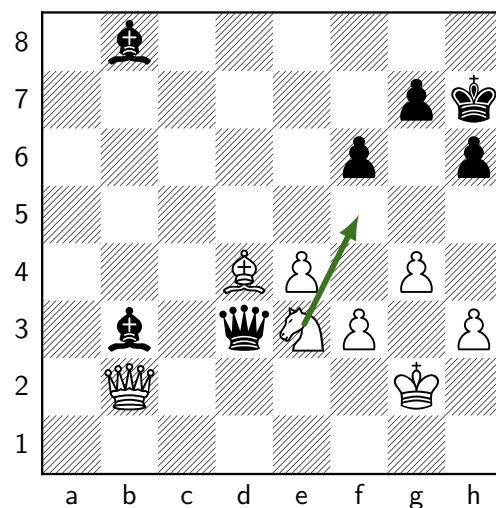
(a)



(b)

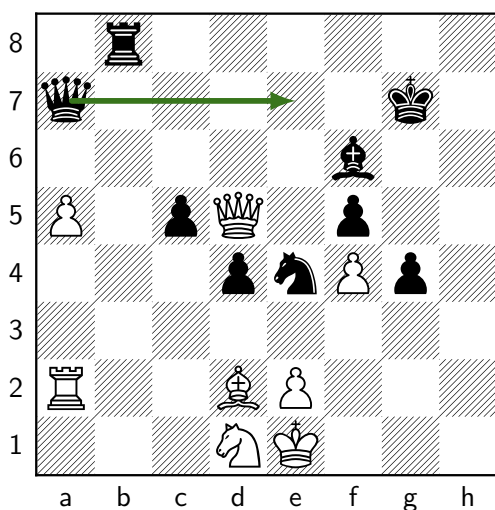


(c)

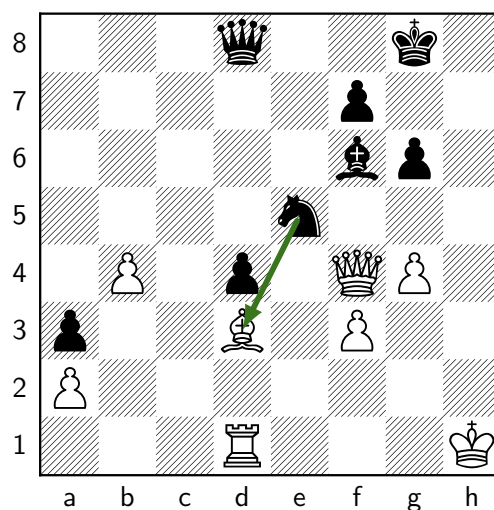


(d)

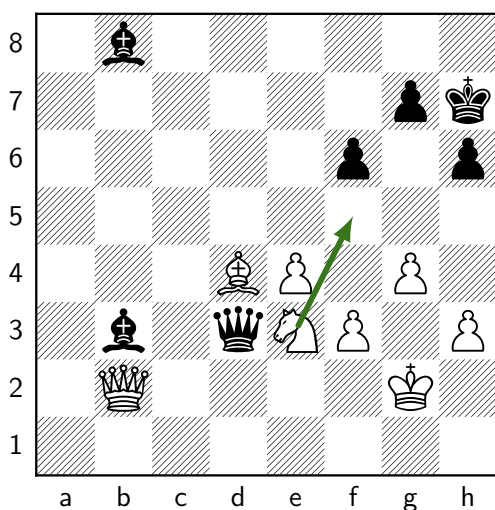
Figure 70: **Move 76, Black: Rb8.** (a) Our position; Melanie plays Rb8 (a8b8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#804` (ply 141) [structural distance $d=99$, QAP obj=45]: the master played Rd8; under μ our Ra1 plays the role of that game’s Rc1; cross-type roles: our Bf6 as their Qe8, our Pf5 as their Bg7; transferred move a8b8 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#2280` (ply 137) [structural distance $d=93$, QAP obj=73]: the master played Nxd3; under μ our Ne5 plays the role of that game’s Ne4; cross-type roles: our Bf6 as their Qd8, our Ra8 as their Bf6; transferred move e4g3. (d) Retrieved from `ccrl_top_p0000#1282` (ply 136) [structural distance $d=87$, QAP obj=66]: the master played Nf5; under μ our Ne5 plays the role of that game’s Ne3; cross-type roles: our Ne4 as their bb3, our Ra8 as their Ph3; transferred move e4g5. Synthesis: 1 retrieved analog transferred to Rb8 (summed QAP objective 45, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



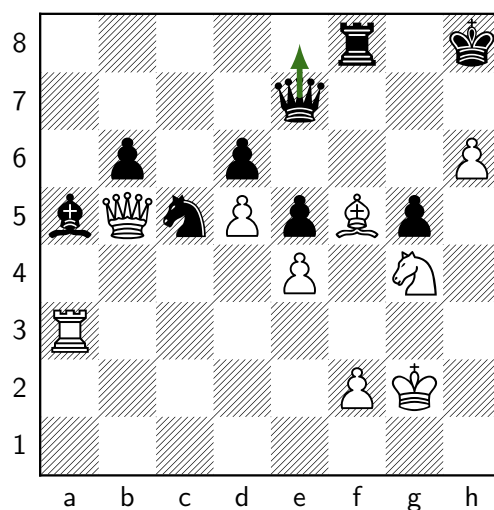
(a)



(b)

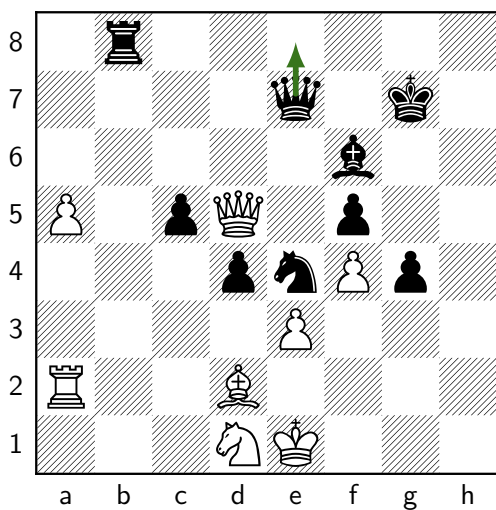


(c)

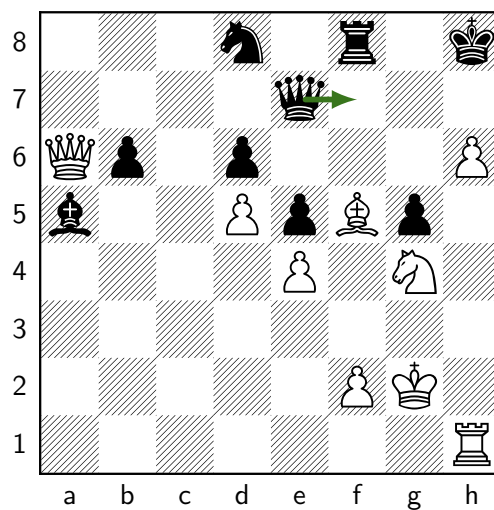


(d)

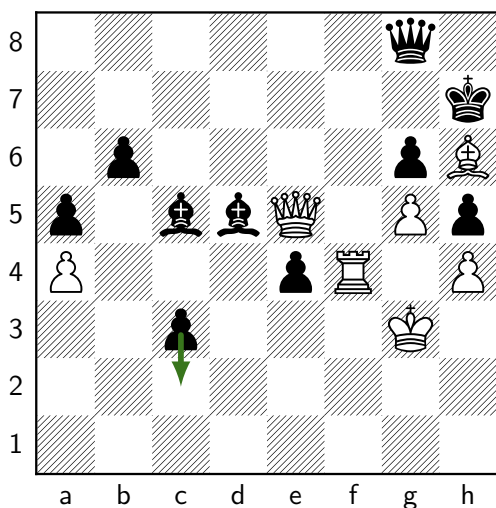
Figure 71: **Move 77, Black: Qe7**. (a) Our position; Melanie plays **Qe7** (a7e7). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2280` (ply 137) [structural distance $d=88$, QAP obj=82]: the master played **Nxd3**; under μ our **Ne5** plays the role of that game's **Ne4**; cross-type roles: our **Bf6** as their **Qd8**, our **Qa7** as their **Pf7**; transferred move **e4g3**. (c) Retrieved from `ccr1_top_p0000#1282` (ply 136) [structural distance $d=86$, QAP obj=78]: the master played **Nf5**; under μ our **Ne5** plays the role of that game's **Ne3**; cross-type roles: our **Ne4** as their **bb3**, our **Ne4** as their **Bd4**; transferred move **e4g5**. (d) Retrieved from `ccr1_top_p0003#2042` (ply 137) [structural distance $d=86$, QAP obj=75]: the master played **Qe8**; under μ our **Qa2** plays the role of that game's **Qe2**; cross-type roles: our **Bf6** as their **Rf8**, our **Ne4** as their **Ba5**; transferred move **a7b7**. Synthesis: no retrieved analog transferred to **Qe7** – the search selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



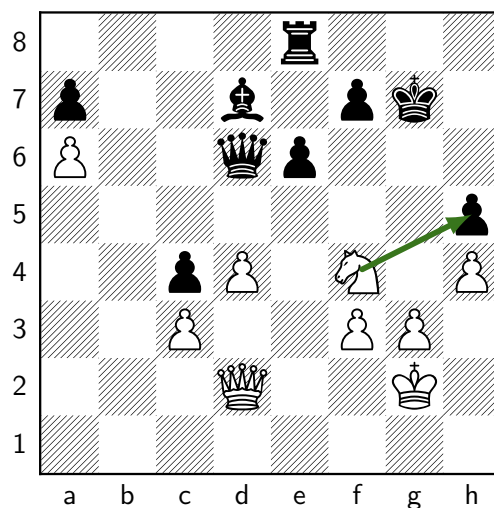
(a)



(b)

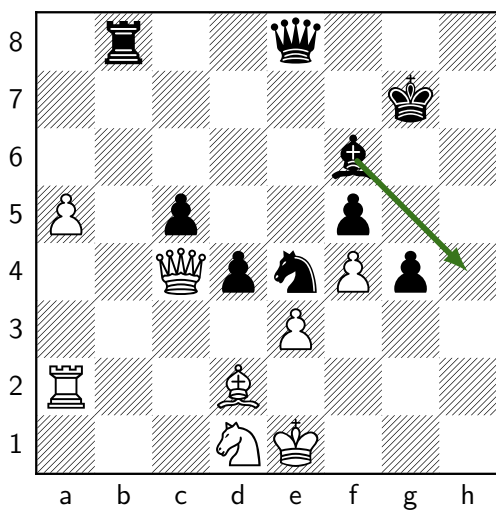


(c)

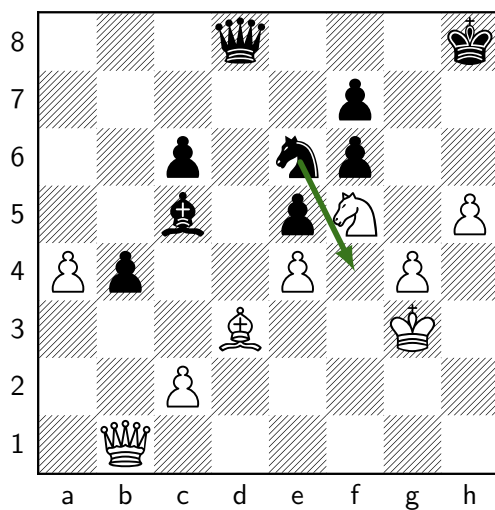


(d)

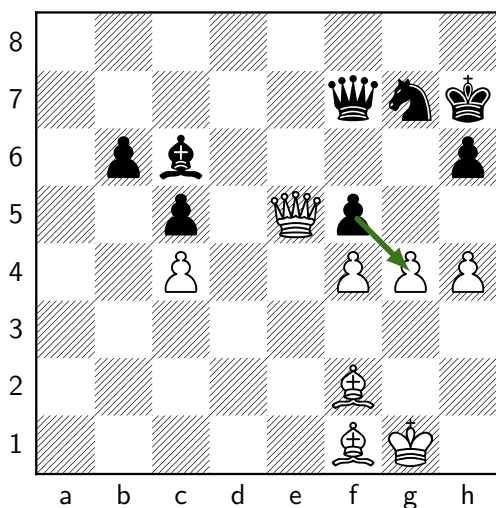
Figure 72: **Move 78, Black: Qe8.** (a) Our position; Melanie plays Qe8 (e7e8). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#2042` (ply 149) [structural distance $d=108$, QAP obj=41]: the master played Qf7; under μ our Qe2 plays the role of that game’s Qe2; cross-type roles: our Bf6 as their Nd8, our Ne4 as their Rf8; transferred move e7e8 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1851` (ply 139) [structural distance $d=103$, QAP obj=50]: the master played c2; under μ our Pc4 plays the role of that game’s Pc6; cross-type roles: our Bf6 as their Qg8, our ra2 as their Pg6; transferred move c5c4. (d) Retrieved from `ccr1_top_p0002#2632` (ply 166) [structural distance $d=105$, QAP obj=41]: the master played Nxh5+; under μ our Ne5 plays the role of that game’s Nf4; cross-type roles: our Bf6 as their Pf3, our bd2 as their pe6; transferred move e4g3. Synthesis: 2 retrieved analogs transferred to Qe8 (summed QAP objective 82, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



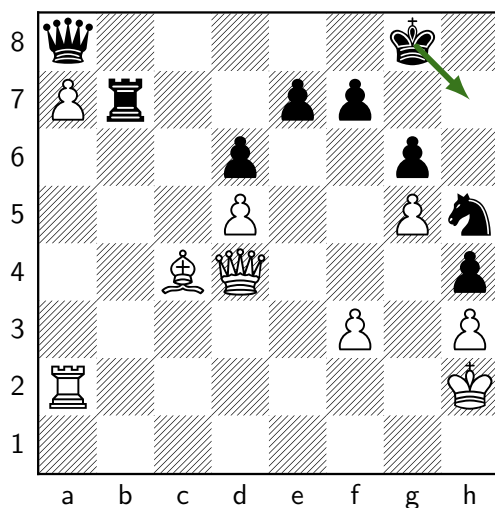
(a)



(b)



(c)



(d)

Figure 73: **Move 79, Black: Bh4+.** (a) Our position; Melanie plays Bh4+ (f6h4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#608` (ply 157) [structural distance $d=88$, QAP obj=13]: the master played Nf4; under μ our Bf3 plays the role of that game’s Ne3 (a **cross-type** correspondence); cross-type roles: our Pd4 as their Qd8, our Qe8 as their Pf7; transferred move f6h4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#1755` (ply 153) [structural distance $d=90$, QAP obj=79]: the master played fxg4; under μ our Pd5 plays the role of that game’s Pf4; cross-type roles: our Ne4 as their Qf7, our Pc5 as their Ng7; transferred move d4d3. (d) Retrieved from `ccr1_top_p0003#2320` (ply 143) [structural distance $d=88$, QAP obj=55]: the master played Kh7; under μ our Kg2 plays the role of that game’s Kg1; cross-type roles: our pe3 as their Rb7, our bd2 as their Pd6; transferred move g7f8. Synthesis: 1 retrieved analog transferred to Bh4+ (summed QAP objective 13, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.

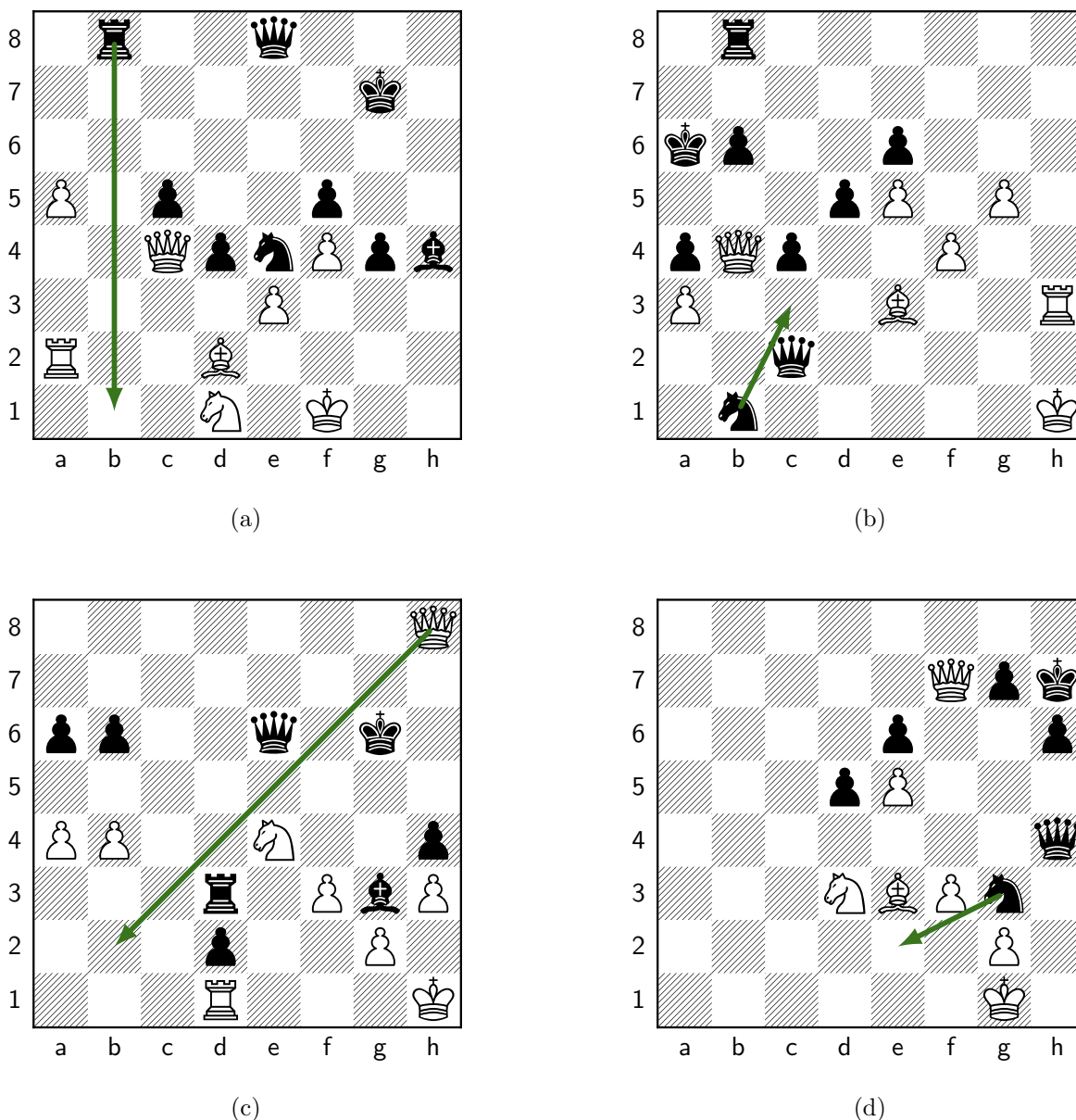


Figure 74: **Move 80, Black: Rb1.** (a) Our position; Melanie plays Rb1 (b8b1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#967` (ply 175) [structural distance $d=82$, QAP obj=91]: the master played Nc3; under μ our Pd5 plays the role of that game's Nb8 (a **cross-type** correspondence); cross-type roles: our Qe8 as their Pd5, our bd2 as their pg5; transferred move d4e3. (c) Retrieved from `ccr1_top_p0003#175` (ply 142) [structural distance $d=75$, QAP obj=74]: the master played Qb2; under μ our Qe1 plays the role of that game's Qh8; cross-type roles: our qc4 as their Rd1, our Rb8 as their Pg2; transferred move e8d7. (d) Retrieved from `ccr1_top_p0000#1643` (ply 145) [structural distance $d=76$, QAP obj=60]: the master played Ne2+; under μ our Bh5 plays the role of that game's Ng6 (a **cross-type** correspondence); cross-type roles: our bd2 as their Ph6, our nd1 as their pe5; transferred move h4g3. Synthesis: no retrieved analog transferred to Rb1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.

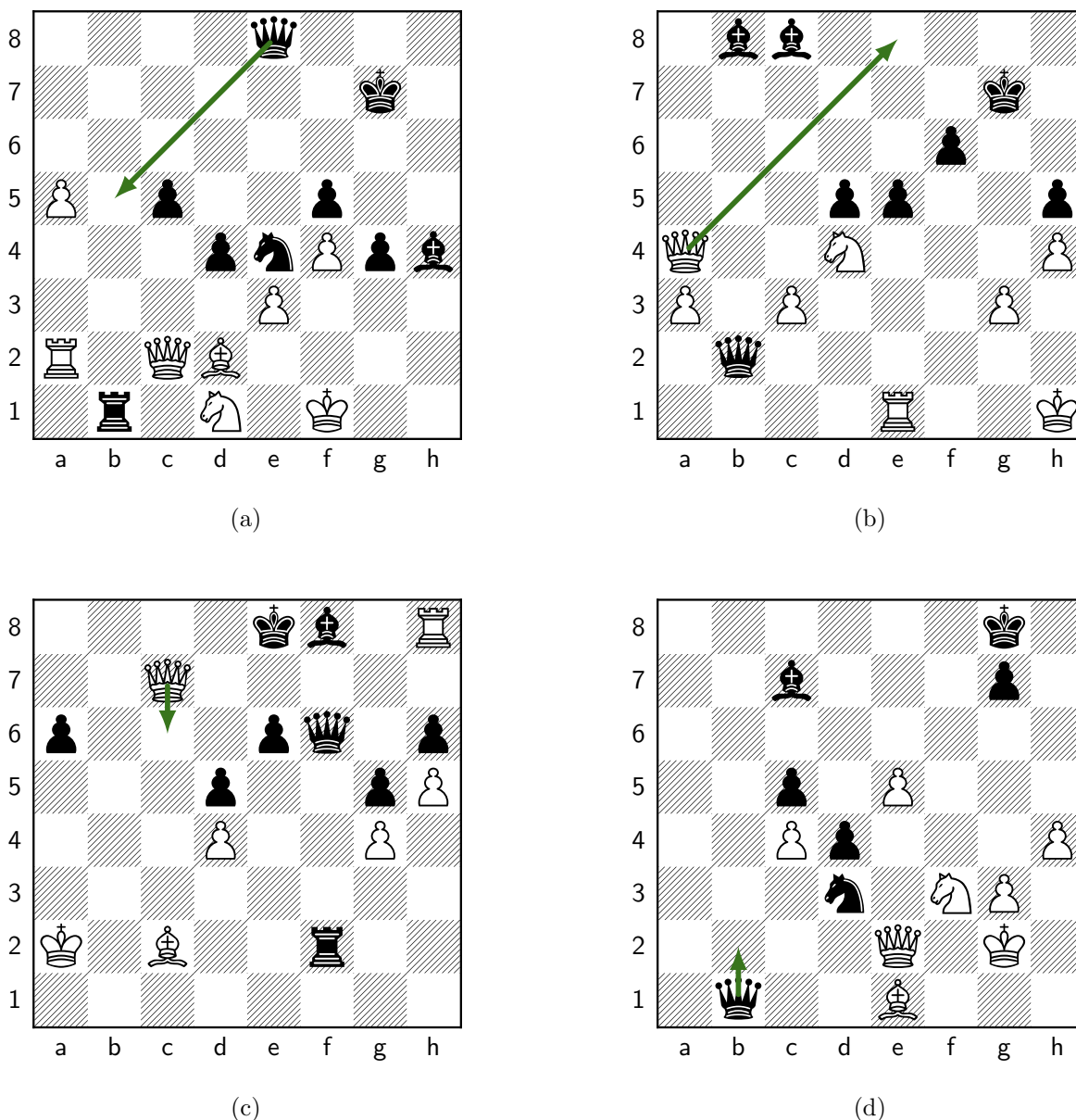
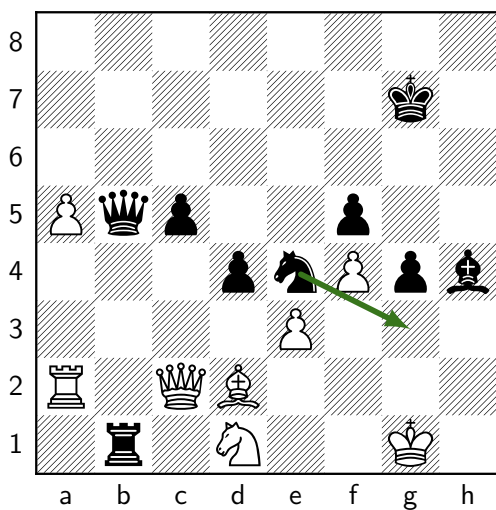
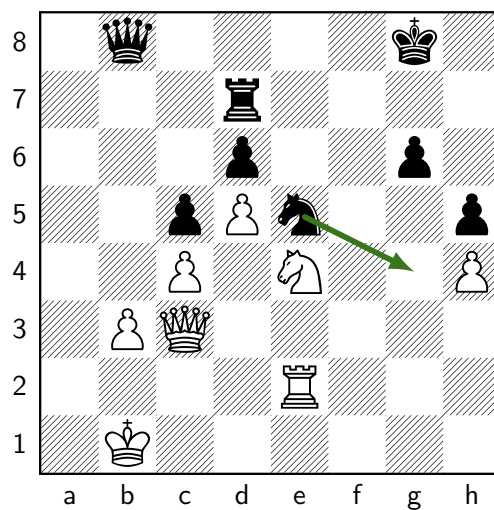


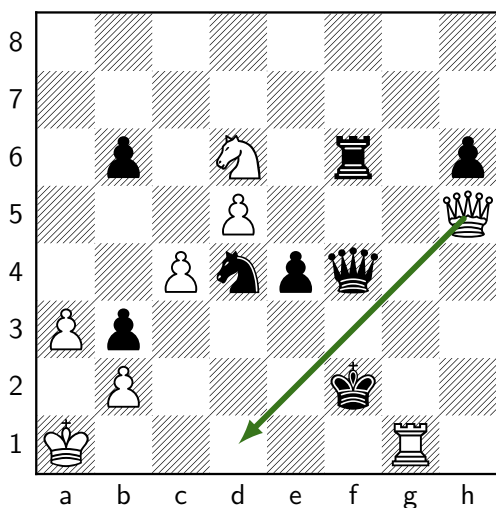
Figure 75: **Move 81, Black: Qb5+.** (a) Our position; Melanie plays Qb5+ (e8b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#239` (ply 158) [structural distance $d=89$, QAP obj=31]: the master played Qe8; under μ our Qe1 plays the role of that game’s Qa4; cross-type roles: our qc2 as their Re1, our Rb1 as their qb2; transferred move e8b5 (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#3186` (ply 146) [structural distance $d=88$, QAP obj=67]: the master played Qc6+; under μ our Ne5 plays the role of that game’s Qc7 (a **cross-type** correspondence); cross-type roles: our nd1 as their Bc2, our bd2 as their pd5; transferred move e4g3. (d) Retrieved from `ccrl_top_p0002#1275` (ply 149) [structural distance $d=92$, QAP obj=42]: the master played Qb2; under μ our Ne5 plays the role of that game’s Qb8 (a **cross-type** correspondence); cross-type roles: our Qe8 as their ph4, our Rb1 as their Nd3; transferred move e4g3. Synthesis: 1 retrieved analog transferred to Qb5+ (summed QAP objective 31, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



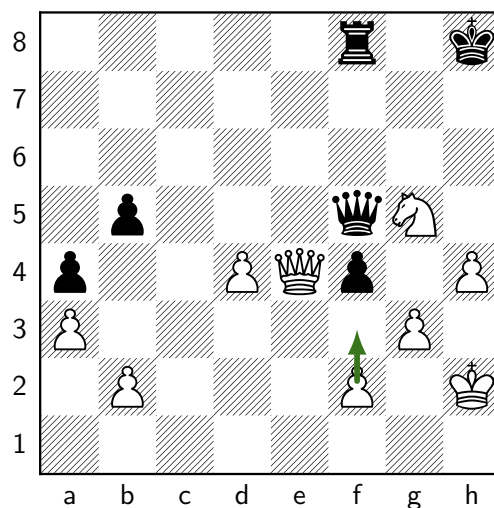
(a)



(b)

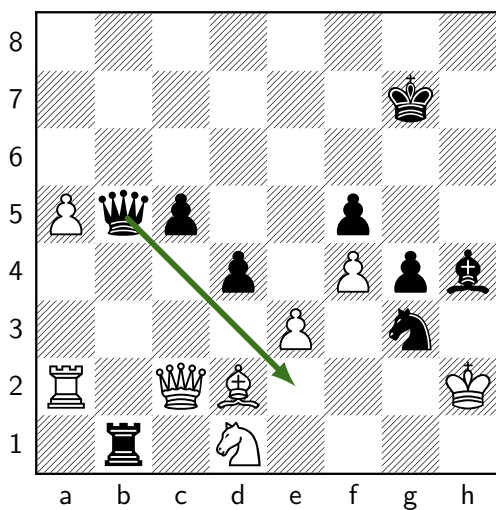


(c)

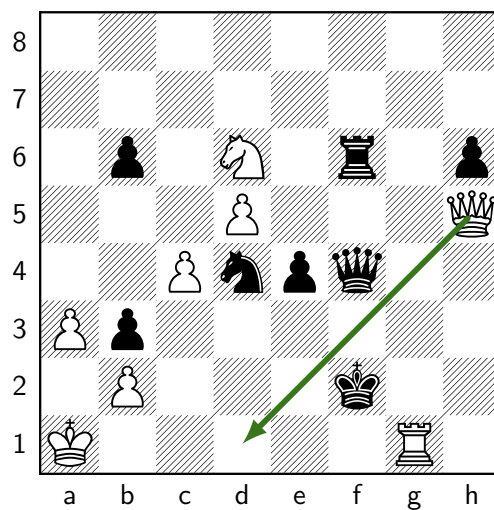


(d)

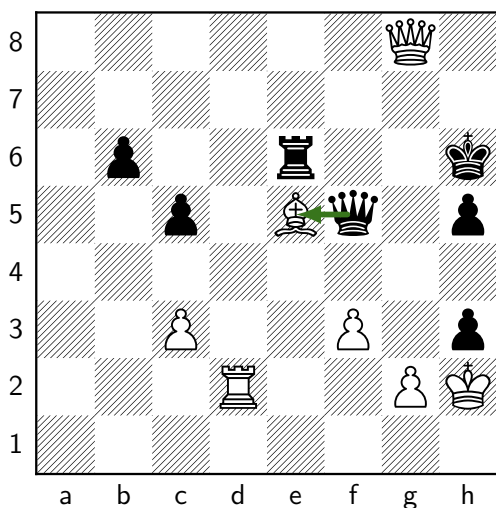
Figure 76: **Move 82, Black: Ng3**. (a) Our position; Melanie plays Ng3 (e4g3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#115` (ply 155) [structural distance $d=77$, QAP obj=19]: the master played Ng4; under μ our Ne5 plays the role of that game's Ne4; cross-type roles: our Rb1 as their Qb8, our Qb5 as their Rd7; transferred move e4g3 (**endorses this move**). (c) Retrieved from `ccrl_top_p0000#1706` (ply 158) [structural distance $d=82$, QAP obj=18]: the master played Qd1; under μ our Rb8 plays the role of that game's Qh5 (a **cross-type** correspondence); cross-type roles: our Pc5 as their Rg1, our Qb5 as their Pb2; transferred move b1b4. (d) Retrieved from `ccrl_top_p0003#2296` (ply 156) [structural distance $d=82$, QAP obj=18]: the master played f3; under μ our Pd5 plays the role of that game's Pf2; cross-type roles: our Bh4 as their Pb2, our Ne4 as their Pg3; transferred move d4d3. Synthesis: 1 retrieved analog transferred to Ng3 (summed QAP objective 19, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



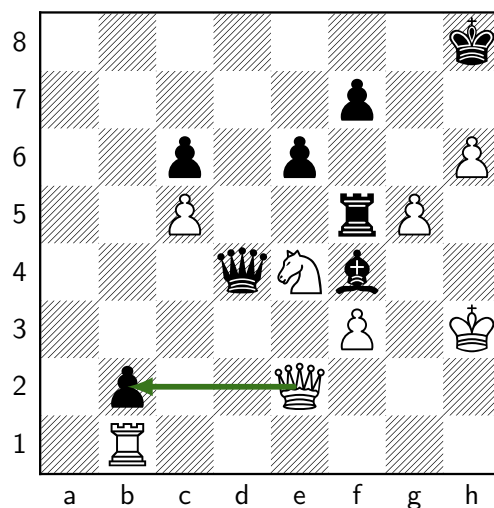
(a)



(b)

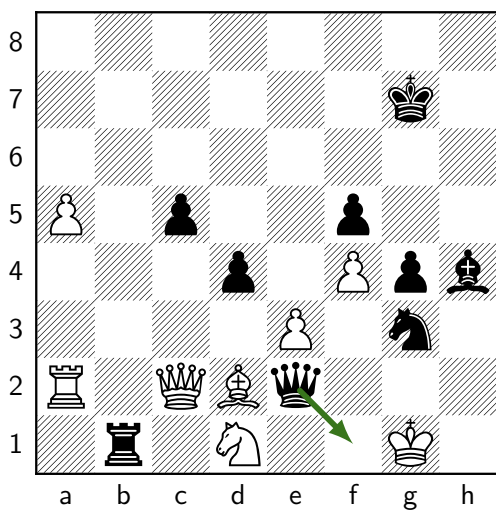


(c)

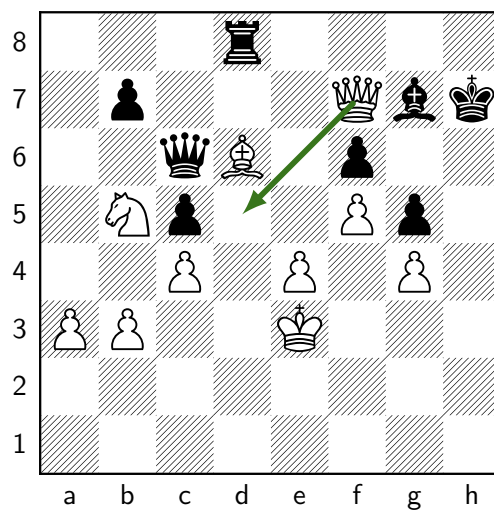


(d)

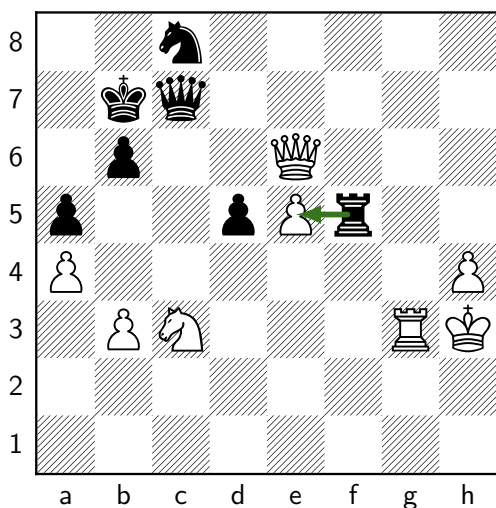
Figure 77: **Move 83, Black: Qe2+.** (a) Our position; Melanie plays Qe2+ (b5e2). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top.p0000#1706` (ply 158) [structural distance $d=83$, QAP obj=27]: the master played Qd1; under μ our Qb4 plays the role of that game's Qh5; cross-type roles: our Ng3 as their Rg1, our bd2 as their nd4; transferred move b5b7. (c) Retrieved from `ccr1_top.p0001#2090` (ply 161) [structural distance $d=86$, QAP obj=23]: the master played Qxe5+; under μ our Rb8 plays the role of that game's Qf4 (a **cross-type** correspondence); cross-type roles: our nd1 as their be5, our Qb5 as their Ph5; transferred move b1b4. (d) Retrieved from `ccr1_top.p0001#3392` (ply 148) [structural distance $d=71$, QAP obj=17]: the master played Qxb2; under μ our Qb4 plays the role of that game's Qe2; cross-type roles: our Bh4 as their Rb1, our Rb1 as their Ne4; transferred move b5b6. Synthesis: no retrieved analog transferred to Qe2+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



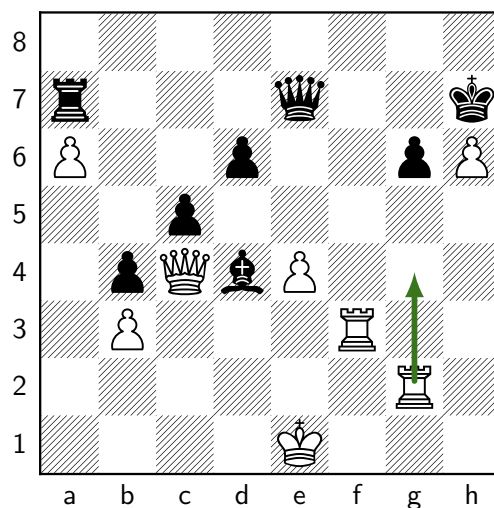
(a)



(b)

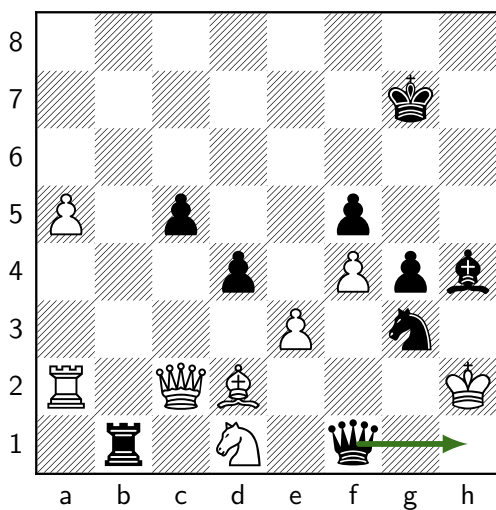


(c)

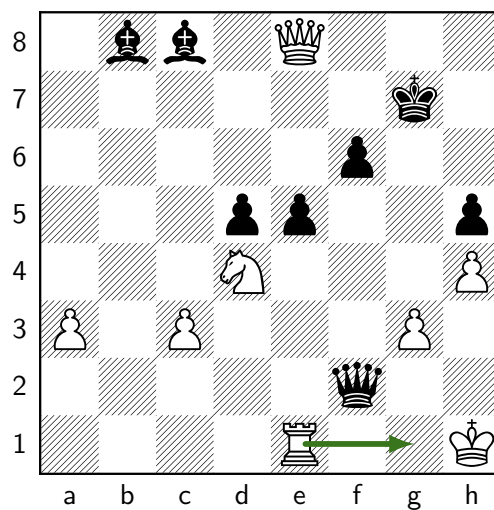


(d)

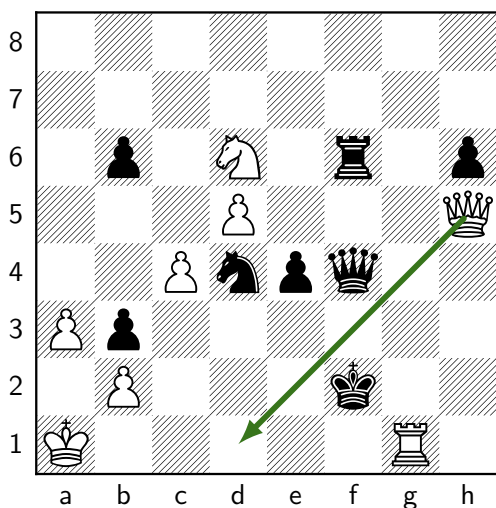
Figure 78: **Move 84, Black: Qf1+.** (a) Our position; Melanie plays Qf1+ (e2f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#536` (ply 158) [structural distance $d=92$, QAP obj=86]: the master played Qd5; under μ our Qe7 plays the role of that game’s Qf7; cross-type roles: our Ng3 as their Pg4, our Rb1 as their Nb5; transferred move e2f3. (c) Retrieved from `ccr1_top_p0000#448` (ply 157) [structural distance $d=99$, QAP obj=44]: the master played Rxe5; under μ our Rb8 plays the role of that game’s Rf4; cross-type roles: our Rb1 as their Nc8, our nd1 as their pe5; transferred move b1b4. (d) Retrieved from `ccr1_top_p0000#3602` (ply 154) [structural distance $d=94$, QAP obj=23]: the master played Rg4; under μ our Bh5 plays the role of that game’s Rg2 (a **cross-type** correspondence); cross-type roles: our nd1 as their pb4, our qc2 as their pd6; transferred move h4e7. Synthesis: no retrieved analog transferred to Qf1+ – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



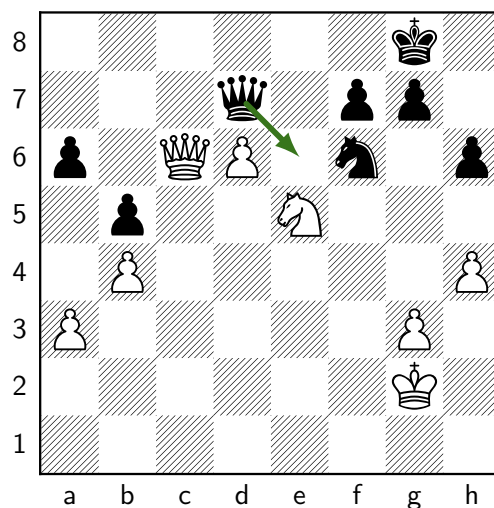
(a)



(b)



(c)



(d)

Figure 79: **Move 85, Black: Qh1#.** (a) Our position; Melanie plays Qh1# (f1h1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#239` (ply 160) [structural distance $d=95$, QAP obj=37]: the master played Rg1; under μ our Rb8 plays the role of that game’s Re1; cross-type roles: our Rb1 as their Nd4, our Ng3 as their Ph4; transferred move b1b7. (c) Retrieved from `ccr1_top_p0000#1706` (ply 158) [structural distance $d=102$, QAP obj=27]: the master played Qd1; under μ our Qf8 plays the role of that game’s Qh5; cross-type roles: our Ng3 as their Rg1, our Bh4 as their pb3; transferred move f1b5. (d) Retrieved from `ccr1_top_p0003#2695` (ply 183) [structural distance $d=102$, QAP obj=19]: the master played Qe6; under μ our Rb8 plays the role of that game’s Qd2 (a **cross-type** correspondence); cross-type roles: our Ng3 as their Pa6, our ra2 as their pd6; transferred move b1b5. Synthesis: no retrieved analog transferred to Qh1# – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.